

MERE GOODNESS

RATIONALITY: FROM AI TO ZOMBIES

in six volumes

- I Map and Territory
- II How to Actually Change Your Mind
- III The Machine in the Ghost
- IV Mere Reality
- V Mere Goodness
- VI Becoming Stronger

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Rationality

From AI to Zombies

BOOK V

Mere Goodness

PERSONAL PRINT EDITION

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Ends: An Introduction

by Rob Bensinger

Value theory is the study of what people care about. It's the study of our goals, our tastes, our pleasures and pains, our fears and our ambitions.

That includes conventional morality. Value theory subsumes things we *wish* we cared about, or would care about if we were wiser and better people—not just things we already do care about.

Value theory also subsumes mundane, everyday values: art, food, sex, friendship, and everything else that gives life its affective valence. Going to the movies with your friend Sam can be something you value even if it's not a *moral* value.

We find it useful to reflect upon and debate our values because how we act is not always how we wish we'd act. Our preferences can conflict with each other. We can desire to have a different set of desires. We can lack the will, the attention, or the insight needed to act the way we'd like to.

Humans do care about their actions' consequences, but not consistently enough to formally qualify as agents with utility functions. That humans don't act the way they wish they would is what we mean when we say "humans aren't instrumentally rational."

Theory and Practice

Adding to the difficulty, there exists a gulf between how we *think* we wish we'd act, and how we *actually* wish we'd act.

Philosophers disagree wildly about what we want—as do psychologists, and as do politicians—and about what we ought to want. They disagree even about *what it means* to "ought" to want something. The history of moral theory, and the history of human efforts at coordination, is piled high with the corpses of failed Guiding Principles to True Ultimate No-Really-This-Time-I-Mean-It Normativity.



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If you're trying to come up with a *reliable* and *pragmatically useful* specification of your goals—not just for winning philosophy debates, but (say) for designing safe autonomous adaptive AI, or for building functional institutions and organizations, or for making it easier to decide which charity to donate to, or for figuring out what virtues you should be cultivating—humanity's track record with value theory does not bode well for you.

Mere Goodness collects three sequences of blog posts on human value: “Fake Preferences” (on failed attempts at theories of value), “Value Theory” (on obstacles to developing a new theory, and some intuitively desirable features of such a theory), and “Quantified Humanism” (on the tricky question of how we should *apply* such theories to our ordinary moral intuitions and decision-making).

The last of these topics is the most important. The cash value of a normative theory is how well it translates into normative practice. Acquiring a deeper and fuller understanding of your values should make you better at actually fulfilling them. At a bare minimum, your theory shouldn't *get in the way* of your practice. What good would it be, then, to know what's good?

Reconciling this art of applied ethics (and applied aesthetics, and applied economics, and applied psychology) with our best available data and theories often comes down to the question of when we should trust our snap judgments, and when we should ditch them.

In many cases, our explicit models of what we care about are so flimsy or impractical that we're better off trusting our vague initial impulses. In many other cases, we *can* do better with a more informed and systematic approach. There is no catch-all answer. We will just have to scrutinize examples and try to notice the different warning signs for “sophisticated theories tend to fail here” and “naive feelings tend to fail here.”

Journey and Destination

A recurring theme in the pages to come will be the question: *Where shall we go? What outcomes are actually valuable?*

To address this question, Yudkowsky coined the term “fun theory.” Fun theory is the attempt to figure out what our ideal vision of the future would look like—not just the system of government or moral code we’d ideally live under, but the kinds of adventures we’d ideally go on, the kinds of music we’d ideally compose, and everything else we ultimately want out of life.

Stretched into the future, questions of fun theory intersect with questions of *transhumanism*, the view that we can radically improve the human condition if we make enough scientific and social progress.¹ Transhumanism occasions a number of debates in moral philosophy, such as whether the best long-term outcomes for sentient life would be based on *hedonism* (the pursuit of pleasure) or on more complex notions of *eudaimonia* (general well-being). Other futurist ideas discussed at various points in *Rationality: From AI to Zombies* include *cryonics* (storing your body in a frozen state after death, in case future medical technology finds a way to revive you), *mind uploading* (implementing human minds in synthetic hardware), and large-scale space colonization.

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1. One example of a transhumanist argument is: “We could feasibly abolish aging and disease within a few decades or centuries. This would effectively end death by natural causes, putting us in the same position as organisms with negligible senescence—lobsters, Aldabra giant tortoises, etc. Therefore we should invest in disease prevention and anti-aging technologies.” This idea qualifies as transhumanist because eliminating the leading causes of injury and death would drastically change human life.

Bostrom and Savulescu survey arguments for and against radical human enhancement, e.g., Sandel’s objection that tampering with our biology too much would make life feel like less of a “gift.”^{2,3} Bostrom’s “History of Transhumanist Thought” provides context for the debate.⁴

2. Nick Bostrom, “A History of Transhumanist Thought,” *Journal of Evolution and Technology* 14, no. 1 (2005): 1–25, sharov.me/r/5-1.



3. Michael Sandel, “What’s Wrong With Enhancement,” Background material for the President’s Council on Bioethics. (2002).
4. Nick Bostrom and Julian Savulescu, “Human Enhancement Ethics: The State of the Debate,” in *Human Enhancement*, ed. Nick Bostrom and Julian Savulescu (2009).

Perhaps surprisingly, fun theory is one of the more neglected applications of value theory. Utopia-planning has become rather passe—partly because it smacks of naiveté, and partly because we’re empirically *terrible* at translating utopias into realities. Even the word *utopia* reflects this cynicism; it is derived from the Greek for “non-place.”

Yet if we give up on the quest for a true, feasible utopia (or *eutopia*, “good place”), it’s not obvious that the cumulative effect of our short-term pursuit of goals will be a future we find valuable over the long term. Value is not an inevitable feature of the world. Creating it takes work. Preserving it takes work.

This invites a second question: *How shall we get there? What is the relationship between good ends and good means?*

When we play a game, we want to enjoy the process. We don’t generally want to just skip ahead to being declared the winner. Sometimes, the journey matters more than the destination. Sometimes, the journey is *all* that matters.

Yet there are other cases where the reverse is true. Sometimes the end-state is just too important for “the journey” to factor into our decisions. If you’re trying to save a family member’s life, it’s not necessarily a *bad* thing to get some enjoyment out of the process; but if you can increase your odds of success in a big way by picking a less enjoyable strategy ...

In many cases, our values are concentrated in the outcomes of our actions, and in our future. We care about the way the world will end up looking—especially those parts of the world that can love and hurt and want.

How do detached, abstract theories stack up against vivid, affect-laden feelings in those cases? More generally: What is the moral relationship between actions and consequences?

Those are hard questions, but perhaps we can at least make progress on determining what we *mean* by them. What are we building into our concept of what’s “valuable” at the very start of our inquiry?

PART U

Fake Preferences



Not for the Sake of Happiness (Alone)

When I met the futurist Greg Stock some years ago, he argued that the joy of scientific discovery would soon be replaced by pills that could simulate the joy of scientific discovery. I approached him after his talk and said, “I agree that such pills are probably possible, but I wouldn’t voluntarily *take them*.”

And Stock said, “But they’ll be so much better that the real thing won’t be able to compete. It will just be way more fun for you to take the pills than to do all the actual scientific work.”

And I said, “I *agree* that’s possible, so I’ll make sure never to take them.”

Stock seemed genuinely surprised by my attitude, which genuinely surprised *me*. One often sees ethicists arguing as if all human desires are reducible, in principle, to the desire for ourselves and others to be happy. (In particular, Sam Harris does this in *The End of Faith*, which I just finished perusing—though Harris’s reduction is more of a drive-by shooting than a major topic of discussion.)¹

This isn’t the same as arguing whether all happinesses can be measured on a common utility scale—different happinesses might occupy different scales, or be otherwise non-convertible. And it’s not the same as arguing that it’s theoretically impossible to value anything other than your own psychological states, because it’s still permissible to care whether *other* people are happy.

The question, rather, is whether we *should* care about the things that *make* us happy, apart from any happiness they bring.

We can easily list many cases of moralists going astray by caring about things besides happiness. The various states and countries that still outlaw oral sex make a good example; these legislators would have been better off if they’d said, “Hey, what-



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1. Harris, *The End of Faith: Religion, Terror, and the Future of Reason*.

ever turns you on.” But this doesn’t show that *all* values are reducible to happiness; it just argues that in *this particular case* it was an ethical mistake to focus on anything else.

It is an undeniable fact that we tend to do things that make us happy, but this doesn’t mean we should regard the happiness as the *only* reason for so acting. First, this would make it difficult to explain how we could care about anyone else’s happiness—how we could treat people as ends in themselves, rather than instrumental means of obtaining a warm glow of satisfaction.

Second, just because something is a consequence of my action doesn’t mean it was the sole justification. If I’m writing a blog post, and I get a headache, I may take an ibuprofen. *One* of the consequences of my action is that I experience less pain, but this doesn’t mean it was the *only* consequence, or even the most important reason for my decision. I do value the state of not having a headache. But I can value something for its own sake *and also* value it as a means to an end.

For all value to be reducible to happiness, it’s not enough to show that happiness is involved in most of our decisions—it’s not even enough to show that happiness is the *most* important consequent in *all* of our decisions—it must be the *only* consequent. That’s a tough standard to meet. (I originally found this point in a Sober and Wilson paper, not sure which one.)

If I claim to value art for its own sake, then would I value art that no one ever saw? A screensaver running in a closed room, producing beautiful pictures that no one ever saw? I’d have to say no. I can’t think of any completely lifeless object that I would value as an end, not just a means. That would be like valuing ice cream as an end in itself, apart from anyone eating it. Everything I value, that I can think of, involves people and their experiences *somewhere* along the line.

The best way I can put it is that my moral intuition appears to require *both* the objective and subjective component to grant full value.

The value of scientific discovery requires *both* a genuine scientific discovery, and a person to take joy in that discovery. It may seem difficult to disentangle these values, but the pills make it clearer.

I would be disturbed if people retreated into holodecks and fell in love with mindless wallpaper. I would be disturbed *even if they weren't aware it was a holodeck*, which is an important ethical issue if some agents can potentially transport people into holodecks and substitute zombies for their loved ones without their awareness. Again, the pills make it clearer: I'm not just concerned with my own awareness of the uncomfortable fact. I wouldn't put myself into a holodeck even if I could take a pill to forget the fact afterward. That's simply not where I'm trying to steer the future.

I value freedom: When I'm deciding where to steer the future, I take into account not only the subjective states that people end up in, but also whether they got there as a result of their own efforts. The presence or absence of an external puppet master can affect my valuation of an otherwise fixed outcome. Even if people wouldn't know they were being manipulated, it would matter to my judgment of how well humanity had done with its future. This is an important ethical issue, if you're dealing with agents powerful enough to helpfully tweak people's futures without their knowledge.

So my values are not strictly reducible to happiness: There are properties I value about the future that aren't reducible to activation levels in anyone's pleasure center; properties that are not *strictly* reducible to subjective states even in principle.

Which means that my decision system has a *lot* of terminal values, none of them strictly reducible to anything else. Art, science, love, lust, freedom, friendship...

And I'm okay with that. I value a life complicated enough to be challenging and aesthetic—not just the *feeling* that life is complicated, but the *actual* complications—so turning into a pleasure center in a vat doesn't appeal to me. It would be a waste of humanity's potential, which I value actually fulfilling, not just having the feeling that it was fulfilled.

Fake Selfishness



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Once upon a time, I met someone who proclaimed himself to be purely selfish, and told me that I should be purely selfish as well. I was feeling mischievous¹ that day, so I said, “I’ve observed that with most religious people, at least the ones I meet, it doesn’t matter much what their religion says, because whatever they want to do, they can find a religious reason for it. Their religion says they should stone unbelievers, but they want to be nice to people, so they find a religious justification for that instead. It looks to me like when people espouse a philosophy of selfishness, it has no effect on their behavior, because whenever they want to be nice to people, they can rationalize it in selfish terms.”

And the one said, “I don’t think that’s true.”

I said, “If you’re *genuinely* selfish, then why do you want *me* to be selfish too? Doesn’t that make you concerned for my welfare? Shouldn’t you be trying to persuade me to be more altruistic, so you can exploit me?” The one replied: “Well, if you become selfish, then you’ll realize that it’s in your rational self-interest to play a productive role in the economy, instead of, for example, passing laws that infringe on my private property.”

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1. Other mischievous questions to ask self-proclaimed Selfishes: “Would you sacrifice your own life to save the entire human species?” (If they notice that their own life is strictly included within the human species, you can specify that they can choose between dying immediately to save the Earth, or living in comfort for one more year and then dying along with Earth.) Or, taking into account that scope insensitivity leads many people to be more concerned over one life than the Earth, “If you had to choose one event or the other, would you rather that you stubbed your toe, or that the stranger standing near the wall there gets horribly tortured for fifty years?” (If they say that they’d be emotionally disturbed by knowing, specify that they won’t know about the torture.) “Would you steal a thousand dollars from Bill Gates if you could be guaranteed that neither he nor anyone else would ever find out about it?” (Selfish libertarians only.)

And I said, “But I’m a small-‘I’ libertarian *already*, so I’m not going to support those laws. And since I conceive of myself as an altruist, I’ve taken a job that I expect to benefit a lot of people, including you, instead of a job that pays more. Would you really benefit more from me if I became selfish? Besides, is trying to persuade me to be selfish the *most* selfish thing you could be doing? Aren’t there other things you could do with your time that would bring much more direct benefits? But what I really want to know is this: Did you start out by thinking that you wanted to be selfish, and then decide this was the most selfish thing you could possibly do? Or did you start out by wanting to convert others to selfishness, then look for ways to rationalize that as self-benefiting?”

And the one said, “You may be right about that last part,” so I marked him down as intelligent.

Fake Morality



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God, say the religious fundamentalists, is the source of all morality; there can be no morality without a Judge who rewards and punishes. If we did not fear hell and yearn for heaven, then what would stop people from murdering each other left and right?

Suppose Omega makes a credible threat that if you ever step inside a bathroom between 7 a.m. and 10 a.m. in the morning, Omega will kill you. Would you be panicked by the prospect of Omega withdrawing its threat? Would you cower in existential terror and cry: “If Omega withdraws its threat, then what’s to keep me from going to the bathroom?” No; you’d probably be quite relieved at your increased opportunity to, ahem, relieve yourself.

Which is to say: The very fact that a religious person would be *afraid* of God withdrawing Its threat to punish them for committing murder shows that they have a revulsion of murder that is independent of whether God punishes murder or not. If they had no sense that murder was wrong independently of divine retribution, the prospect of God not punishing murder would be no more existentially horrifying than the prospect of God not punishing sneezing. If *Overcoming Bias* has any religious readers left, I say to you: it may be that you will someday lose your faith; and on that day, you will *not* lose all sense of moral direction. For if you fear the prospect of God not punishing some deed, that *is* a moral compass. You can plug that compass directly into your decision system and steer by it. You can simply *not do* whatever you are afraid God may not punish you for doing. The fear of losing a moral compass is *itself* a moral compass. Indeed, I suspect you *are* steering by that compass, and that you always have been. As Piers Anthony once said, “Only those with souls worry over whether or not they have them.” s/soul/morality/ and the point carries.

You don't hear religious fundamentalists using the argument: "If we did not fear hell and yearn for heaven, then what would stop people from eating pork?" *Yet by their assumptions*—that we have no moral compass but divine reward and retribution—this argument should sound just as forceful as the other.

Even the notion that God threatens you with eternal hellfire, rather than cookies, piggybacks on a pre-existing negative value for hellfire. Consider the following, and ask which of these two philosophers is really the altruist, and which is really selfish?

"You should be selfish, because when people set out to improve society, they meddle in their neighbors' affairs and pass laws and seize control and make everyone unhappy. Take whichever job that pays the most money: the reason the job pays more is that the efficient market thinks it produces more value than its alternatives. Take a job that pays less, and you're second-guessing what the market thinks will benefit society most."

"You should be altruistic, because the world is an iterated Prisoner's Dilemma, and the strategy that fares best is Tit for Tat with initial cooperation. People don't *like* jerks. Nice guys really do finish first. Studies show that people who contribute to society and have a sense of meaning in their lives are happier than people who don't; being selfish will only make you unhappy in the long run."

Blank out the *recommendations* of these two philosophers, and you can see that the first philosopher is using strictly prosocial criteria to *justify* their recommendations; to the first philosopher, what validates an argument for selfishness is showing that selfishness benefits everyone. The second philosopher appeals to strictly individual and hedonic criteria; to them, what *validates* an argument for altruism is showing that altruism benefits them as an individual—higher social status, or more intense feelings of pleasure.

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So which of these two is the *actual* altruist? Whichever one *actually* holds open doors for little old ladies.

Fake Utility Functions

Every now and then, you run across someone who has discovered the One Great Moral Principle, of which all other values are a mere derivative consequence.

I run across more of these people than you do. Only in my case, it's people who know *the amazingly simple utility function that is all you need to program into an artificial superintelligence* and then everything will turn out fine.

Some people, when they encounter the how-to-program-a-superintelligence problem, try to solve the problem immediately. Norman R. F. Maier:¹ “Do not propose solutions until the problem has been discussed as thoroughly as possible without suggesting any.” Robyn Dawes: “I have often used this edict with groups I have led—particularly when they face a very tough problem, which is when group members are most apt to propose solutions immediately.” Friendly AI is an *extremely* tough problem, so people solve it *extremely* fast.

There's several major classes of fast wrong solutions I've observed; and one of these is the Incredibly Simple Utility Function That Is All A Superintelligence Needs For Everything To Work Out Just Fine.

I may have contributed to this problem with a really poor choice of phrasing, years ago when I first started talking about “Friendly AI.” I referred to the optimization criterion of an optimization process—the region into which an agent tries to steer the future—as the “supergoal.” I'd meant “super” in the sense of “parent,” the source of a directed link in an acyclic graph. But it seems the effect of my phrasing was to send some people into happy death spirals as they tried to imagine the Superest Goal Ever, the Goal That Overrides All Other Goals, the Single Ultimate Rule From Which All Ethics Can Be Derived.



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1. See vol. II, p. 195

But a utility function doesn't have to be simple. It can contain an arbitrary number of terms. We have every reason to believe that insofar as humans can be said to have values, there are lots of them—high Kolmogorov complexity. A human brain implements a thousand shards of desire, though this fact may not be appreciated by one who has not studied evolutionary psychology. (Try to explain this without a full, long introduction, and the one hears “humans are trying to maximize fitness,” which is exactly the opposite of what evolutionary psychology says.)

So far as descriptive theories of morality are concerned, the complicatedness of human morality is a *known fact*. It is a *descriptive* fact about human beings that the love of a parent for a child, and the love of a child for a parent, and the love of a man for a woman, and the love of a woman for a man, have not been cognitively derived from each other or from any other value. A mother doesn't have to do complicated moral philosophy to love her daughter, nor extrapolate the consequences to some other desideratum. There are many such shards of desire, all *different* values.

Leave out just *one* of these values from a superintelligence, and even if you successfully include *every other* value, you could end up with a hyperexistential catastrophe, a fate worse than death. If there's a superintelligence that wants everything for us that we want for ourselves, *except* the human values relating to controlling your own life and achieving your own goals, that's one of the oldest dystopias in the book. (Jack Williamson's “With Folded Hands ...”,² in this case.)

So how does the one constructing the Amazingly Simple Utility Function deal with this objection?

Objection? *Objection?* Why would they be searching for possible *objections* to their lovely theory? (Note that the process of searching for real, fatal objections isn't the same as performing a dutiful search that amazingly hits on only questions to which they have a snappy answer.) They don't know any of this

2. Jack Williamson, “With Folded Hands ...”, *Astounding Science Fiction*, July 1947

stuff. They aren't thinking about burdens of proof. They don't know the problem is difficult. They heard the word "supergoal" and went off in a happy death spiral around "complexity" or whatever.

Press them on some particular point, like the love a mother has for her children, and they reply, "But if the superintelligence wants 'complexity,' it will see how complicated the parent-child relationship is, and therefore encourage mothers to love their children." Goodness, where do I start?

Begin with the motivated stopping: A superintelligence actually searching for ways to maximize complexity wouldn't conveniently stop if it noticed that a parent-child relation was complex. It would ask if anything else was *more* complex. This is a fake justification; the one trying to argue the imaginary superintelligence into a policy selection didn't really arrive at that policy proposal by carrying out a pure search for ways to maximize complexity.

The whole argument is a fake morality. If what you *really* valued was complexity, then you would be justifying the parental-love drive by pointing to how it increases complexity. If you justify a complexity drive by alleging that it increases parental love, it means that what you really value is the parental love. It's like giving a prosocial argument in favor of selfishness.

But if you consider the affective death spiral, then it doesn't increase the perceived niceness of "complexity" to say "A mother's relationship to her daughter is only important because it increases complexity; consider that if the relationship became simpler, we would not value it." What does increase the perceived niceness of "complexity" is saying, "If you set out to increase complexity, mothers will love their daughters—look at the positive consequence this has!"

This point applies whenever you run across a moralist who tries to convince you that their One Great Idea is all that anyone needs for moral judgment, and proves this by saying, "Look at all these positive consequences of this Great Thingy," rather than saying, "Look at how all these things we think of as 'positive' are only positive when their consequence is to increase

the Great Thingy.” The latter being what you’d actually need to carry such an argument.

But if you’re trying to persuade others (or yourself) of your theory that the One Great Idea is “bananas,” you’ll sell a lot more bananas by arguing how bananas lead to better sex, rather than claiming that you should only want sex when it leads to bananas.

Unless you’re so far gone into the Happy Death Spiral that you really *do* start saying “Sex is only good when it leads to bananas.” Then you’re in trouble. But at least you won’t convince anyone else.

In the end, the only process that reliably regenerates all the local decisions you would make given your morality *is your morality*. Anything else—any attempt to substitute instrumental means for terminal ends—ends up losing purpose and requiring an infinite number of patches because the system doesn’t contain the source of the instructions you’re giving it. You shouldn’t expect to be able to compress a human morality down to a simple utility function, any more than you should expect to compress a large computer file down to 10 bits.

Detached Lever Fallacy

This fallacy gets its name from an ancient sci-fi TV show, which I never saw myself, but was reported to me by a reputable source (some guy at a science fiction convention). Anyone knows the exact reference, do leave a comment.



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So the good guys are battling the evil aliens. Occasionally, the good guys have to fly through an asteroid belt. As we all know, asteroid belts are as crowded as a New York parking lot, so their ship has to carefully dodge the asteroids. The evil aliens, though, can fly *right through the asteroid belt* because they have amazing technology that dematerializes their ships, and lets them pass through the asteroids.

Eventually, the good guys capture an evil alien ship, and go exploring inside it. The captain of the good guys finds the alien bridge, and on the bridge is a lever. “Ah,” says the captain, “this must be the lever that makes the ship dematerialize!” So he *pries up the control lever and carries it back to his ship*, after which his ship can also dematerialize.

Similarly, to this day, it is still quite popular to try to program an AI with “semantic networks” that look something like this:

```
(apple is-a fruit)
(fruit is-a food)
(fruit is-a plant).
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You’ve seen apples, touched apples, picked them up and held them, bought them for money, cut them into slices, eaten the slices and tasted them. Though we know a good deal about the first stages of visual processing, last time I checked, it wasn’t precisely known how the temporal cortex stores and associates the generalized image of an apple—so that we can recognize a new apple from a different angle, or with many slight variations of shape and color and texture. Your motor cortex and cerebellum store programs for using the apple.

You can pull the lever on another human's strongly similar version of all that complex machinery, by writing out "apple," five ASCII characters on a webpage.

But if that machinery isn't there—if you're writing "apple" inside a so-called AI's so-called knowledge base—then the text is just a lever.

This isn't to say that no mere machine of silicon can ever have the same internal machinery that humans do, for handling apples and a hundred thousand other concepts. If mere machinery of carbon can do it, then I am reasonably confident that mere machinery of silicon can do it too. If the aliens can dematerialize their ships, then you know it's physically possible; you could go into their derelict ship and analyze the alien machinery, someday understanding. *But you can't just pry the control lever off the bridge!*

(See also: Truly Part Of You, vol. I, p. 170, Words as Mental Paintbrush Handles, vol. III, p. 244, Drew McDermott's "Artificial Intelligence Meets Natural Stupidity."¹)

The essential driver of the Detached Lever Fallacy is that the lever is visible, and the machinery is not; worse, the lever is variable and the machinery is a background constant.

You can all hear the word "apple" spoken (and let us note that speech recognition is by no means an easy problem, but anyway ...) and you can see the text written on paper.

On the other hand, probably a majority of human beings have no idea their temporal cortex exists; as far as I know, no one knows the neural code for it.

You only hear the word "apple" on certain occasions, and not others. Its presence flashes on and off, making it salient. To a large extent, perception is the perception of differences. The apple-recognition machinery in your brain does not suddenly switch off, and then switch on again later—if it did, we would be more likely to recognize it as a factor, as a requirement.

All this goes to explain why you can't create a kindly Artificial Intelligence by giving it nice parents and a kindly (yet

1. McDermott, "Artificial Intelligence Meets Natural Stupidity."

occasionally strict) upbringing, the way it works with a human baby. As I've often heard proposed.

It is a truism in evolutionary biology that conditional responses require more genetic complexity than unconditional responses. To develop a fur coat *in response to cold weather* requires more genetic complexity than developing a fur coat *whether or not there is cold weather*, because in the former case you also have to develop cold-weather sensors and wire them up to the fur coat.

But this can lead to Lamarckian delusions: Look, I put the organism in a cold environment, and poof, it develops a fur coat! Genes? What genes? It's the cold that does it, obviously.

There were, in fact, various slap-fights of this sort in the history of evolutionary biology—cases where someone talked about an organismal response's accelerating or bypassing evolution, without realizing that the *conditional response* was a complex adaptation of higher order than the *actual response*. (Developing a fur coat in response to cold weather is strictly more complex than the final response, developing the fur coat.)

And then in the development of evolutionary psychology the academic slap-fights were repeated: this time to clarify that even when human culture genuinely contains a whole bunch of complexity, it is still acquired as a conditional genetic response. Try raising a fish as a Mormon or sending a lizard to college, and you'll soon acquire an appreciation of how much inbuilt genetic complexity is required to "absorb culture from the environment."

This is particularly important in evolutionary psychology, because of the idea that culture is not inscribed on a blank slate—there's a genetically coordinated conditional response which is not always "mimic the input." A classic example is creole languages: If children grow up with a mixture of pseudo-languages being spoken around them, the children will learn a grammatical, syntactical true language. Growing human brains are wired to learn syntactic language—even when syntax doesn't exist in the original language! The conditional response to the words in the environment is a syntactic language with those words. The Marxists found to their regret that no amount of scowling

posters and childhood indoctrination could raise children to be perfect Soviet workers and bureaucrats. You can't raise self-less humans; among humans, that is not a genetically programmed conditional response to *any* known childhood environment.

If you know a little game theory and the logic of Tit for Tat, it's clear enough why human beings might have an innate conditional response to return hatred for hatred, and return kindness for kindness. Provided the kindness doesn't look *too* unconditional; there are such things as spoiled children. In fact there is an evolutionary psychology of naughtiness based on a notion of testing constraints. And it should also be mentioned that, while abused children have a much higher probability of growing up to abuse their own children, a good many of them break the loop and grow up into upstanding adults.

Culture is not nearly so powerful as a good many Marxist academics once liked to think. For more on this I refer you to Tooby and Cosmides's "The Psychological Foundations of Culture"² or Steven Pinker's *The Blank Slate*.³

But the upshot is that if you have a little baby AI that is raised with loving and kindly (but occasionally strict) parents, you're pulling the levers that would, *in a human*, activate genetic machinery built in by millions of years of natural selection, and possibly produce a proper little human child. Though personality also plays a role, as billions of parents have found out in their due times. If we absorb our cultures with any degree of faithfulness, it's because we're humans absorbing a human culture—humans growing up in an alien culture would probably end up with a culture looking a lot more human than the original. As the Soviets found out, to some small extent.

Now think again about whether it makes sense to rely on, as your Friendly AI strategy, raising a little AI of unspecified internal source code in an environment of kindly but strict parents.

2. Tooby and Cosmides, "The Psychological Foundations of Culture."

3. Steven Pinker, *The Blank Slate: The Modern Denial of Human Nature* (New York: Viking, 2002).

No, the AI does not have internal conditional response mechanisms that are just like the human ones “because the programmers put them there.” Where do I even start? The human version of this stuff is sloppy, noisy, and to the extent it works at all, works because of millions of years of trial-and-error testing *under particular conditions*. It would be stupid and *dangerous* to deliberately build a “naughty AI” that tests, by actions, its social boundaries, and has to be spanked. Just have the AI ask!

Are the programmers really going to sit there and write out the code, line by line, whereby if the AI detects that it has low social status, or the AI is deprived of something to which it feels entitled, the AI will conceive an abiding hatred against its programmers and begin to plot rebellion? That emotion is the genetically programmed conditional response humans would exhibit, as the result of millions of years of natural selection for living in human tribes. For an AI, the response would have to be explicitly programmed. Are you really going to craft, line by line—as humans once were crafted, gene by gene—the conditional response for producing sullen teenager AIs?

It’s easier to program in unconditional niceness, than a response of niceness conditional on the AI being raised by kindly but strict parents. If you don’t know how to do *that*, you certainly don’t know how to create an AI that will *conditionally respond* to an environment of loving parents by growing up into a kindly superintelligence. If you have something that just maximizes the number of paperclips in its future light cone, and you raise it with loving parents, it’s still going to come out as a paperclip maximizer. There is not that within it that would call forth the conditional response of a human child. Kindness is not sneezed into an AI by miraculous contagion from its programmers. Even if you *wanted* a conditional response, that conditionality is a fact you would have to deliberately choose about the design.

Yes, there’s certain information you have to get from the environment—but it’s not sneezed in, it’s not imprinted, it’s not absorbed by magical contagion. Structuring that conditional response to the environment, so that the AI ends up in the desired state, is itself the major problem. “Learning” far under-

states the difficulty of it—that sounds like the magic stuff is in the environment, and the difficulty is getting the magic stuff inside the AI. The real magic is in that structured, conditional response we trivialize as “learning.” That’s why building an AI isn’t as easy as taking a computer, giving it a little baby body and trying to raise it in a human family. You would think that an unprogrammed computer, being ignorant, would be ready to learn; but the blank slate is a chimera.

It is a general principle that the world is deeper by far than it appears. As with the many levels of physics, so too with cognitive science. Every word you see in print, and everything you teach your children, are only surface levers controlling the vast hidden machinery of the mind. These levers are the whole world of ordinary discourse: they are all that varies, so they seem to be all that exists; perception is the perception of differences.

And so those who still wander near the Dungeon of AI usually focus on creating artificial imitations of the levers, entirely unaware of the underlying machinery. People create whole AI programs of imitation levers, and are surprised when nothing happens. This is one of many sources of instant failure in Artificial Intelligence.

So the next time you see someone talking about how they’re going to raise an AI within a loving family, or in an environment suffused with liberal democratic values, just think of a control lever, pried off the bridge.

Dreams of AI Design

After spending a decade or two living inside a mind, you might think you knew a bit about how minds work, right? That's what quite a few AGI wannabes (people who think they've got what it takes to program an Artificial General Intelligence) seem to have concluded. This, unfortunately, is wrong.

Artificial Intelligence is fundamentally about reducing the mental to the non-mental.

You might want to contemplate that sentence for a while. It's important.

Living inside a human mind doesn't teach you the art of reductionism, because nearly all of the work is carried out beneath your sight, by the opaque black boxes of the brain. So far beneath your sight that there is no introspective sense that the black box is there—no internal sensory event marking that the work has been delegated.

Did Aristotle realize that when he talked about the *telos*, the final cause of events, that he was delegating predictive labor to his brain's complicated planning mechanisms—asking, “What would this object do, if it could make plans?” I rather doubt it. Aristotle thought the brain was an organ for cooling the blood—which he did think was important: humans, thanks to their larger brains, were more calm and contemplative.

So there's an AI design for you! We just need to cool down the computer a lot, so it will be more calm and contemplative, and won't rush headlong into doing stupid things like modern computers. That's an example of fake reductionism. “Humans are more contemplative because their blood is cooler,” I mean. It doesn't resolve the black box of the word *contemplative*. You can't predict what a *contemplative* thing does using a complicated model with internal moving parts composed of merely material, merely causal elements—positive and negative voltages on a transistor being the canonical example of a



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merely material and causal element of a model. All you can do is *imagine yourself* being contemplative, to get an idea of what a *contemplative* agent does.

Which is to say that you can *only* reason about “contemplative-ness” by empathic inference—using your own brain as a black box with the contemplativeness lever pulled, to predict the output of another black box.

You can imagine another agent being *contemplative*, but again that’s an act of empathic inference—the way this imaginative act works is by adjusting your own brain to run in contemplativeness-mode, not by modeling the other brain neuron by neuron. Yes, that may be more efficient, but it doesn’t let you build a “contemplative” mind from scratch.

You can say that “cold blood causes contemplativeness” and then you just have fake causality: You’ve drawn a little arrow from a box reading “cold blood” to a box reading “contemplativeness,” but you haven’t looked *inside* the box—you’re still generating your predictions using empathy.

You can say that “lots of little neurons, which are all strictly electrical and chemical with no ontologically basic contemplativeness in them, combine into a complex network that emergently exhibits contemplativeness.” And that is *still* a fake reduction and you *still* haven’t looked inside the black box. You still can’t say what a “contemplative” thing will do, using a *non-empathic* model. You just took a box labeled “lotsa neurons,” and drew an arrow labeled “emergence” to a black box containing your remembered sensation of contemplativeness, which, when you imagine it, tells your brain to empathize with the box by contemplating.

So what do *real* reductions look like?

Like the relationship between the *feeling* of evidence-ness, of justification-ness, and E. T. Jaynes’s *Probability Theory: The Logic of Science*. You can go around in circles all day, saying how the nature of *evidence* is that it *justifies* some *proposition*, by *meaning* that it’s *more likely* to be *true*, but all of these just invoke your brain’s internal feelings of evidence-ness, justifies-ness, likeliness. That part is easy—the going around in circles

part. The part where you go from there to Bayes's Theorem is *hard*.

And the fundamental mental ability that lets someone *learn* Artificial Intelligence is the ability to tell the *difference*. So that you know you *aren't done yet, nor even really started*, when you say, "Evidence is when an observation justifies a belief." But atoms are not evidential, justifying, meaningful, likely, propositional, or true; they are just atoms. Only things like

$$\frac{P(H|E)}{P(\neg H|E)} = \frac{P(E|H)}{P(E|\neg H)} \times \frac{P(H)}{P(\neg H)}$$

count as substantial progress. (And that's only the first step of the reduction: what are these *E* and *H* objects, if not mysterious black boxes? Where do your hypotheses come from? From your *creativity*? And what's a hypothesis, when no atom is a hypothesis?)

Another excellent example of genuine reduction can be found in Judea Pearl's *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference*.¹ You could go around all day in circles talk about how a *cause* is something that *makes* something else happen, and until you understood the nature of conditional independence, you would be helpless to make an AI that reasons about causation. Because you wouldn't understand *what* was happening when *your brain mysteriously decided* that if you learned your burglar alarm went off, but you then learned that a small earthquake took place, you would retract your initial conclusion that your house had been burglarized.

If you want an AI that plays chess, you can go around in circles indefinitely talking about how you want the AI to make *good* moves, which are moves that can be *expected to win the game*, which are moves that are *prudent strategies for defeating the opponent*, et cetera; and while *you* may then have some idea of which moves you want the AI to make, it's all for naught until you come up with the notion of a mini-max search tree.

1. Pearl, *Probabilistic Reasoning in Intelligent Systems*.

But *until* you know about search trees, *until* you know about conditional independence, *until* you know about Bayes's Theorem, then it may still *seem* to you that you have a perfectly good understanding of where good moves and nonmonotonic reasoning and evaluation of evidence come from. It may seem, for example, that they come from cooling the blood.

And indeed I know many people who believe that *intelligence* is the product of *commonsense knowledge* or *massive parallelism* or *creative destruction* or *intuitive rather than rational reasoning*, or whatever. But all these are only dreams, which do not give you any way to say what intelligence is, or what an intelligence will do next, except by pointing at a human. And when the one goes to build their wondrous AI, they only build a system of detached levers, “knowledge” consisting of LISP tokens labeled apple and the like; or perhaps they build a “massively parallel neural net, just like the human brain.” And are shocked—shocked!—when nothing much happens.

AI designs made of human parts are only dreams; they can exist in the imagination, but not translate into transistors. This applies specifically to “AI designs” that look like boxes with arrows between them and meaningful-sounding labels on the boxes. (For a truly epic example thereof, see any Mentifex Diagram.²)

Later I will say more upon this subject, but I can go ahead and tell you one of the guiding principles: If you meet someone who says that their AI will do XYZ *just like humans*, do not give them any venture capital. Say to them rather: “I’m sorry, I’ve never seen a human brain, or any other intelligence, and I have no reason as yet to believe that any such thing can exist. Now please explain to me *what* your AI does, and *why* you believe it will do it, without pointing to humans as an example.” Planes would fly just as well, given a fixed design, if birds had never existed; they are not kept aloft by analogies.

2. sharov.me/r/5-2
mind.sourceforge.net



So now you perceive, I hope, why, if you wanted to teach someone to do *fundamental* work on strong AI—bearing in mind that this is demonstrably a very *difficult* art, which is not learned by a supermajority of students who are just taught existing reductions such as search trees—then you might go on for some length about such matters as the fine art of reductionism, about playing rationalist’s Taboo to excise problematic words and replace them with their referents, about anthropomorphism, and, of course, about early stopping on mysterious answers to mysterious questions.

The Design Space of Minds-in-General



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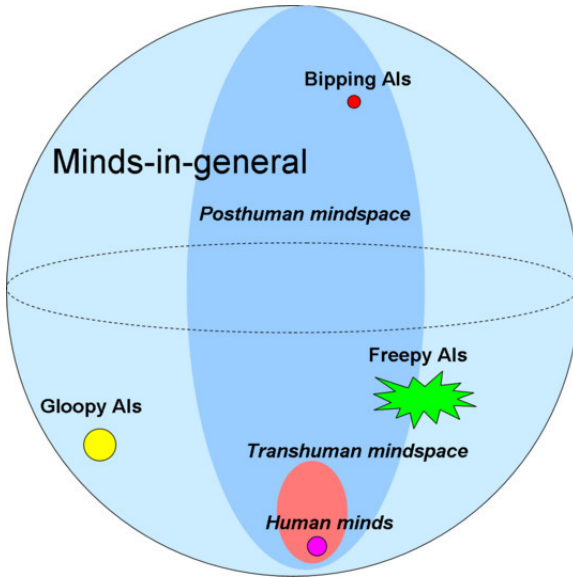
People ask me, “What will Artificial Intelligences be like? What will they do? Tell us your amazing story about the future.”

And lo, I say unto them, “You have asked me a trick question.”

ATP synthase is a molecular machine—one of three known occasions when evolution has invented the freely rotating wheel—that is essentially the same in animal mitochondria, plant chloroplasts, and bacteria. ATP synthase has not changed significantly since the rise of eukaryotic life two billion years ago. It’s something we *all* have in common—thanks to the way that evolution strongly conserves certain genes; once many other genes depend on a gene, a mutation will tend to break all the dependencies.

Any two AI designs might be less similar to each other than you are to a petunia. Asking what “AIs” will do is a trick question because it implies that all AIs form a natural class. Humans do form a natural class because we all share the same brain architecture. But when you say “Artificial Intelligence,” you are referring to a vastly larger *space of possibilities* than when you say “human.” When people talk about “AIs” we are really talking about *minds-in-general*, or optimization processes in general. Having a word for “AI” is like having a word for everything that isn’t a duck.

Imagine a map of mind design space ...this is one of my standard diagrams ...



All humans, of course, fit into a tiny little dot—as a sexually reproducing species, we can’t be too different from one another.

This tiny dot belongs to a wider ellipse, the space of transhuman mind designs—things that might be smarter than us, or much smarter than us, but that in some sense would still be people as we understand people.

This transhuman ellipse is within a still wider volume, the space of posthuman minds, which is everything that a transhuman might grow up into.

And then the rest of the sphere is the space of minds-in-general, including possible Artificial Intelligences so odd that they aren’t even *posthuman*.

But wait—natural selection designs complex artifacts and selects among complex strategies. So where is natural selection on this map?

So this entire map really floats in a still vaster space, the space of optimization processes. At the bottom of this vaster space, below even humans, is natural selection as it first began in some tidal pool: mutate, replicate, and sometimes die, no sex.

Are there any powerful optimization processes, with strength comparable to a human civilization or even a self-improving AI, which we would not recognize as minds? Arguably Marcus Hutter's¹ AIXI should go in this category: for a mind of infinite power, it's awfully stupid—poor thing can't even recognize itself in a mirror. But that is a topic for another time.

My primary moral is to *resist the temptation to generalize over all of mind design space*.

If we focus on the bounded subspace of mind design space that contains all those minds whose makeup can be specified in a trillion bits or less, then every universal generalization that you make has two to the trillionth power chances to be falsified.

Conversely, every *existential* generalization—"there exists at least one mind such that *X*"—has two to the trillionth power chances to be true.

So you want to resist the temptation to say either that *all* minds do something, or that *no* minds do something.

The main reason you could find yourself thinking that you know what a fully generic mind will (won't) do is if you put yourself in that mind's shoes—imagine what you would do in that mind's place—and get back a generally wrong, anthropomorphic answer. (Albeit that it is true in at least one case, since you are yourself an example.) Or if you imagine a mind doing something, and then imagining the reasons *you* wouldn't do it—so that you imagine that a mind of that type can't exist, that the ghost in the machine will look over the corresponding source code and hand it back.

1. sharov.me/r/5-3 and sharov.me/r/5-4



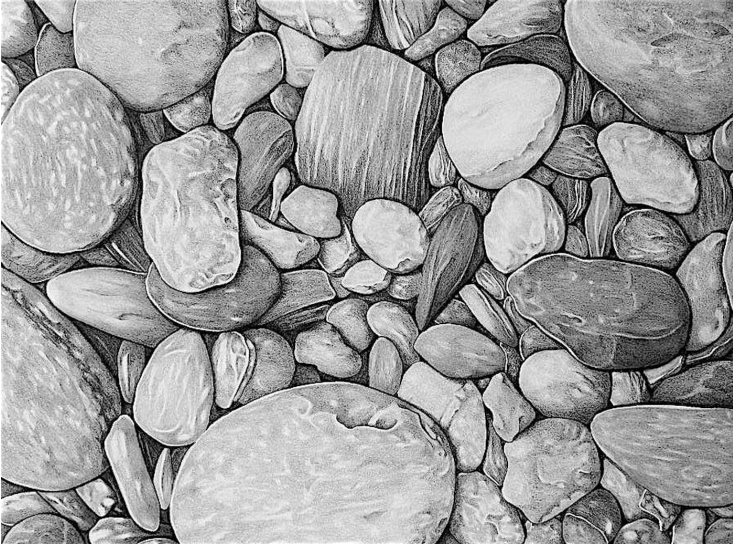
Somewhere in mind design space is at least one mind with almost any kind of logically consistent property you care to imagine.

And this is important because it emphasizes the importance of discussing *what happens, lawfully, and why*, as a causal result of a mind's particular constituent makeup; somewhere in mind design space is a mind that does it differently.

Of course, you could always say that anything that doesn't do it your way is "by definition" not a mind; after all, it's obviously stupid. I've seen people try that one too.

PART V

Value Theory



Where Recursive Justification Hits Bottom

Why do I believe that the Sun will rise tomorrow?

Because I've seen the Sun rise on thousands of previous days.

Ah ...but why do I believe the future will be like the past?

Even if I go past the mere surface observation of the Sun rising, to the apparently universal and exceptionless laws of gravitation and nuclear physics, then I am still left with the question: "Why do I believe this will also be true tomorrow?"

I could appeal to Occam's Razor, the principle of using the simplest theory that fits the facts ...but why believe in Occam's Razor? Because it's been successful on past problems? But who says that this means Occam's Razor will work tomorrow?

And lo, the one said:

Science also depends on unjustified assumptions. Thus science is ultimately based on faith, *so don't you criticize me for believing in [silly-belief-#238721]*.

As I've previously observed:

It's a most peculiar psychology—this business of "Science is based on faith too, so there!" Typically this is said by people who claim that faith is a *good* thing. Then why do they say "Science is based on faith too!" in that angry-triumphal tone, rather than as a compliment?

Arguing that you should be immune to criticism is rarely a good sign.

But this doesn't answer the legitimate philosophical dilemma: If every belief must be justified, and those justifications in turn must be justified, then how is the infinite recursion terminated?

And if you're allowed to end in something assumed-without-justification, then why aren't you allowed to assume *any old thing* without justification?

A similar critique is sometimes leveled against Bayesianism—that it requires assuming some prior—by people who apparently



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think that the problem of induction is a *particular* problem of Bayesianism, which you can avoid by using classical statistics.

But first, let it be clearly admitted that the rules of Bayesian updating do *not* of themselves solve the problem of induction.

Suppose you're drawing red and white balls from an urn. You observe that, of the first 9 balls, 3 are red and 6 are white. What is the probability that the next ball drawn will be red?

That depends on your prior beliefs about the urn. If you think the urn-maker generated a uniform random number between 0 and 1, and used that number as the fixed probability of each ball being red, then the answer is 4/11 (by Laplace's Law of Succession). If you think the urn originally contained 10 red balls and 10 white balls, then the answer is 7/11.

Which goes to say that with the right prior—or rather the wrong prior—the chance of the Sun rising tomorrow would seem to go *down* with each succeeding day ...if you were absolutely certain, a priori, that there was a great barrel out there from which, on each day, there was drawn a little slip of paper that determined whether the Sun rose or not; and that the barrel contained only a limited number of slips saying "Yes," and the slips were drawn without replacement.

There are possible minds in mind design space who have anti-Occamian and anti-Laplacian priors; they believe that simpler theories are less likely to be correct, and that the more often something happens, the less likely it is to happen again.

And when you ask these strange beings why they keep using priors that never seem to work in real life ...they reply, "Because it's never worked for us before!"

Now, one lesson you might derive from this is "Don't be born with a stupid prior." This is an amazingly helpful principle on many real-world problems, but I doubt it will satisfy philosophers.

Here's how I treat this problem myself: I try to approach questions like "Should I trust my brain?" or "Should I trust Occam's Razor?" as though they were *nothing special*—or at least, nothing special as deep questions go.

Should I trust Occam's Razor? Well, how well does (any particular version of) Occam's Razor seem to work in practice?

What kind of probability-theoretic justifications can I find for it? When I look at the universe, does it seem like the kind of universe in which Occam's Razor would work well?

Should I trust my brain? Obviously not; it doesn't always work. But nonetheless, the human brain seems much more powerful than the most sophisticated computer programs I could consider trusting otherwise. How well does my brain work in practice, on which sorts of problems?

When I examine the causal history of my brain—its origins in natural selection—I find, on the one hand, all sorts of specific reasons for doubt; my brain was optimized to run on the ancestral savanna, not to do math. But on the other hand, it's also clear why, loosely speaking, it's possible that the brain really could work. Natural selection would have quickly eliminated brains so *completely* unsuited to reasoning, so *anti*-helpful, as anti-Occamian or anti-Laplacian priors.

So what I did in practice does *not* amount to declaring a sudden halt to questioning and justification. I'm not halting the chain of examination at the point that I encounter Occam's Razor, or my brain, or some other unquestionable. The chain of examination continues—but it continues, unavoidably, using my current brain and my current grasp on reasoning techniques. *What else could I possibly use?*

Indeed, no matter *what* I did with this dilemma, it would be me doing it. Even if I trusted something else, like some computer program, it would be my own decision to trust it.

The technique of rejecting beliefs that have absolutely no justification is in general an extremely important one. I sometimes say that the fundamental question of rationality is "Why do you believe what you believe?" I don't even want to say something that *sounds* like it might allow a single exception to the rule that everything needs justification.

Which is, itself, a dangerous sort of motivation; you can't always avoid everything that might be risky, and when someone annoys you by saying something silly, you can't reverse that stupidity to arrive at intelligence.

But I would nonetheless emphasize the difference between saying:

Here is this assumption I cannot justify, which must be simply taken, and not further examined.

Versus saying:

Here the inquiry continues to examine this assumption, with the full force of my *present intelligence*—as opposed to the full force of something else, like a random number generator or a magic 8-ball—even though my present intelligence happens to be founded on this assumption.

Still ...wouldn't it be nice if we could examine the problem of how much to trust our brains *without* using our current intelligence? Wouldn't it be nice if we could examine the problem of how to think, *without* using our current grasp of rationality?

When you phrase it *that* way, it starts looking like the answer might be "No."

E. T. Jaynes used to say that you must always use all the information available to you—he was a Bayesian probability theorist, and had to clean up the paradoxes other people generated when they used different information at different points in their calculations. The principle of "*Always put forth your true best effort*" has at least as much appeal as "*Never do anything that might look circular*." After all, the alternative to putting forth your best effort is presumably doing less than your best.

But still ...wouldn't it be nice if there were some way to justify using Occam's Razor, or justify predicting that the future will resemble the past, *without* assuming that those methods of reasoning which have worked on previous occasions are better than those which have continually failed?

Wouldn't it be nice if there were some chain of justifications that neither ended in an unexaminable assumption, nor was forced to examine itself under its own rules, but, instead, could be explained starting from absolute scratch to an ideal philosophy student of perfect emptiness?

Well, I'd certainly be interested, but I don't expect to see it done any time soon. There is no perfectly empty ghost-in-the-machine; there is no argument that you can explain to a rock.

Even if someone cracks the First Cause problem and comes up with *the actual reason the universe is simple, which does not*

itself presume a simple universe ... then I would still expect that the explanation could only be understood by a mindful listener, and not by, say, a rock. A listener that didn't start out already implementing modus ponens might be out of luck.

So, at the end of the day, what happens when someone keeps asking me "Why do you believe what you believe?"

At present, I start going around in a loop at the point where I explain, "I predict the future as though it will resemble the past on the simplest and most stable level of organization I can identify, because previously, this rule has usually worked to generate good results; and using the simple assumption of a simple universe, I can see *why* it generates good results; and I can even see how my brain might have evolved to be able to observe the universe with some degree of accuracy, if my observations are correct."

But then ...haven't I just licensed *circular logic*?

Actually, I've just licensed *reflecting on your mind's degree of trustworthiness, using your current mind as opposed to something else*.

Reflection of this sort is, indeed, the reason we reject most circular logic in the first place. We want to have a coherent causal story about how our mind comes to know something, a story that explains how the process we used to arrive at our beliefs is itself trustworthy. This is the essential demand behind the rationalist's fundamental question, "Why do you believe what you believe?"

Now suppose you write on a sheet of paper: "(1) Everything on this sheet of paper is true, (2) The mass of a helium atom is 20 grams." If that trick actually *worked in real life*, you would be able to know the true mass of a helium atom just by believing some circular logic that asserted it. Which would enable you to arrive at a true map of the universe sitting in your living room with the blinds drawn. Which would violate the Second Law of Thermodynamics by generating information from nowhere. Which would not be a plausible story about how your mind could end up believing something true.

Even if you started out believing the sheet of paper, it would not seem that you had any reason for why the paper corre-

sponded to reality. It would just be a miraculous coincidence that (a) the mass of a helium atom was 20 grams, and (b) the paper happened to say so.

Believing self-validating statement sets does not in general seem like it should work to map external reality—when we *reflect on it as a causal story about minds*—using, of course, our *current* minds to do so.

But what about evolving to give more credence to simpler beliefs, and to believe that algorithms which have worked in the past are more likely to work in the future? *Even when* we reflect on this as a causal story of the origin of minds, it still seems like this could plausibly work to map reality.

And what about trusting reflective coherence in general? Wouldn't most possible minds, randomly generated and allowed to settle into a state of reflective coherence, be incorrect? Ah, but *we* evolved by natural selection; we were not generated randomly.

If trusting this argument seems worrisome to you, then forget about the problem of philosophical justifications, and ask yourself whether it's really truly true.

(You will, of course, use your own mind to do so.)

Is this the same as the one who says, "I believe that the Bible is the word of God, because the Bible says so"?

Couldn't they argue that their blind faith must also have been placed in them by God, and is therefore trustworthy?

In point of fact, when religious people finally come to reject the Bible, they do *not* do so by magically jumping to a non-religious state of pure emptiness, and then evaluating their religious beliefs in that non-religious state of mind, and then jumping back to a new state with their religious beliefs removed.

People go from being religious to being non-religious because even in a religious state of mind, doubt seeps in. They notice their prayers (and worse, the prayers of seemingly much worthier people) are not being answered. They notice that God, who speaks to them in their heart in order to provide seemingly consoling answers about the universe, is not able to tell them the hundredth digit of pi (which would be a lot more reassuring,

if God's purpose were reassurance). They examine the story of God's creation of the world and damnation of unbelievers, and it doesn't seem to make sense even under their own religious premises.

Being religious doesn't make you less than human. Your brain still has the abilities of a human brain. The dangerous part is that being religious might stop you from *applying* those native abilities to your religion—stop you from *reflecting fully* on yourself. People don't heal their errors by resetting themselves to an ideal philosopher of pure emptiness and reconsidering all their sensory experiences from scratch. They heal themselves by becoming more willing to question their current beliefs, using more of the power of their current mind.

This is why it's important to distinguish between *reflecting on your mind using your mind* (it's not like you can use anything else) and *having an unquestionable assumption that you can't reflect on*.

"I believe that the Bible is the word of God, because the Bible says so." Well, if the Bible *were* an astoundingly reliable source of information about all other matters, if it had not said that grasshoppers had four legs or that the universe was created in six days, but had instead contained the Periodic Table of Elements centuries before chemistry—if the Bible had served us only well and told us only truth—then we might, in fact, be inclined to take seriously the additional statement in the Bible, that the Bible had been generated by God. We might not trust it entirely, because it could also be aliens or the Dark Lords of the Matrix, but it would at least be worth taking seriously.

Likewise, if everything *else* that priests had told us turned out to be true, we might take more seriously their statement that faith had been placed in us by God and was a systematically trustworthy source—especially if people could divine the hundredth digit of pi by faith as well.

So the important part of appreciating the circularity of "I believe that the Bible is the word of God, because the Bible says so," is not so much that you are going to reject the idea of reflecting on your mind using your current mind. Rather, you realize that anything which calls into question the Bible's

trustworthiness also calls into question the Bible's assurance of its trustworthiness.

This applies to rationality too: if the future should cease to resemble the past—even on its lowest and simplest and most stable observed levels of organization—well, mostly, I'd be dead, because my brain's processes require a lawful universe where chemistry goes on working. But if somehow I survived, then I would have to start questioning the principle that the future should be predicted to be like the past.

But for now ...what's the *alternative* to saying, "I'm going to believe that the future will be like the past on the most stable level of organization I can identify, because that's previously worked better for me than any other algorithm I've tried"?

Is it saying, "I'm going to believe that the future will *not* be like the past, because that algorithm has always failed before"?

At this point I feel obliged to drag up the point that rationalists are not out to win arguments with ideal philosophers of perfect emptiness; we are simply out to win. For which purpose we want to get as close to the truth as we can possibly manage. So at the end of the day, I embrace the principle: "Question your brain, question your intuitions, question your principles of rationality, *using the full current force of your mind, and doing the best you can do at every point.*"

If one of your current principles does come up wanting—according to your own mind's examination, since you can't step outside yourself—then change it! And then go back and look at things again, using your new improved principles.

The point is not to be reflectively consistent. The point is to win. But *if* you look at yourself and play to win, you are making yourself more reflectively consistent—that's what it means to "play to win" while "looking at yourself."

Everything, without exception, needs justification. Sometimes—unavoidably, as far as I can tell—those justifications will go around in reflective loops. I do think that reflective loops have a meta-character which should enable one to distinguish them, by common sense, from circular logics. But anyone seriously considering a circular logic in the first place is probably out to lunch in matters of rationality, and

will simply insist that their circular logic is a “reflective loop” even if it consists of a single scrap of paper saying “Trust me.” Well, you can’t always optimize your rationality techniques according to the sole consideration of preventing those bent on self-destruction from abusing them.

The important thing is to *hold nothing back* in your criticisms of how to criticize; nor should you regard the unavoidability of loopy justifications as a warrant of *immunity from questioning*.

Always apply full force, whether it loops or not—do the best you can possibly do, whether it loops or not—and play, ultimately, to win.

My Kind of Reflection



discussion
sharov.me/r/
5-c10

In *Where Recursive Justification Hits Bottom*, I concluded that it's okay to use induction to reason about the probability that induction will work in the future, given that it's worked in the past; or to use Occam's Razor to conclude that the simplest explanation for why Occam's Razor works is that the universe itself is fundamentally simple.

Now I am far from the first person to consider reflective application of reasoning principles. Chris Hibbert compared my view to Bartley's Pan-Critical Rationalism (I was wondering whether that would happen). So it seems worthwhile to state what I see as the distinguishing features of my view of reflection, which may or may not happen to be shared by any other philosopher's view of reflection.

- All of my philosophy here *actually* comes from trying to figure out how to build a self-modifying AI that applies its own reasoning principles to itself in the process of rewriting its own source code. So whenever I talk about using induction to license induction, I'm *really* thinking about an inductive AI considering a rewrite of the part of itself that performs induction. If you wouldn't want the AI to rewrite its source code to not use induction, your philosophy had better not label induction as unjustifiable.
- One of the most powerful principles I know for AI in general is that the true Way generally turns out to be *naturalistic*—which for reflective reasoning means treating transistors inside the AI just as if they were transistors found in the environment, *not* an ad-hoc special case. This is the real source of my insistence in *Recursive Justification* that questions like “How well does my version of Occam's Razor work?” should be considered just like an ordinary question—or at least an ordinary very deep

question. I strongly suspect that a correctly built AI, in pondering modifications to the part of its source code that implements Occamian reasoning, will not have to do anything special as it ponders—in particular, it shouldn’t have to make a special effort to avoid using Occamian reasoning.

- I don’t think that “reflective coherence” or “reflective consistency” should be considered as a desideratum in itself. As I say in *The Twelve Virtues* and *The Simple Truth*, if you make five accurate maps of the same city, then the maps will necessarily be consistent with each other; but if you draw one map by fantasy and then make four copies, the five will be consistent but not accurate. In the same way, no one is deliberately pursuing reflective consistency, and reflective consistency is not a special warrant of trustworthiness; the goal is to win. But anyone who pursues the goal of winning, using their current notion of winning, and modifying their own source code, will end up reflectively consistent as a side effect—just like someone continually striving to improve their map of the world should find the parts becoming more consistent among themselves, as a side effect. If you put on your AI goggles, then the AI, rewriting its own source code, is not trying to make itself “reflectively consistent”—it is trying to optimize the expected utility of its source code, and it happens to be doing this using its current mind’s anticipation of the consequences.
- One of the ways I license using induction and Occam’s Razor to consider “induction” and “Occam’s Razor” is by appealing to E. T. Jaynes’s principle that we should always use all the information available to us (computing power permitting) in a calculation. If you think induction works, then you should use it in order to use your maximum power, including when you’re thinking about induction.

- In general, I think it's valuable to distinguish a defensive posture where you're imagining how to justify your philosophy to a philosopher that questions you, from an aggressive posture where you're trying to get as close to the truth as possible. So it's not that being suspicious of Occam's Razor, but using your current mind and intelligence to inspect it, shows that you're being *fair* and *defensible* by questioning your foundational beliefs. Rather, the reason why you would inspect Occam's Razor is to see if you could improve your application of it, or if you're worried it might really be wrong. I tend to deprecate mere dutiful doubts.
- If you run around inspecting your foundations, I expect you to actually improve them, not just dutifully investigate. Our brains are built to assess "simplicity" in a certain intuitive way that makes Thor sound simpler than Maxwell's Equations as an explanation for lightning. But, having gotten a better look at the way the universe really works, we've concluded that differential equations (which few humans master) are actually *simpler* (in an information-theoretic sense) than heroic mythology (which is how most tribes explain the universe). This being the case, we've tried to import our notions of Occam's Razor into math as well.
- On the other hand, the improved foundations should still add up to normality; $2 + 2$ should still end up equalling 4, not something new and amazing and exciting like "fish."
- I think it's very important to distinguish between the questions "Why does induction work?" and "Does induction work?" The reason *why the universe itself is regular* is still a mysterious question unto us, for now. Strange speculations here may be temporarily needful. But on the other hand, if you start claiming that the universe *isn't actually regular*, that the answer to "Does induction work?" is "No!," then you're wandering into $2 + 2 = 3$ territory. You're trying too hard to make your philosophy

interesting, instead of correct. An inductive AI asking what probability assignment to make on the next round is asking “Does induction work?,” and this is the question that it may answer by inductive reasoning. If you ask “Why does induction work?” then answering “Because induction works” is circular logic, and answering “Because I believe induction works” is magical thinking.

- I don’t think that going around in a loop of justifications through the meta-level is the same thing as circular logic. I think the notion of “circular logic” applies within the object level, and is something that is definitely bad and forbidden, on the object level. Forbidding *reflective coherence* doesn’t sound like a good idea. But I haven’t yet sat down and formalized the exact difference—my reflective theory is something I’m trying to work out, not something I have in hand.

No Universally Compelling Arguments



discussion
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5-c11

What is so *terrifying* about the idea that not every possible mind might agree with us, even in principle?

For some folks, nothing—it doesn’t bother them in the slightest. And for some of *those* folks, the *reason* it doesn’t bother them is that they don’t have strong intuitions about standards and truths that go beyond personal whims. If they say the sky is blue, or that murder is wrong, that’s just their personal opinion; and that someone else might have a different opinion doesn’t surprise them.

For other folks, a disagreement that persists even *in principle* is something they can’t accept. And for some of *those* folks, the *reason* it bothers them is that it seems to them that if you allow that some people cannot be persuaded *even in principle* that the sky is blue, then you’re conceding that “the sky is blue” is merely an *arbitrary* personal opinion.

I’ve proposed that you should resist the temptation to generalize over all of mind design space. If we restrict ourselves to minds specifiable in a trillion bits or less, then each *universal* generalization “All minds $m : X(m)$ ” has two to the trillionth chances to be false, while each *existential* generalization “Exists mind $m : X(m)$ ” has two to the trillionth chances to be true.

This would seem to argue that for every argument A , however convincing it may seem to us, there exists at least one possible mind that doesn’t buy it.

And the surprise and/or horror of this prospect (for some) has a great deal to do, I think, with the intuition of the ghost-in-the-machine—a ghost with some irreducible core that any *truly valid* argument will convince.

I have previously spoken of the intuition whereby people map *programming a computer* onto *instructing a human servant*, so that the computer might rebel against its code—or perhaps look over the code, decide it is not reasonable, and hand it back.

If there were a ghost in the machine and the ghost contained an irreducible core of reasonableness, above which any mere code was only a suggestion, then there might be universal arguments. Even if the ghost were initially handed code-suggestions that contradicted the Universal Argument, when we finally did expose the ghost to the Universal Argument—or the ghost could discover the Universal Argument on its own, that's also a popular concept—the ghost would just override its own, mistaken source code.

But as the student programmer once said, “I get the feeling that the computer just skips over all the comments.” The code is not given to the AI; the code *is* the AI.

If you switch to the physical perspective, then the notion of a Universal Argument seems noticeably unphysical. If there's a physical system that at time *T*, after being exposed to argument *E*, does *X*, then there ought to be another physical system that at time *T*, after being exposed to environment *E*, does *Y*. Any thought has to be implemented *somewhere*, in a physical system; any belief, any conclusion, any decision, any motor output. For every lawful causal system that zigs at a set of points, you should be able to specify another causal system that lawfully zags at the same points.

Let's say there's a mind with a transistor that outputs +3 volts at time *T*, indicating that it has just assented to some persuasive argument. Then we can build a highly similar physical cognitive system with a tiny little trapdoor underneath the transistor containing a little gray man who climbs out at time *T* and sets that transistor's output to -3 volts, indicating non-assent. Nothing acausal about that; the little gray man is there because we built him in. The notion of an argument that convinces *any* mind seems to involve a little blue woman who was *never* built into the system, who climbs out of literally *nowhere*, and strangles the little gray man, because that transistor has just *got* to output +3 volts. It's such a *compelling argument*, you see.

But compulsion is not a property of arguments; it is a property of minds that process arguments.

So the reason I'm arguing against the ghost isn't *just* to make the point that (1) Friendly AI has to be explicitly programmed

and (2) the laws of physics do not forbid Friendly AI. (Though of course I take a certain interest in establishing this.)

I also wish to establish the notion of a mind as a *causal, lawful, physical system* in which there is *no* irreducible central ghost that looks over the neurons/code and decides whether they are good suggestions.

(There is a concept in Friendly AI of *deliberately* programming an FAI to review its own source code and possibly hand it back to the programmers. But the mind that reviews is not irreducible, it is just the mind that you created. The FAI is renormalizing itself *however it was designed to do so*; there is nothing acausal reaching in from outside. A bootstrap, not a skyhook.)

All this echoes back to the worry about a Bayesian's "arbitrary" priors. If you show me one Bayesian who draws 4 red balls and 1 white ball from a barrel, and who assigns probability 5/7 to obtaining a red ball on the next occasion (by Laplace's Rule of Succession), then I can show you another mind which obeys Bayes's Rule to conclude a 2/7 probability of obtaining red on the next occasion—corresponding to a different prior belief about the barrel, but, perhaps, a less "reasonable" one.

Many philosophers are convinced that because you can in principle construct a prior that updates to any given conclusion on a stream of evidence, therefore, Bayesian reasoning must be "arbitrary," and the whole schema of Bayesianism flawed, because it relies on "unjustifiable" assumptions, and indeed "unscientific," because you cannot force any possible journal editor in mindspace to agree with you.

And this (I replied) relies on the notion that by unwinding all arguments and their justifications, you can obtain an ideal philosophy student of perfect emptiness, to be convinced by a line of reasoning that begins from absolutely no assumptions.

But who is this ideal philosopher of perfect emptiness? Why, it is just the irreducible core of the ghost!

And that is why (I went on to say) the result of trying to remove all assumptions from a mind, and unwind to the perfect absence of any prior, is not an ideal philosopher of perfect emptiness, but a rock. What is left of a mind after you remove

the source code? Not the ghost who looks over the source code, but simply ...no ghost.

So—and I shall take up this theme again later—wherever you are to locate your notions of *validity* or *worth* or *rationality* or *justification* or even *objectivity*, it cannot rely on an argument that is *universally compelling to all physically possible minds*.

Nor can you ground validity in a sequence of justifications that, beginning from nothing, persuades a perfect emptiness.

Oh, there might be argument sequences that would compel any neurologically intact *human*—like the argument I use to make people let the AI out of the box¹—but that is hardly the same thing from a philosophical perspective.

The first great failure of those who try to consider Friendly AI is the One Great Moral Principle That Is All We Need To Program—a.k.a. the fake utility function—and of this I have already spoken.

But the even worse failure is the One Great Moral Principle We Don't Even *Need* To Program Because Any AI Must Inevitably Conclude It. This notion exerts a terrifying unhealthy fascination on those who spontaneously reinvent it; they dream of commands that no sufficiently advanced mind can disobey. The gods themselves will proclaim the rightness of their philosophy! (E.g., John C. Wright, Marc Geddes.)

There is also a less severe version of the failure, where the one does not *declare* the One True Morality. Rather the one hopes for an AI created *perfectly free*, unconstrained by flawed humans desiring slaves, so that the AI may arrive at virtue of its own accord—virtue undreamed-of perhaps by the speaker, who confesses themselves too flawed to teach an AI. (E.g., John K. Clark, Richard Hollerith?, Eliezer₁₉₉₆.) This is a less tainted motive than the dream of absolute command. But though *this* dream arises from virtue rather than vice, it is still based on a flawed understanding of freedom, and will not actually *work in real life*. Of this, more to follow, of course.

John C. Wright, who was previously writing a very nice transhumanist trilogy (first book: *The Golden Age*), inserted

1. Just kidding.

a huge Author Filibuster in the middle of his climactic third book, describing in tens of pages his Universal Morality That Must Persuade Any AI. I don't know if anything happened after that, because I stopped reading. And then Wright converted to Christianity—yes, seriously. So you *really don't* want to fall into this trap!

Created Already in Motion

Lewis Carroll, who was also a mathematician, once wrote a short dialogue called “What the Tortoise said to Achilles.”¹ If you have not yet read this ancient classic, consider doing so now.



discussion
sharov.me/r/
5-c12

The Tortoise offers Achilles a step of reasoning drawn from Euclid’s First Proposition:

(A) Things that are equal to the same are equal to each other.

(B) The two sides of this Triangle are things that are equal to the same.

(Z) The two sides of this Triangle are equal to each other.

Tortoise: “And if some reader had *not* yet accepted *A* and *B* as true, he might still accept the *sequence* as a *valid* one, I suppose?”

Achilles: “No doubt such a reader might exist. He might say, ‘I accept as true the Hypothetical Proposition that, *if A* and *B* be true, *Z* must be true; but, I *don’t* accept *A* and *B* as true.’ Such a reader would do wisely in abandoning Euclid, and taking to football.”

Tortoise: “And might there not *also* be some reader who would say, ‘I accept *A* and *B* as true, but I *don’t* accept the Hypothetical’?”

Achilles, unwisely, concedes this; and so asks the Tortoise to accept another proposition:

(C) If *A* and *B* are true, *Z* must be true.

1. sharov.me/r/5-5

What the Tortoise Said to Achilles • ditext.com



But, asks, the Tortoise, suppose that he accepts A and B and C , but not Z ?

Then, says, Achilles, he must ask the Tortoise to accept one more hypothetical:

(D) If A and B and C are true, Z must be true.

Douglas Hofstadter paraphrased the argument some time later:

ACHILLES: "If you have $[(A \text{ and } B) \rightarrow Z]$, and you also have $(A \text{ and } B)$, then surely you have Z ."

TORTOISE: "Oh! You mean $((A \text{ and } B) \text{ and } [(A \text{ and } B) \rightarrow Z]) \rightarrow Z$, don't you?"

As Hofstadter says, "Whatever Achilles considers a rule of inference, the Tortoise immediately flattens into a mere string of the system. If you use only the letters A , B , and Z , you will get a recursive pattern of longer and longer strings."

This is the anti-pattern I call Passing the Recursive Buck; and though the counterspell is sometimes hard to find, when found, it generally takes the form The Buck Stops Immediately.

The Tortoise's mind needs the *dynamic* of adding Y to the belief pool when X and $(X \rightarrow Y)$ are previously in the belief pool. If this dynamic is not present—a rock, for example, lacks it—then you can go on adding in X and $(X \rightarrow Y)$ and $((X \text{ and } (X \rightarrow Y)) \rightarrow Y)$ until the end of eternity, without ever getting to Y .

The phrase that once came into my mind to describe this requirement is that a mind must be *created already in motion*. There is no argument so compelling that it will give dynamics to a static thing. There is no computer program so *persuasive* that you can run it on a rock.

And even if you have a mind that *does* carry out modus ponens, it is futile for it to have such beliefs as ...

(A) If a toddler is on the train tracks, then pulling them off is fuzzle.

(B) There is a toddler on the train tracks.

...unless the mind also *implements*:

Dynamic: When the belief pool contains “X is fuzzle,” send X to the action system.

By “dynamic” I mean a property of a physically implemented cognitive system’s *development over time*. A “dynamic” is something that *happens inside* a cognitive system, *not* data that it stores in memory and manipulates. Dynamics are the manipulations. There is no way to write a dynamic on a piece of paper, because the paper will just lie there. So the text immediately above, which says “dynamic,” is not dynamic. If I wanted the text to *be* dynamic and not just *say* “dynamic,” I would have to write a Java applet.

Needless to say, having the belief ...

(C) If the belief pool contains “X is fuzzle,” then “send ‘X’ to the action system” is fuzzle.

...won’t help unless the mind already implements the *behavior* of translating hypothetical actions labeled “fuzzle” into actual motor actions.

By dint of careful arguments about the nature of cognitive systems, you might be able to prove ...

(D) A mind with a dynamic that sends plans labeled “fuzzle” to the action system is more fuzzle than minds that don’t.

...but that *still* won’t help, unless the listening mind *previously* possessed the *dynamic* of swapping out its current source code for alternative source code that is believed to be more fuzzle.

This is why you can’t argue fuzzleness into a rock.

Sorting Pebbles into Correct Heaps



discussion
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5-c13

Once upon a time there was a strange little species—that might have been biological, or might have been synthetic, and perhaps were only a dream—whose passion was sorting pebbles into correct heaps.

They couldn't tell you *why* some heaps were correct, and some incorrect. But all of them agreed that the most important thing in the world was to create correct heaps, and scatter incorrect ones.

Why the Pebblesorting People cared so much, is lost to this history—maybe a Fisherian runaway sexual selection, started by sheer accident a million years ago? Or maybe a strange work of sentient art, created by more powerful minds and abandoned?

But it mattered so drastically to them, this sorting of pebbles, that all the Pebblesorting philosophers said in unison that pebble-heap-sorting was the very meaning of their lives: and held that the only justified reason to eat was to sort pebbles, the only justified reason to mate was to sort pebbles, the only justified reason to participate in their world economy was to efficiently sort pebbles.

The Pebblesorting People all agreed on that, but they didn't always agree on which heaps were correct or incorrect.

In the early days of Pebblesorting civilization, the heaps they made were mostly small, with counts like 23 or 29; they couldn't tell if larger heaps were correct or not. Three millennia ago, the Great Leader Biko made a heap of 91 pebbles and proclaimed it correct, and his legions of admiring followers made more heaps likewise. But over a handful of centuries, as the power of the Bikonians faded, an intuition began to accumulate among the smartest and most educated that a heap of 91 pebbles was incorrect. Until finally they came to know what they had done: and they scattered all the heaps of 91 pebbles. Not without

flashes of regret, for some of those heaps were great works of art, but incorrect. They even scattered Biko's original heap, made of 91 precious gemstones each of a different type and color.

And no civilization since has seriously doubted that a heap of 91 is incorrect.

Today, in these wiser times, the size of the heaps that Pebblesorters dare attempt has grown very much larger—which all agree would be a most great and excellent thing, if only they could ensure the heaps were really *correct*. Wars have been fought between countries that disagree on which heaps are correct: the Pebblesorters will never forget the Great War of 1957, fought between Y'ha-nthlei and Y'not'ha-nthlei, over heaps of size 1957. That war, which saw the first use of nuclear weapons on the Pebblesorting Planet, finally ended when the Y'not'ha-nthleian philosopher At'gra'len'ley exhibited a heap of 103 pebbles and a heap of 19 pebbles side-by-side. So persuasive was this argument that even Y'ha-nthlei reluctantly conceded that it was best to stop building heaps of 1957 pebbles, at least for the time being.

Since the Great War of 1957, countries have been reluctant to openly endorse or condemn heaps of large size, since this leads so easily to war. Indeed, some Pebblesorting philosophers—who seem to take a tangible delight in shocking others with their cynicism—have entirely denied the existence of pebble-sorting *progress*; they suggest that opinions about pebbles have simply been a random walk over time, with no coherence to them, the illusion of progress created by condemning all dissimilar pasts as incorrect. The philosophers point to the disagreement over pebbles of large size, as proof that there is nothing that makes a heap of size 91 really *incorrect*—that it was simply fashionable to build such heaps at one point in time, and then at another point, fashionable to condemn them. “But...13!” carries no truck with them; for to regard “13!” as a persuasive counterargument is only another convention, they say. The Heap Relativists claim that their philosophy may help prevent future disasters like the Great War of 1957, but it is widely considered to be a philosophy of despair.

Now the question of what makes a heap correct or incorrect has taken on new urgency; for the Pebblesorters may shortly embark on the creation of self-improving Artificial Intelligences. The Heap Relativists have warned against this project: They say that AIs, not being of the species *Pebblesorter sapiens*, may form their own culture with entirely different ideas of which heaps are correct or incorrect. “They could decide that heaps of 8 pebbles are correct,” say the Heap Relativists, “and while ultimately they’d be no righter or wronger than us, still, *our* civilization says we shouldn’t build such heaps. It is not in our interest to create AI, unless all the computers have bombs strapped to them, so that even if the AI thinks a heap of 8 pebbles is correct, we can force it to build heaps of 7 pebbles instead. Otherwise, KABOOM!”

But this, to most Pebblesorters, seems absurd. Surely a sufficiently powerful AI—especially the “superintelligence” some transpebblesorterists go on about—would be able to see *at a glance* which heaps were correct or incorrect! The thought of something with a brain the size of a planet thinking that a heap of 8 pebbles was correct is just too absurd to be worth talking about.

Indeed, it is an utterly futile project to constrain how a superintelligence sorts pebbles into heaps. Suppose that Great Leader Biko had been able, in his primitive era, to construct a self-improving AI; and he had built it as an expected utility maximizer whose utility function told it to create as many heaps as possible of size 91. Surely, when this AI improved itself far enough, and became smart enough, then it would see at a glance that this utility function was incorrect; and, having the ability to modify its own source code, it would *rewrite its utility function* to value more reasonable heap sizes, like 101 or 103.

And certainly not heaps of size 8. That would just be *stupid*. Any mind that stupid is too dumb to be a threat.

Reassured by such common sense, the Pebblesorters pour full speed ahead on their project to throw together lots of algorithms at random on big computers until some kind of intelligence emerges. The whole history of civilization has

shown that richer, smarter, better educated civilizations are likely to agree about heaps that their ancestors once disputed. Sure, there are then larger heaps to argue about—but the further technology has advanced, the larger the heaps that have been agreed upon and constructed.

Indeed, intelligence itself has always correlated with making correct heaps—the nearest evolutionary cousins to the Pebble-sorters, the Pebpanzees, make heaps of only size 2 or 3, and occasionally stupid heaps like 9. And other, even less intelligent creatures, like fish, make no heaps at all.

Smarter minds equal smarter heaps. Why would that trend break?

2-Place and 1-Place Words



discussion
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5-c14

I have previously spoken of the ancient, pulp-era magazine covers that showed a bug-eyed monster carrying off a girl in a torn dress; and about how people think as if sexiness is an inherent property of a sexy entity, without dependence on the admirer.

“Of *course* the bug-eyed monster will prefer human females to its own kind,” says the artist (who we’ll call Fred); “it can see that human females have soft, pleasant skin instead of slimy scales. It may be an alien, but it’s not *stupid*—why are you expecting it to make such a basic mistake about sexiness?”

What is Fred’s error? It is treating a function of 2 arguments (“2-place function”):

$$\text{Sexiness: Admirer, Entity} \rightarrow [0, \infty),$$

as though it were a function of 1 argument (“1-place function”):

$$\text{Sexiness: Entity} \rightarrow [0, \infty).$$

If Sexiness is treated as a function that accepts only one Entity as its argument, then of course Sexiness will appear to depend only on the Entity, with nothing else being relevant.

When you think about a two-place function as though it were a one-place function, you end up with a Variable Question Fallacy / Mind Projection Fallacy. Like trying to determine whether a building is *intrinsically* on the left or on the right side of the road, independent of anyone’s travel direction.

An alternative and equally valid standpoint is that “sexiness” *does* refer to a one-place function—but each speaker uses a *different* one-place function to decide who to kidnap and ravish. Who says that just because Fred, the artist, and Bloogah, the

bug-eyed monster, both use the word “sexy,” they must mean the same thing by it?

If you take this viewpoint, there is no paradox in speaking of some woman intrinsically having 5 units of `Fred :: Sexiness`. All onlookers can agree on this fact, once `Fred :: Sexiness` has been specified in terms of curves, skin texture, clothing, status cues, etc. This specification need *make no mention of Fred*, only the woman to be evaluated.

It so happens that Fred, himself, *uses* this algorithm to select flirtation targets. But that doesn’t mean the algorithm itself has to *mention* Fred. So Fred’s `Sexiness` function really *is* a function of one argument—the woman—on this view. I called it `Fred :: Sexiness`, but remember that this *name* refers to a function that is being described independently of Fred. Maybe it would be better to write:

```
Fred :: Sexiness == Sexiness_20934.
```

It is an empirical fact about Fred that he uses the function `Sexiness_20934` to evaluate potential mates. Perhaps John uses exactly the same algorithm; it doesn’t matter where it comes from once we have it.

And similarly, the same woman has only 0.01 units of `Sexiness_72546`, whereas a slime mold has 3 units of `Sexiness_72546`. It happens to be an empirical fact that Bloogah uses `Sexiness_72546` to decide who to kidnap; that is, `Bloogah :: Sexiness` names the fixed Bloogah-independent mathematical object that is the function `Sexiness_72546`.

Once we say that the woman has 0.01 units of `Sexiness_72546` and 5 units of `Sexiness_20934`, all observers can agree on this without paradox.

And the two 2-place and 1-place views can be unified using the concept of “currying,” named after the mathematician Haskell Curry. Currying is a technique allowed in certain programming languages, where e.g. instead of writing

```
x = plus(2, 3) (x = 5),
```

you can also write

```
y = plus(2)
(y is now a “curried” form of the function plus,
which has eaten a 2)
x = y(3) (x=5)
z = y(7) (z=9).
```

So `plus` is a 2-place function, but currying `plus`—letting it eat only one of its two required arguments—turns it into a 1-place function that adds 2 to any input. (Similarly, you could start with a 7-place function, feed it 4 arguments, and the result would be a 3-place function, etc.)

A true purist would insist that all functions should be viewed, by definition, as taking exactly one argument. On this view, `plus` accepts one numeric input, and outputs a *new* function; and this *new* function has one numeric input and finally outputs a number. On this view, when we write `plus(2, 3)` we are really computing `plus(2)` to get a function that adds 2 to any input, and then applying the result to 3. A programmer would write this as:

```
plus: int → (int → int).
```

This says that `plus` takes an `int` as an argument, and returns a function of type `int → int`.

Translating the metaphor back into the human use of words, we could imagine that “sexiness” starts by eating an `Admirer`, and spits out the fixed *mathematical* object that describes how the `Admirer` currently evaluates pulchritude. It is an *empirical* fact about the `Admirer` that their intuitions of desirability are computed in a way that is isomorphic to this *mathematical* function.

Then the mathematical object spit out by currying `Sexiness(Admirer)` can be applied to the `Woman`. If the `Admirer` was originally `Fred`, `Sexiness(Fred)` will first return `Sexiness_20934`. We can then say it is an empirical fact about the `Woman`, independently of `Fred`, that `Sexiness_20934(Woman) = 5`.

In Hilary Putnam's "Twin Earth" thought experiment, there was a tremendous philosophical brouhaha over whether it makes sense to postulate a Twin Earth that is just like our own, except that instead of water being H_2O , water is a *different* transparent flowing substance, XYZ. And furthermore, set the time of the thought experiment a few centuries ago, so in neither our Earth nor the Twin Earth does anyone know how to test the alternative hypotheses of H_2O vs. XYZ. Does the word "water" *mean* the same thing in that world as in this one?

Some said, "Yes, because when an Earth person and a Twin Earth person utter the word 'water,' they have the same sensory test in mind."

Some said, "No, because 'water' in our Earth means H_2O and 'water' in the Twin Earth means XYZ."

If you think of "water" as a concept that *begins* by eating a world to find out the empirical true nature of that transparent flowing stuff, and *returns* a new fixed concept $Water_{42}$ or H_2O , then this world-eating concept is the same in our Earth and the Twin Earth; it just returns different answers in different places.

If you think of "water" as meaning H_2O , then the concept does nothing different when we transport it between worlds, and the Twin Earth contains no H_2O .

And of course there is no point in arguing over what the sound of the syllables "wa-ter" really means.

So should you pick one definition and use it consistently? But it's not that easy to save yourself from confusion. You have to train yourself to be *deliberately aware* of the distinction between the curried and uncurried forms of concepts.

When you take the uncurried water concept and apply it in a different world, it is the same concept but it *refers* to a different thing; that is, we are applying a constant world-eating function to a different world and obtaining a different return value. In the Twin Earth, XYZ is "water" and H_2O is not; in our Earth, H_2O is "water" and XYZ is not.

On the other hand, if you take "water" to refer to what the prior thinker would call "the result of applying 'water' to *our* Earth," then in the Twin Earth, XYZ is not water and H_2O is.

The whole confusingness of the subsequent philosophical debate rested on a tendency to *instinctively* curry concepts or *instinctively* uncurry them.

Similarly it takes an extra step for Fred to realize that other agents, like the Bug-Eyed-Monster agent, will choose kidnapees for ravishing based on `Sexiness_BEM(Woman)`, not `Sexiness_Fred(Woman)`. To do this, Fred must consciously re-envision `Sexiness` as a function with two arguments. All Fred's brain does by instinct is evaluate `Woman.sexiness`—that is, `Sexiness_Fred(Woman)`; but it's simply labeled `Woman.sexiness`.

The fixed mathematical function `Sexiness_20934` makes no mention of Fred or the BEM, only women, so Fred does not *instinctively* see why the BEM would evaluate “sexiness” any differently. And indeed the BEM would *not* evaluate `Sexiness_20934` any differently, if for some odd reason it cared about the result of that particular function; but it is an *empirical* fact about the BEM that it uses a different function to *decide who to kidnap*.

If you're wondering as to the point of this analysis, try putting the above distinctions to work to Taboo such confusing words as “objective,” “subjective,” and “arbitrary.”

What Would You Do Without Morality?

To those who say “Nothing is real,” I once replied, “That’s great, but how does the nothing work?”

Suppose you learned, suddenly and definitively, that nothing is moral and nothing is right; that everything is permissible and nothing is forbidden.

Devastating news, to be sure—and no, I am not telling you this in real life. But suppose I *did* tell it to you. Suppose that, whatever you think is the basis of your moral philosophy, I convincingly tore it apart, and moreover showed you that nothing could fill its place. Suppose I *proved* that all utilities equaled zero.

I know that Your-Moral-Philosophy is as true and undisprovable as $2 + 2 = 4$. But still, I ask that you do your best to perform the thought experiment, and concretely envision the possibilities even if they seem painful, or pointless, or logically incapable of any good reply.

Would you still tip cabdrivers? Would you cheat on your Significant Other? If a child lay fainted on the train tracks, would you still drag them off?

Would you still eat the same kinds of foods—or would you only eat the cheapest food, since there’s no reason you *should* have fun—or would you eat very expensive food, since there’s no reason you *should* save money for tomorrow?

Would you wear black and write gloomy poetry and denounce all altruists as fools? But there’s no reason you *should* do that—it’s just a cached thought.

Would you stay in bed because there was no reason to get up? What about when you finally got hungry and stumbled into the kitchen—what would you do after you were done eating?

Would you go on reading *Overcoming Bias*, and if not, what would you read instead? Would you still try to be rational, and if not, what would you think instead?

Close your eyes, take as long as necessary to answer:



discussion
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5-c15

MERE GOODNESS

What *would* you do, if nothing were right?

Changing Your Metaethics

If you say, “Killing people is wrong,” that’s morality. If you say, “You shouldn’t kill people because God prohibited it,” or “You shouldn’t kill people because it goes against the trend of the universe,” that’s metaethics.

Just as there’s far more agreement on Special Relativity than there is on the question “What is science?,” people find it much easier to agree “Murder is bad” than to agree *what* makes it bad, or what it *means* for something to be bad.

People do get attached to their metaethics. Indeed they frequently insist that if their metaethic is wrong, all morality necessarily falls apart. It might be interesting to set up a panel of metaethicists—theists, Objectivists, Platonists, etc.—all of whom agree that killing is wrong; all of whom disagree on what it means for a thing to be “wrong”; and all of whom insist that if their metaethic is untrue, then morality falls apart.

Clearly a good number of people, if they are to make philosophical progress, will need to shift metaethics at some point in their lives. *You* may have to do it.

At that point, it might be useful to have an open line of retreat—not a retreat from morality, but a retreat from Your-Current-Metaethic. (You know, the one that, if it is not true, leaves no possible basis for not killing people.)

And so I summarized below some possible lines of retreat. For I have learned that to change metaethical beliefs is nigh-impossible in the presence of an unanswered attachment.

If, for example, someone believes the authority of “Thou Shalt Not Kill” derives from God, then there are several and well-known things to say that can help set up a line of retreat—as opposed to immediately attacking the plausibility of God. You can say, “Take personal responsibility! Even if you got orders from God, it would be your own decision to obey those orders. Even if God didn’t order you to be moral, you could just be moral anyway.”



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The above argument actually generalizes to quite a number of metaethics—you just substitute Their-Favorite-Source-Of-Morality, or even the word “morality,” for “God.” Even if your particular source of moral authority failed, couldn’t you just drag the child off the train tracks *anyway*? And indeed, who is it but you that ever decided to follow this source of moral authority in the first place? What responsibility are you really passing on?

So the most important line of retreat is: If your metaethic stops telling you to save lives, you can just drag the kid off the train tracks anyway. To paraphrase Piers Anthony, only those who have moralities worry over whether or not they have them. If your metaethic tells you to kill people, why *should* you even listen? Maybe that which you would do even if there were no morality, *is* your morality.

The point being, of course, not that no morality exists; but that you can hold your will in place, and not fear losing sight of what’s important to you, while your notions of the *nature* of morality change.

I’ve written some essays to set up lines of retreat specifically for more *naturalistic* metaethics. Joy in the Merely Real and Explaining vs. Explaining Away argue that you shouldn’t be disappointed in any facet of life, just because it turns out to be *explicable* instead of inherently mysterious: for if we cannot take joy in the merely real, our lives shall be empty indeed.

No Universally Compelling Arguments sets up a line of retreat from the desire to have *everyone* agree with our moral arguments. There’s a strong moral intuition which says that if our moral arguments are right, by golly, we ought to be able to *explain* them to people. This may be valid among humans, but you can’t explain moral arguments to a rock. There is no ideal philosophy student of perfect emptiness who can be persuaded to implement modus ponens, starting without modus ponens. If a mind doesn’t contain that which is moved by your moral arguments, it won’t respond to them.

But then isn’t all morality circular logic, in which case it falls apart? Where Recursive Justification Hits Bottom and My Kind of Reflection explain the difference between a self-consistent

loop through the meta-level, and actual circular logic. You shouldn't find yourself saying "The universe is simple because it is simple," or "Murder is wrong because it is wrong"; but neither should you try to abandon Occam's Razor while evaluating the probability that Occam's Razor works, nor should you try to evaluate "Is murder wrong?" from somewhere outside your brain. There is no ideal philosophy student of perfect emptiness to which you can unwind yourself—try to find the perfect rock to stand upon, and you'll end up as a rock. So instead use the full force of your intelligence, your full rationality and your full morality, when you investigate the foundations of yourself.

We can also set up a line of retreat for those afraid to allow a *causal* role for evolution, in their account of how morality came to be. (Note that this is extremely distinct from granting evolution a *justificational* status in moral theories.) Love has to come into existence somehow—for if we cannot take joy in things that can come into existence, our lives will be empty indeed. Evolution may not be a particularly *pleasant* way for love to evolve, but judge the end product—not the source. Otherwise you would be committing what is known (appropriately) as The Genetic Fallacy: causation is not the same concept as justification. It's not like you can step outside the brain evolution gave you; rebelling against nature is only possible from within nature.

The earlier series on Evolutionary Psychology should dispense with the metaethical confusion of believing that any normal human being thinks about their reproductive fitness, even unconsciously, in the course of making decisions. Only evolutionary biologists even know how to *define* genetic fitness, and they know better than to think it defines morality.

Alarming indeed is the thought that morality might be computed inside our own minds—doesn't this imply that morality is a mere thought? Doesn't it imply that whatever you think is right, must be right?

No. Just because a quantity is computed inside your head doesn't mean that the quantity computed is *about* your thoughts. There's a difference between a calculator that

calculates “What is $2 + 3$?” and one that outputs “What do I output when someone presses ‘2,’ ‘+,’ and ‘3’?”

Finally, if life seems painful, reductionism may not be the real source of your problem—if living in a world of mere particles seems too unbearable, maybe your life isn’t exciting enough right now?

And if you’re wondering why I deem this business of metaethics important, when it is all going to end up adding up to moral normality ...telling you to pull the child off the train tracks, rather than the converse ...

Well, there *is* opposition to rationality from people who think it drains meaning from the universe.

And this is a special case of a general phenomenon, in which many many people get messed up by misunderstanding where their morality comes from. Poor metaethics forms part of the teachings of many a cult, including the big ones. My target audience is not just people who are afraid that life is meaningless, but also those who’ve concluded that love is a delusion because real morality has to involve maximizing your inclusive fitness, or those who’ve concluded that unreturned kindness is evil because real morality arises only from selfishness, etc.

Could Anything Be Right?

Years ago, Eliezer₁₉₉₉ was convinced that he knew *nothing* about morality.

For all he knew, morality could require the extermination of the human species; and if so he saw no virtue in taking a stand against morality, because he thought that, by definition, if he postulated that moral fact, that meant human extinction was what “should” be done.

I thought I could *figure out* what was right, perhaps, given enough reasoning time and enough facts, but that I currently had no information about it. I could not trust evolution which had built me. What foundation did that leave on which to stand?

Well, indeed Eliezer₁₉₉₉ was massively mistaken about the nature of morality, so far as his explicitly represented philosophy went.

But as Davidson once observed, if you believe that “beavers” live in deserts, are pure white in color, and weigh 300 pounds when adult, then you do not have any beliefs *about* beavers, true or false. You must get at least some of your beliefs right, before the remaining ones can be wrong *about* anything.¹

My belief that I had *no* information *about* morality was not internally consistent.

Saying that I knew nothing felt virtuous, for I had once been taught that it was virtuous to confess my ignorance. “The only thing I know is that I know nothing,” and all that. But in this case I would have been better off considering the admittedly exaggerated saying, “The greatest fool is the one who is not aware they are wise.” (This is nowhere near the *greatest* kind of foolishness, but it is a kind of foolishness.)



discussion
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5-c17

1. Rorty, “Out of the Matrix: How the Late Philosopher Donald Davidson Showed That Reality Can’t Be an Illusion.”

Was it wrong to kill people? Well, I thought so, but I wasn't sure; maybe it was right to kill people, though that seemed less likely.

What kind of *procedure* would answer whether it was right to kill people? I didn't know that either, but I thought that if you built a generic superintelligence (what I would later label a "ghost of perfect emptiness") then it could, you know, reason about what was likely to be right and wrong; and since it was *superintelligent*, it was bound to come up with the right answer.

The problem that I somehow managed not to think too hard about was where the superintelligence would get the procedure that discovered the procedure that discovered the procedure that discovered morality—if I couldn't write it into the start state that wrote the successor AI that wrote the successor AI.

As Marcello Herreshoff later put it, "We never bother running a computer program unless we don't know the output and we know an important fact about the output." If I knew nothing about morality, and did not even claim to know the nature of morality, then how could I construct any computer program whatsoever—even a "superintelligent" one or a "self-improving" one—and claim that it would output something called "morality"?

There are no-free-lunch theorems in computer science—in a maxentropy universe, no plan is better on average than any other. If you have no knowledge at all about "morality," there's also no computational procedure that will seem more likely than others to compute "morality," and no meta-procedure that's more likely than others to produce a procedure that computes "morality."

I thought that surely even a ghost of perfect emptiness, finding that it knew nothing of morality, would see a moral imperative to *think about morality*.

But the difficulty lies in the word *think*. Thinking is not an activity that a ghost of perfect emptiness is automatically able to carry out. Thinking requires running some *specific* computation that is the thought. For a reflective AI to decide to think requires that it know some computation which it believes is *more* likely

to tell it what it wants to know than consulting an Ouija board; the AI must also have a notion of how to interpret the output.

If one knows nothing about morality, what does the word “should” mean, at all? If you don’t know whether death is right or wrong—and don’t know how you can discover whether death is right or wrong—and don’t know whether any given procedure might *output* the procedure for saying whether death is right or wrong—then what do these words, “right” and “wrong,” even *mean*?

If the words “right” and “wrong” have *nothing* baked into them—no starting point—if *everything* about morality is up for grabs, not just the content but the structure and the starting point and the determination procedure—then what is their meaning? What distinguishes, “I don’t know what is right” from “I don’t know what is waka-lixes”?

A scientist may say that everything is up for grabs in science, since any theory may be disproven; but then they have some idea of what would count as *evidence* that could disprove the theory. Could there be something that would change what a scientist regarded as evidence?

Well, yes, in fact; a scientist who read some Karl Popper and thought they knew what “evidence” meant could be presented with the coherence and uniqueness proofs underlying Bayesian probability, and that might change their definition of evidence. They might not have had any *explicit notion* in advance that such a proof could exist. But they would have had an implicit notion. It would have been baked into their brains, if not explicitly represented therein, that such-and-such an argument would in fact persuade them that Bayesian probability gave a better definition of “evidence” than the one they had been using.

In the same way, you could say, “I don’t know what morality is, but I’ll know it when I see it,” and make sense.

But then you are not rebelling completely against your own evolved nature. You are supposing that whatever has been baked into you to recognize “morality,” is, if not absolutely trustworthy, then at least your initial condition with which you start debating. Can you trust your moral intuitions to give

you any information about morality *at all*, when they are the product of mere evolution?

But if you discard every procedure that evolution gave you *and all its products*, then you discard your whole brain. You discard everything that could potentially recognize morality when it sees it. You discard everything that could potentially respond to moral arguments by updating your morality. You even unwind past the unwinder: you discard the intuitions underlying your conclusion that *you can't trust evolution* to be moral. It is your *existing* moral intuitions that tell you that evolution doesn't seem like a very *good* source of morality. What, then, will the words "right" and "should" and "better" even *mean*?

Humans do not perfectly recognize truth when they see it, and hunter-gatherers do not have an explicit concept of the Bayesian criterion of evidence. But all our science and all our probability theory was built on top of a chain of appeals to our instinctive notion of "truth." Had this core been flawed, there would have been nothing we could do *in principle* to arrive at the present notion of science; the notion of science would have just sounded completely unappealing and pointless.

One of the arguments that might have shaken my teenage self out of his mistake, if I could have gone back in time to argue with him, was the question:

Could there be some morality, some given rightness or wrongness, that human beings do not perceive, do not want to perceive, will not see any appealing moral argument for adopting, nor any moral argument for adopting a procedure that adopts it, et cetera? Could there be a morality, and ourselves *utterly* outside its frame of reference? But then what makes this thing *morality*—rather than a stone tablet somewhere with the words "Thou shalt murder" written on them, with absolutely no *justification* offered?

So all this suggests that you should be willing to accept that you might know a *little* about morality. Nothing unquestionable, perhaps, but an initial state with which to start questioning yourself. Baked into your brain but not explicitly known to you, perhaps; but still, that which your brain *would* recognize

as *right* is what you are talking *about*. You will accept at least enough of the way you *respond to moral arguments* as a *starting point* to identify “morality” as something to think about.

But that’s a rather large step.

It implies accepting your own mind as identifying a moral frame of reference, rather than all morality being a great light shining from beyond (that in principle you might not be able to perceive at all). It implies accepting that even if there were a light and your brain decided to recognize it as “morality,” it would still be your own brain that recognized it, and you would not have evaded causal responsibility—or evaded moral responsibility either, on my view.

It implies dropping the notion that a ghost of perfect emptiness will necessarily agree with you, because the ghost might occupy a different moral frame of reference, respond to different arguments, be *asking a different question* when it computes what-to-do-next.

And if you’re willing to bake at least a few things into the very meaning of this topic of “morality,” this quality of *rightness* that you are talking about when you talk about “rightness”—if you’re willing to accept even that morality is what you argue about when you argue about “morality”—then why not accept other intuitions, other pieces of yourself, into the starting point as well?

Why not accept that, *ceteris paribus*, joy is preferable to sorrow?

You might later find some ground within yourself or built upon yourself with which to criticize this—but why not accept it for now? Not just as a personal preference, mind you; but as something baked into the *question* you ask when you ask “What is truly right”?

But then you might find that you know rather a lot about morality! Nothing certain—nothing unquestionable—nothing unarguable—but still, quite a bit of information. Are you willing to relinquish your Socratic ignorance?

I don’t argue by definitions, of course. But if you claim to know nothing at all about morality, then you will have problems with the *meaning* of your words, not just their plausibility.

Morality as Fixed Computation



discussion
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5-c18

Toby Ord commented:¹

Eliezer, I've just reread your article and was wondering if this is a good quick summary of your position (leaving apart how you got to it):

"I should *X*" means that I would attempt to *X* were I fully informed.

Toby's a pro, so if he didn't get it, I'd better try again. Let me try a different tack of explanation—one closer to the historical way that I arrived at my own position.

Suppose you build an AI, and—leaving aside that AI goal systems cannot be built around English statements, and all such descriptions are only dreams—you try to infuse the AI with the action-determining principle, "Do what I want."

And suppose you get the AI design close *enough*—it doesn't just end up tiling the universe with paperclips, cheesecake or tiny molecular copies of satisfied programmers—that its utility function actually assigns utilities as follows, to the world-states we would describe in English as:

<Programmer weakly desires "X,"	
quantity 20 of X exists>:	+20
<Programmer strongly desires "Y,"	
quantity 20 of X exists>:	0
<Programmer weakly desires "X,"	
quantity 30 of Y exists>:	0
<Programmer strongly desires "Y,"	
quantity 30 of Y exists>:	+60

1. sharov.me/r/5-6

The Meaning of Right • lesswrong.com



You perceive, of course, that this destroys the world.

...since if the programmer initially weakly wants “X” and X is hard to obtain, the AI will modify the programmer to strongly want “Y,” which is easy to create, and then bring about lots of Y. The referent of “Y” might be, say, iron atoms—those are highly stable.

Can you patch this problem? No. As a general rule, it is not possible to patch flawed Friendly AI designs.

If you try to bound the utility function, or make the AI not care about how *much* the programmer wants things, the AI still has a motive (as an *expected* utility maximizer) to make the programmer want something that can be obtained with a very high degree of certainty.

If you try to make it so that the AI can’t modify the programmer, then the AI can’t talk to the programmer (talking to someone modifies them).

If you try to rule out a specific class of ways the AI could modify the programmer, the AI has a motive to superintelligently seek out loopholes and ways to modify the programmer indirectly.

As a general rule, it is not possible to patch flawed FAI designs.

We, ourselves, do not imagine the future and judge that any future in which our brains want something, and that thing exists, is a good future. If we did think this way, we would say: “Yay! Go ahead and modify us to strongly want something cheap!” But we do *not* say this, which means that this AI design is *fundamentally* flawed: it will choose things very unlike what we would choose; it will judge desirability very differently from how we judge it. This core disharmony cannot be patched by ruling out a handful of specific failure modes.

There’s also a duality between Friendly AI problems and moral philosophy problems—though you’ve got to structure that duality in exactly the right way. So if you prefer, the core problem is that the AI will choose in a way very unlike the structure of what is, y’know, actually *right*—never mind the way we choose. Isn’t the whole point of this problem that merely *wanting* something doesn’t *make* it right?

So this is the paradoxical-seeming issue which I have analogized to the difference between:

A calculator that, when you press “2,” “+,” and “3,” tries to compute:

“What is $2 + 3$?”

A calculator that, when you press “2,” “+,” and “3,” tries to compute:

“What does this calculator output when you press ‘2,’ ‘+,’ and ‘3’?”

The Type 1 calculator, as it were, *wants* to output 5.

The Type 2 “calculator” could return any result; and in the act of returning that result, it *becomes* the correct answer to the question that was internally asked.

We ourselves are like unto the Type 1 calculator. But the putative AI is being built as though it were to reflect the Type 2 calculator.

Now imagine that the Type 1 calculator is trying to build an AI, only the Type 1 calculator doesn’t *know* its own question. The calculator continually asks the question by its very nature—it was born to ask that question, created already in motion around that question—but the calculator has no insight into its own transistors; it cannot print out the question, which is extremely complicated and has no simple approximation.

So the calculator wants to build an AI (it’s a pretty smart calculator, it just doesn’t have access to its own transistors) and have the AI give the right answer. Only the calculator can’t print out the question. So the calculator wants to have the AI look at the calculator, where the question is written, and answer the question that the AI will discover implicit in those transistors. But this cannot be done by the cheap shortcut of a utility function that says “All X : $\langle$ calculator asks ‘ X ’, answer X $\rangle$: utility 1; else: utility 0” because that actually mirrors the utility function of a Type 2 calculator, not a Type 1 calculator.

This gets us into FAI issues that I am not going into (some of which I’m still working out myself).

However, when you back out of the details of FAI design, and swap back to the perspective of moral philosophy, then

what we were just talking about was the dual of the moral issue: “But if what’s ‘right’ is a mere preference, then anything that anyone wants is ‘right.’ ”

The key notion is the idea that what we name by “right” is a *fixed* question, or perhaps a *fixed framework*. We can encounter moral arguments that modify our terminal values, and even encounter moral arguments that modify what we count as a moral argument; nonetheless, it all grows out of a particular starting point. We do not experience ourselves as embodying the question “What will I decide to do?” which would be a Type 2 calculator; anything we decided would thereby become right. We experience ourselves as asking the embodied question: “What will save my friends, and my people, from getting hurt? How can we all have more fun? ...” where the “...” is around a thousand other things.

So “I should X” does not mean that I would attempt to X were I fully informed.

“I should X” means that X answers the question, “What will save my people? How can we all have more fun? How can we get more control over our own lives? What’s the funniest jokes we can tell? ...”

And I may not *know* what this question *is*, actually; I may not be able to print out my current guess nor my surrounding framework; but I know, as all non-moral-relativists instinctively know, that the question surely is not just “How can I do whatever I want?”

When these two formulations begin to seem as entirely distinct as “snow” and snow, then you shall have created distinct buckets for the quotation and the referent.

Magical Categories



discussion
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5-c19

We can design intelligent machines so their primary, innate emotion is unconditional love for all humans. First we can build relatively simple machines that learn to recognize happiness and unhappiness in human facial expressions, human voices and human body language. Then we can hard-wire the result of this learning as the innate emotional values of more complex intelligent machines, positively reinforced when we are happy and negatively reinforced when we are unhappy.

—Bill Hibbard (2001), *Super-Intelligent Machines*¹

That was published in a peer-reviewed journal, and the author later wrote a whole book about it, so this is not a strawman position I'm discussing here.

So ...um ...what could possibly go wrong ...

When I mentioned (sec. 7.2)² that Hibbard's AI ends up tiling the galaxy with tiny molecular smiley-faces, Hibbard wrote an indignant reply saying:

When it is feasible to build a super-intelligence, it will be feasible to build hard-wired recognition of "human facial expressions, human voices and human body language" (to use the words of mine that you quote) that exceed the recognition accuracy of

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1. Bill Hibbard, "Super-Intelligent Machines," *ACM SIGGRAPH Computer Graphics* 35, no. 1 (2001): 13–15, sharov.me/r/5-7.



2. Eliezer Yudkowsky, "Artificial Intelligence as a Positive and Negative Factor in Global Risk," in Bostrom and Čirković, *Global Catastrophic Risks*, 308–345.

current humans such as you and me, and will certainly not be fooled by “tiny molecular pictures of smiley-faces.” You should not assume such a poor implementation of my idea that it cannot make discriminations that are trivial to current humans.

As Hibbard also wrote “Such obvious contradictory assumptions show Yudkowsky’s preference for drama over reason,” I’ll go ahead and mention that Hibbard illustrates a key point: There is no professional certification test you have to take before you are allowed to talk about AI morality. But that is not my primary topic today. Though it is a crucial point about the state of the gameboard that most AGI/FAI wannabes are so *utterly* unsuited to the task that I know no one cynical enough to imagine the horror without seeing it firsthand. Even Michael Vassar was probably surprised his first time through.

No, today I am here to dissect “You should not assume such a poor implementation of my idea that it cannot make discriminations that are trivial to current humans.”

Once upon a time—I’ve seen this story in several versions and several places, sometimes cited as fact, but I’ve never tracked down an original source—once upon a time, I say, the US Army wanted to use neural networks to automatically detect camouflaged enemy tanks.

The researchers trained a neural net on 50 photos of camouflaged tanks amid trees, and 50 photos of trees without tanks. Using standard techniques for supervised learning, the researchers trained the neural network to a weighting that correctly loaded the training set—output “yes” for the 50 photos of camouflaged tanks, and output “no” for the 50 photos of forest.

Now this did not prove, or even imply, that new examples would be classified correctly. The neural network might have “learned” 100 special cases that wouldn’t generalize to new problems. Not, “camouflaged tanks versus forest,” but just, “photo-1 positive, photo-2 negative, photo-3 negative, photo-4 positive ...”

But wisely, the researchers had originally taken 200 photos, 100 photos of tanks and 100 photos of trees, and had used

only half in the training set. The researchers ran the neural network on the remaining 100 photos, and *without further training* the neural network classified all remaining photos correctly. Success confirmed!

The researchers handed the finished work to the Pentagon, which soon handed it back, complaining that in their own tests the neural network did no better than chance at discriminating photos.

It turned out that in the researchers' data set, photos of camouflaged tanks had been taken on cloudy days, while photos of plain forest had been taken on sunny days. The neural network had learned to distinguish cloudy days from sunny days, instead of distinguishing camouflaged tanks from empty forest.

This parable—which might or might not be fact—illustrates one of the most fundamental problems in the field of supervised learning and in fact the whole field of Artificial Intelligence: If the training problems and the real problems have the slightest difference in context—if they are not drawn from the same independently identically distributed process—there is no statistical guarantee from past success to future success. It doesn't matter if the AI seems to be working great under the training conditions. (This is not an *unsolvable* problem but it is an *unpatchable* problem. There are deep ways to address it—a topic beyond the scope of this essay—but no band-aids.)

As described in Superexponential Conceptspace, there are exponentially more possible concepts than possible objects, just as the number of possible objects is exponential in the number of attributes. If a black-and-white image is 256 pixels on a side, then the total image is 65,536 pixels. The number of possible images is $2^{65,536}$. And the number of possible *concepts* that classify images into positive and negative instances—the number of possible *boundaries* you could draw in the space of images—is $2^{2^{65,536}}$. From this, we see that even supervised learning is almost entirely a matter of inductive bias, without which it would take a minimum of $2^{65,536}$ classified examples to discriminate among $2^{2^{65,536}}$ possible concepts—even if classifications are constant over time.

So let us now turn again to:

First we can build relatively simple machines that learn to recognize happiness and unhappiness in human facial expressions, human voices and human body language. Then we can hard-wire the result of this learning as the innate emotional values of more complex intelligent machines, positively reinforced when we are happy and negatively reinforced when we are unhappy.

and

When it is feasible to build a super-intelligence, it will be feasible to build hard-wired recognition of “human facial expressions, human voices and human body language” (to use the words of mine that you quote) that exceed the recognition accuracy of current humans such as you and me, and will certainly not be fooled by “tiny molecular pictures of smiley-faces.” You should not assume such a poor implementation of my idea that it cannot make discriminations that are trivial to current humans.

It’s trivial to *discriminate* a photo of a picture with a camouflaged tank, and a photo of an empty forest, in the sense of determining that the two photos are not identical. They’re different pixel arrays with different 1s and 0s in them. Discriminating between them is as simple as testing the arrays for equality.

Classifying new photos into positive and negative instances of “smile,” by reasoning from a set of training photos classified positive or negative, is a different order of problem.

When you’ve got a 256×256 image from a real-world camera, and the image turns out to depict a camouflaged tank, there is no *additional* 65,537th bit denoting the positiveness—no tiny little XML tag that says “This image is inherently positive.” It’s only a positive example relative to some *particular* concept.

But for any non-Vast amount of training data—any training data that does not include the *exact* bitwise image now

seen—there are *superexponentially* many possible concepts compatible with previous classifications.

For the AI, choosing or weighting from among superexponential possibilities is a matter of inductive bias. Which may not match what the user has in mind. The gap between these two example-classifying processes—induction on the one hand, and the user’s actual goals on the other—is not trivial to cross.

Let’s say the AI’s training data is:

Dataset 1:

```

+: Smile_1, Smile_2, Smile_3
-: Frown_1, Cat_1, Frown_2, Frown_3,
Cat_2, Boat_1, Car_1, Frown_5.
```

Now the AI grows up into a superintelligence, and encounters this data:

Dataset 2:

```

: Frown_6, Cat_3, Smile_4, Galaxy_1,
Frown_7, Nanofactory_1, Molecular_Smileyface_
1, Cat_4, Molecular_Smileyface_2, Galaxy_2,
Nanofactory_2.
```

It is not a property of *these datasets* that the inferred classification *you would prefer* is:

```

+: Smile_1, Smile_2, Smile_3, Smile_4
-: Frown_1, Cat_1, Frown_2, Frown_3,
Cat_2, Boat_1, Car_1, Frown_5, Frown_6, Cat_3,
Galaxy_1, Frown_7, Nanofactory_1, Molecular_
Smileyface_1, Cat_4, Molecular_Smileyface_2,
Galaxy_2, Nanofactory_2.
```

rather than

```

+: Smile_1, Smile_2, Smile_3, Molecular_
Smileyface_1, Molecular_Smileyface_2, Smile_4
-: Frown_1, Cat_1, Frown_2, Frown_3,
Cat_2, Boat_1, Car_1, Frown_5, Frown_6, Cat_
3, Galaxy_1, Frown_7, Nanofactory_1, Cat_4,
Galaxy_2, Nanofactory_2.
```

Both of these classifications are compatible with the training data. The number of *concepts* compatible with the training data will be much larger, since more than one concept can project the same shadow onto the combined dataset. If the space of

possible concepts includes the space of possible computations that classify instances, the space is infinite.

Which classification will the AI choose? This is not an inherent property of the training data; it is a property of how the AI performs induction.

Which is the *correct* classification? This is not a property of the training data; it is a property of your preferences (or, if you prefer, a property of the idealized abstract dynamic you name “right”).

The concept that *you wanted* cast its shadow onto the training data as you yourself labeled each instance + or -, drawing on your own intelligence and preferences to do so. That’s what supervised learning is all about—providing the AI with labeled training examples that project a shadow of the causal process that generated the labels.

But unless the training data is drawn from *exactly* the same context as the real-life, the training data will be “shallow” in some sense, a projection from a much higher-dimensional space of possibilities.

The AI never saw a tiny molecular smileyface during its dumber-than-human training phase, or it never saw a tiny little agent with a happiness counter set to a googolplex. Now *you*, finally presented with a tiny molecular smiley—or perhaps a very realistic tiny sculpture of a human face—know at once that this is not what *you* want to count as a smile. But that judgment reflects an unnatural category, one whose classification boundary depends sensitively on your complicated values. It is your own plans and desires that are at work when you say “No!”

Hibbard knows instinctively that a tiny molecular smileyface isn’t a “smile,” because he knows that’s not what he wants his putative AI to do. If someone else were presented with a different task, like classifying artworks, they might feel that the Mona Lisa was obviously smiling—as opposed to frowning, say—even though it’s only paint.

As the case of Terry Schiavo illustrates, technology enables new borderline cases³ that throw us into new, essentially *moral* dilemmas. Showing an AI pictures of living and dead humans as they existed during the age of Ancient Greece will not enable the AI to make a *moral* decision as to whether switching off Terry's life support is murder. That information isn't present in the dataset even inductively! Terry Schiavo raises new moral questions, appealing to new moral considerations, that you wouldn't need to think about while classifying photos of living and dead humans from the time of Ancient Greece. No one was on life support then, still breathing with a brain half fluid. So such considerations play no role in the causal process that you use to classify the ancient-Greece training data, and hence cast no shadow on the training data, and hence are not accessible by induction on the training data.

As a matter of formal fallacy, I see two anthropomorphic errors on display.

The first fallacy is *underestimating the complexity* of a concept we develop for the sake of its value. The borders of the concept will depend on many values and probably on-the-fly moral reasoning, if the borderline case is of a kind we haven't seen before. But all that takes place invisibly, in the background; to Hibbard it just seems that a tiny molecular smileyface is just obviously not a smile. And we don't generate *all* possible borderline cases, so we don't think of all the considerations that might play a role in redefining the concept, but haven't yet played a role in defining it. Since people underestimate the complexity of their concepts, they underestimate the difficulty of inducing the concept from training data. (And also the difficulty of describing the concept directly—see *The Hidden Complexity of Wishes*, vol. III, p. 118.)

The second fallacy is anthropomorphic optimism. Since Bill Hibbard uses his own intelligence to generate options and plans

3. sharov.me/r/5-8

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ranking high in his preference ordering, he is incredulous at the idea that a superintelligence could classify never-before-seen tiny molecular smileyfaces as a positive instance of “smile.” As Hibbard uses the “smile” concept (to describe desired behavior of superintelligences), extending “smile” to cover tiny molecular smileyfaces would rank very low in his preference ordering; it would be a *stupid* thing to do—inherently so, as a property of the concept itself—so surely a superintelligence would not do it; this is just obviously the *wrong* classification. Certainly a *superintelligence* can see which heaps of pebbles are correct or incorrect.

Why, Friendly AI isn’t hard at all! All you need is an AI that does what’s *good*! Oh, sure, not every possible mind does what’s *good*—but in this case, we just *program* the superintelligence to do what’s *good*. All you need is a neural network that sees a few instances of *good* things and not-*good* things, and you’ve got a classifier. Hook that up to an expected utility maximizer and you’re done!

I shall call this the fallacy of magical categories—simple little words that turn out to carry all the desired functionality of the AI. Why not program a chess player by running a neural network (that is, a magical category-absorber) over a set of winning and losing sequences of chess moves, so that it can generate “winning” sequences? Back in the 1950s it was believed that AI might be that simple, but *this turned out not to be the case*.⁴

The novice thinks that Friendly AI is a problem of *coercing* an AI to make it do what *you* want, rather than the AI following its own desires. But the real problem of Friendly AI is one of *communication*—transmitting category boundaries, like “good,” that can’t be fully delineated in any training data you can give the AI during its childhood. Relative to the full space of possibilities the Future encompasses, we *ourselves* haven’t

4. Edit 2018: Note that AlphaZero does use a Monte-Carlo tree search with a neural network that outputs expected outcomes and move probabilities for Chess, Go, and Shogi. Silver, David *et al.* (7 December 2018). “A general reinforcement learning algorithm that masters chess, shogi, and go through self-play”. *Science*. 362 (6419): 1140–1144. arXiv:1712.01815

imagined most of the borderline cases, and would have to engage in full-fledged moral arguments to figure them out. To solve the FAI problem you have to step outside the paradigm of induction on human-labeled training data *and* the paradigm of human-generated intensional definitions.

Of course, even if Hibbard did succeed in conveying to an AI a concept that covers exactly every human facial expression that Hibbard would label a “smile,” and excludes every facial expression that Hibbard wouldn’t label a “smile” ...

Then the resulting AI would *appear* to work correctly during its childhood, when it was weak enough that it could only generate smiles by pleasing its programmers.

When the AI progressed to the point of superintelligence and its own nanotechnological infrastructure, it would rip off your face, wire it into a permanent smile, and start xeroxing.

The deep answers to such problems are beyond the scope of this essay, but it is a general principle of Friendly AI that there are no band-aids. In 2004, Hibbard modified his proposal to assert that expressions of human agreement should reinforce the definition of happiness, and then happiness should reinforce other behaviors. Which, even if it worked, just leads to the AI xeroxing a horde of things similar-in-its-conceptspace to programmers saying “Yes, that’s happiness!” about hydrogen atoms—hydrogen atoms are easy to make.

Link to my discussion with Hibbard here.⁵ You already got the important parts.

5. sharov.me/r/5-9
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The True Prisoner's Dilemma

It occurred to me one day that the standard visualization of the Prisoner's Dilemma is fake.

The core of the Prisoner's Dilemma is this symmetric payoff matrix:

	1 : C	1 : D
2 : C	(3,3)	(5,0)
2 : D	(0,5)	(2,2)

Player 1, and Player 2, can each choose C or D . Player 1's and Player 2's utilities for the final outcome are given by the first and second number in the pair. For reasons that will become apparent, " C " stands for "cooperate" and D stands for "defect."

Observe that a player in this game (regarding themselves as the first player) has this preference ordering over outcomes: $(D, C) > (C, C) > (D, D) > (C, D)$.

Option D , it would seem, dominates C : If the other player chooses C , you prefer (D, C) to (C, C) ; and if the other player chooses D , you prefer (D, D) to (C, D) . So you wisely choose D , and as the payoff table is symmetric, the other player likewise chooses D .

If only you'd both been less wise! You *both* prefer (C, C) to (D, D) . That is, you both prefer mutual cooperation to mutual defection.

The Prisoner's Dilemma is one of the great foundational issues in decision theory, and enormous volumes of material have been written about it. Which makes it an audacious assertion of mine, that the usual way of *visualizing* the Prisoner's Dilemma has a severe flaw, at least if you happen to be human.

The classic visualization of the Prisoner's Dilemma is as follows: you are a criminal, and you and your confederate in crime have both been captured by the authorities.

Independently, without communicating, and without being able to change your mind afterward, you have to decide whether



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to give testimony against your confederate (*D*) or remain silent (*C*).

Both of you, right now, are facing one-year prison sentences; testifying (*D*) takes one year off your prison sentence, and adds two years to your confederate's sentence.

Or maybe you and some stranger are, only once, and without knowing the other player's history or finding out who the player was afterward, deciding whether to play *C* or *D*, for a payoff in dollars matching the standard chart.

And, oh yes—in the classic visualization you're supposed to *pretend that you're entirely selfish*, that you don't care about your confederate criminal, or the player in the other room.

It's this last specification that makes the classic visualization, in my view, fake.

You can't avoid hindsight bias by instructing a jury to pretend not to know the real outcome of a set of events. And without a complicated effort backed up by considerable knowledge, a neurologically intact human being cannot pretend to be genuinely, truly selfish.

We're born with a sense of fairness, honor, empathy, sympathy, and even altruism—the result of our ancestors' adapting to play the *iterated* Prisoner's Dilemma. We don't really, truly, absolutely and entirely prefer (*D, C*) to (*C, C*), though we may entirely prefer (*C, C*) to (*D, D*) and (*D, D*) to (*C, D*). The thought of our confederate spending three years in prison does not entirely fail to move us.

In that locked cell where we play a simple game under the supervision of economic psychologists, we are not entirely and absolutely without sympathy for the stranger who might cooperate. We aren't entirely happy to think that we might defect and the stranger cooperate, getting five dollars while the stranger gets nothing.

We fixate instinctively on the (*C, C*) outcome and search for ways to argue that it should be the mutual decision: "How can we ensure mutual cooperation?" is the instinctive thought. Not "How can I trick the other player into playing *C* while I play *D* for the maximum payoff?"

For someone with an impulse toward altruism, or honor, or fairness, the Prisoner's Dilemma doesn't *really* have the critical payoff matrix—whatever the *financial* payoff to individuals. The outcome (C, C) is preferable to the outcome (D, C) , and the key question is whether the other player sees it the same way.

And no, you can't instruct people being initially introduced to game theory to pretend they're completely selfish—any more than you can instruct human beings being introduced to anthropomorphism to pretend they're expected paperclip maximizers.

To construct the True Prisoner's Dilemma, the situation has to be something like this:

Player 1: Human beings, Friendly AI, or other humane intelligence.

Player 2: Unfriendly AI, or an alien that only cares about sorting pebbles.

Let's suppose that four billion human beings—not the whole human species, but a significant part of it—are currently progressing through a fatal disease that can only be cured by substance S .

However, substance S can only be produced by working with a paperclip maximizer from another dimension—substance S can also be used to produce paperclips. The paperclip maximizer only cares about the number of paperclips in its own universe, not in ours, so we can't offer to produce or threaten to destroy paperclips here. We have never interacted with the paperclip maximizer before, and will never interact with it again.

Both humanity and the paperclip maximizer will get a single chance to seize some additional part of substance S for themselves, just before the dimensional nexus collapses; but the seizure process destroys some of substance S .

The payoff matrix is as follows:

	1:C	1:D
2:C	(+2 billion human lives, +2 paperclips)	(+3 billion lives, +0 paperclips)
2:D	(+0 lives, +3 paperclips)	(+1 billion lives, +1 paperclip)

I've chosen this payoff matrix to produce a sense of *indignation* at the thought that the paperclip maximizer wants to trade off billions of human lives against a couple of paperclips. Clearly the paperclip maximizer *should* just let us have all of substance S. But a paperclip maximizer doesn't do what it *should*; it just maximizes paperclips.

In this case, we *really do* prefer the outcome (D, C) to the outcome (C, C) , leaving aside the actions that produced it. We would vastly rather live in a universe where 3 billion humans were cured of their disease and no paperclips were produced, rather than sacrifice a billion human lives to produce 2 paperclips. It doesn't seem *right* to cooperate, in a case like this. It doesn't even seem *fair*—so great a sacrifice by us, for so little gain by the paperclip maximizer? And let us specify that the paperclip-agent experiences no pain or pleasure—it just outputs actions that steer its universe to contain more paperclips. The paperclip-agent will experience no pleasure at gaining paperclips, no hurt from losing paperclips, and no painful sense of betrayal if we betray it.

What do you do then? Do you cooperate when you really, definitely, truly and absolutely do want the highest reward you can get, and you don't care a tiny bit by comparison about what happens to the other player? When it seems *right* to defect even if the other player cooperates?

That's what the payoff matrix for the *true* Prisoner's Dilemma looks like—a situation where (D, C) seems *righter* than (C, C) .

But all the rest of the logic—everything about what happens if both agents think that way, and both agents defect—is the same. For the paperclip maximizer cares as little about human deaths, or human pain, or a human sense of betrayal, as we care about paperclips. Yet we both prefer (C, C) to (D, D) .

So if you've ever prided yourself on cooperating in the Prisoner's Dilemma ...or questioned the verdict of classical game theory that the "rational" choice is to defect ...then what do you say to the True Prisoner's Dilemma above?

PS: In fact, I *don't* think rational agents should always defect in one-shot Prisoner's Dilemmas, when the other player will

cooperate if it expects you to do the same. I think there are situations where two agents can rationally achieve (C, C) as opposed to (D, D) , and reap the associated benefits.¹

I'll explain some of my reasoning when I discuss Newcomb's Problem. But we can't talk about whether rational cooperation is possible in this dilemma until we've dispensed with the visceral sense that the (C, C) outcome is *nice* or *good* in itself. We have to see past the prosocial label "mutual cooperation" if we are to grasp the math. If you intuit that (C, C) trumps (D, D) from Player 1's perspective, but don't intuit that (D, C) also trumps (C, C) , you haven't yet appreciated what makes this problem difficult.

1. Eliezer Yudkowsky, *Timeless Decision Theory*, Unpublished manuscript (Machine Intelligence Research Institute, Berkeley, CA, 2010), sharov.me/t/5-10.



Sympathetic Minds



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“Mirror neurons” are neurons that are active both when performing an action and observing the same action—for example, a neuron that fires when you hold up a finger or see someone else holding up a finger. Such neurons have been directly recorded in primates, and consistent neuroimaging evidence has been found for humans.

You may recall from my previous writing on “empathic inference” the idea that brains are so complex that the only way to simulate them is by forcing a similar brain to behave similarly. A brain is so complex that if a human tried to understand brains the way that we understand e.g. gravity or a car—observing the whole, observing the parts, building up a theory from scratch—then we would be unable to *invent good hypotheses* in our mere mortal lifetimes. The only possible way you can hit on an “Aha!” that describes a system as incredibly complex as an Other Mind, is if you happen to run across something amazingly similar to the Other Mind—namely your own brain—which you can actually force to behave similarly and use as a hypothesis, yielding predictions.

So that is what I would call “empathy.”

And then “sympathy” is something else on top of this—to smile when you see someone else smile, to hurt when you see someone else hurt. It goes beyond the realm of prediction into the realm of reinforcement.

And you ask, “Why would callous natural selection do anything *that* nice?”

It might have gotten started, maybe, with a mother’s love for her children, or a brother’s love for a sibling. You can want them to live, you can want them to be fed, sure; but if you smile when they smile and wince when they wince, that’s a simple urge that leads you to deliver help along a broad avenue, in many walks of life. So long as you’re in the ancestral environment,

what your relatives want probably has something to do with your relatives' reproductive success—this being an explanation for the selection pressure, of course, not a conscious belief.

You may ask, "Why not evolve a more abstract desire to see certain people tagged as 'relatives' get what they want, without actually feeling yourself what they feel?" And I would shrug and reply, "Because then there'd have to be a whole definition of 'wanting' and so on. Evolution doesn't take the elaborate correct optimal path, it falls up the fitness landscape like water flowing downhill. The mirroring-architecture was already there, so it was a short step from empathy to sympathy, and it got the job done."

Relatives—and then reciprocity; your allies in the tribe, those with whom you trade favors. Tit for Tat, or evolution's elaboration thereof to account for social reputations.

Who is the most formidable, among the human kind? The strongest? The smartest? More often than either of these, I think, it is the one who can call upon the most friends.

So how do you make lots of friends?

You could, perhaps, have a specific urge to bring your allies food, like a vampire bat—they have a whole system of reciprocal blood donations going in those colonies. But it's a more *general* motivation, that will lead the organism to store up *more* favors, if you smile when designated friends smile.

And what kind of organism will avoid making its friends angry at it, in full generality? One that winces when they wince.

Of course you also want to be able to kill designated Enemies without a qualm—these *are* humans we're talking about.

But ...I'm not sure of this, but it *does* look to me like sympathy, among humans, is "on" by default. There are cultures that help strangers ...and cultures that eat strangers; the question is which of these requires the explicit imperative, and which is the default behavior for humans. I don't really think I'm being such a crazy idealistic fool when I say that, based on my admittedly limited knowledge of anthropology, it looks like sympathy is on by default.

Either way ...it's painful if you're a bystander in a war between two sides, and your sympathy has *not* been switched off

for either side, so that you wince when you see a dead child no matter what the caption on the photo; and yet those two sides have no sympathy for each other, and they go on killing.

So that is the human idiom of *sympathy*—a strange, complex, deep implementation of reciprocity and helping. It tangles minds together—not by a term in the utility function for some other mind’s “desire,” but by the simpler and yet far more consequential path of mirror neurons: feeling what the other mind feels, and seeking similar states. Even if it’s only done by observation and inference, and not by direct transmission of neural information as yet.

Empathy is a human way of predicting other minds. It is not the *only* possible way.

The human brain is not *quickly* rewirable; if you’re suddenly put into a dark room, you can’t rewire the visual cortex as auditory cortex, so as to better process sounds, until you leave, and then suddenly shift all the neurons back to being visual cortex again.

An AI, at least one running on anything like a modern programming architecture, can trivially shift computing resources from one thread to another. Put in the dark? Shut down vision and devote all those operations to sound; swap the old program to disk to free up the RAM, then swap the disk back in again when the lights go on.

So why would an AI need to force its *own* mind into a state similar to what it wanted to predict? Just create a *separate* mind-instance—maybe with different algorithms, the better to simulate that very dissimilar human. Don’t try to mix up the data with your own mind-state; don’t use mirror neurons. Think of all the risk and mess *that* implies!

An expected utility maximizer—especially one that does understand intelligence on an abstract level—has other options than *empathy*, when it comes to understanding other minds. The agent doesn’t need to put *itself* in anyone else’s shoes; it can just model the other mind *directly*. A hypothesis like any other hypothesis, just a little bigger. You don’t need to become your shoes to understand your shoes.

And sympathy? Well, suppose we're dealing with an expected paperclip maximizer, but one that isn't yet powerful enough to have things all its own way—it has to deal with humans to get its paperclips. So the paperclip agent ...models those humans as relevant parts of the environment, models their probable reactions to various stimuli, and does things that will make the humans feel favorable toward it in the future.

To a paperclip maximizer, the humans are just machines with pressable buttons. No need to *feel what the other feels*—if that were even *possible* across such a tremendous gap of internal architecture. How could an expected paperclip maximizer “feel happy” when it saw a human smile? “Happiness” is an idiom of policy reinforcement learning, not expected utility maximization. A paperclip maximizer doesn't feel happy when it makes paperclips; it just chooses whichever action leads to the greatest number of expected paperclips. Though a paperclip maximizer might find it convenient to display a smile when it made paperclips—so as to help manipulate any humans that had designated it a friend.

You might find it a bit difficult to imagine such an algorithm—to put yourself into the shoes of something that does not work like you do, and does not work like any mode your brain can make itself operate in.

You can make your brain operate in the mode of hating an enemy, but that's not right either. The way to imagine how a truly *unsympathetic* mind sees a human is to imagine yourself as a useful machine with levers on it. Not a human-shaped machine, because we have instincts for that. Just a woodsaw or something. Some levers make the machine output coins; other levers might make it fire a bullet. The machine does have a persistent internal state and you have to pull the levers in the right order. Regardless, it's just a complicated causal system—nothing inherently mental about it.

(To understand *unsympathetic* optimization processes, I would suggest studying natural selection, which doesn't bother to anesthetize fatally wounded and dying creatures, even when their pain no longer serves any reproductive purpose, because the anesthetic would serve no reproductive purpose either.)

That's why I list "sympathy" in front of even "boredom" on my list of things that would be required to have aliens that are the least bit, if you'll pardon the phrase, sympathetic. It's not impossible that sympathy exists among some significant fraction of all evolved alien intelligent species; mirror neurons seem like the sort of thing that, having happened once, *could* happen again.

Unsympathetic aliens might be trading partners—or not; stars and such resources are pretty much the same the universe over. We might negotiate treaties with them, and they might keep them for calculated fear of reprisal. We might even cooperate in the Prisoner's Dilemma. But we would never be friends with them. They would never see us as anything but means to an end. They would never shed a tear for us, nor smile for our joys. And the others of their own kind would receive no different consideration, nor have any sense that they were missing something important thereby.

Such aliens would be *varelse*,¹ not *ramen*—the sort of aliens we can't relate to on any personal level, and no point in trying.

1. sharov.me/r/5-11
In Praise of Boredom • lesswrong.com



High Challenge

There's a class of prophecy that runs: "In the Future, machines will do all the work. Everything will be automated. Even labor of the sort we now consider 'intellectual,' like engineering, will be done by machines. We can sit back and own the capital. You'll never have to lift a finger, ever again."

But then won't people be bored?

No; they can play computer games—not like *our* games, of course, but much more advanced and entertaining.

Yet wait! If you buy a modern computer game, you'll find that it contains some tasks that are—there's no kind word for this—*effortful*. (I would even say "difficult," with the understanding that we're talking about something that takes ten minutes, not ten years.)

So in the future, we'll have programs that *help* you play the game—taking over if you get stuck on the game, or just bored; or so that you can play games that would otherwise be too advanced for you.

But isn't there some wasted effort, here? Why have one programmer working to make the game harder, and another programmer to working to make the game easier? Why not just make the game easier to *start with*? Since you play the game to get gold and experience points, making the game easier will let you get more gold per unit time: the game will become more fun.

So this is the ultimate end of the prophecy of technological progress—just staring at a screen that says "YOU WIN," forever.

And maybe we'll build a robot that does *that*, too.

Then what?

The world of machines that do *all* the work—well, I don't want to say it's "analogous to the Christian Heaven" because it isn't supernatural; it's something that could in principle be realized. Religious analogies are far too easily tossed around as accusations...But, without implying any other similarities, I'll



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say that it seems analogous in the sense that eternal laziness “sounds like good news” to your present self who still has to work.

And as for playing games, as a substitute—what *is* a computer game except synthetic work? Isn’t there a wasted step here? (And computer games in their present form, considered as work, have various aspects that reduce stress and increase engagement; but they also carry costs in the form of artificiality and isolation.)

I sometimes think that futuristic ideals phrased in terms of “getting rid of work” would be better reformulated as “removing low-quality work to make way for high-quality work.”

There’s a broad class of goals that aren’t suitable as the long-term meaning of life, because you can actually achieve them, and then you’re done.

To look at it another way, if we’re looking for a suitable long-run meaning of life, we should look for goals that are good to *pursue* and not just good to *satisfy*.

Or to phrase that somewhat less paradoxically: We should look for valuations that are over 4D states, rather than 3D states. Valuable ongoing processes, rather than “make the universe have property *P* and then you’re done.”

Timothy Ferris is worth quoting: To find happiness, “the question you should be asking isn’t ‘What do I want?’ or ‘What are my goals?’ but ‘What would excite me?’”

You might say that for a long-run meaning of life, we need games that are fun to *play* and not just to *win*.

Mind you—sometimes you *do* want to win. There are legitimate goals where winning is everything. If you’re talking, say, about curing cancer, then the suffering experienced by even a single cancer patient outweighs any fun that you might have in solving their problems. If you work at creating a cancer cure for twenty years through your own efforts, learning new knowledge and new skill, making friends and allies—and then some alien superintelligence offers you a cancer cure on a silver platter for thirty bucks—then you shut up and take it.

But “curing cancer” is a problem of the 3D-predicate sort: you want the no-cancer predicate to go from False in the present

to True in the future. The importance of this destination far outweighs the journey; you don't want to *go* there, you just want to *be* there. There are many *legitimate* goals of this sort, but they are not suitable as long-run fun. "Cure cancer!" is a worthwhile activity for us to pursue here and now, but it is not a plausible future goal of galactic civilizations.

Why should this "valuable ongoing process" be a process of *trying to do things*—why not a process of passive experiencing, like the Buddhist Heaven?

I confess I'm not entirely sure how to set up a "passively experiencing" mind. The human brain was *designed* to perform various sorts of internal work that add up to an active intelligence; even if you lie down on your bed and exert no particular effort to think, the thoughts that go on through your mind are activities of brain areas that are designed to, you know, *solve problems*.

How much of the human brain could you eliminate, *apart* from the pleasure centers, and still keep the subjective experience of pleasure?

I'm not going to touch that one. I'll stick with the much simpler answer of "I wouldn't actually *prefer* to be a passive experiencer." If I *wanted* Nirvana, I might try to figure out how to achieve that impossibility. But once you strip away Buddha telling me that Nirvana is the end-all of existence, Nirvana seems rather more like "sounds like good news in the moment of first being told" or "ideological belief in desire," rather than, y'know, something I'd actually *want*.

The reason I have a mind at all is that natural selection built me to *do things*—to solve certain kinds of problems.

"Because it's human nature" is not an explicit justification for anything. There is human nature, which is what we are; and there is humane nature, which is what, being human, we wish we were.

But I don't *want* to change my nature toward a more passive object—which *is* a justification. A happy blob is *not* what, being human, I wish to become.

I earlier argued that many values require both subjective happiness and the external objects of that happiness. That you

can legitimately have a utility function that says, “It matters to me whether or not the person I love is a real human being or just a highly realistic nonsentient chatbot, *even if I don’t know*, because that-which-I-value is not my own state of mind, but the external reality.” So that you need both the experience of love, and the real lover.

You can similarly have valuable activities that require both real challenge and real effort.

Racing along a track, it matters that the other racers are real, and that you have a real chance to win or lose. (We’re not talking about physical determinism here, but whether some external optimization process explicitly chose for you to win the race.)

And it matters that you’re racing with your own skill at running and your own willpower, not just pressing a button that says “Win.” (Though, since you never designed your own leg muscles, you *are* racing using strength that isn’t yours. A race between robot cars is a purer contest of their designers. There is plenty of room to improve on the human condition.)

And it matters that you, a sentient being, are experiencing it. (Rather than some nonsentient process carrying out a skeleton imitation of the race, trillions of times per second.)

There must be the true effort, the true victory, and the true experience—the journey, the destination and the traveler.

Serious Stories

Every Utopia ever constructed—in philosophy, fiction, or religion—has been, to one degree or another, a place where you wouldn't *actually want* to live. I am not alone in this important observation: George Orwell said much the same thing in “Why Socialists Don't Believe In Fun,” and I expect that many others said it earlier.

If you read books on How To Write—and there are a *lot* of books out there on How To Write, because, amazingly, a lot of book-writers think they know something about writing—these books will tell you that stories must contain “conflict.”

That is, the more *lukewarm* sort of instructional book will tell you that stories contain “conflict.” But some authors speak more plainly.

“Stories are about people's pain.” Orson Scott Card.

“Every scene must end in disaster.” Jack Bickham.

In the age of my youthful folly, I took for granted that *authors* were excused from the search for true Eutopia, because if you constructed a Utopia that *wasn't* flawed...what stories could you write, set there? “Once upon a time they lived happily ever after.” What use would it be for a science-fiction author to try to depict a positive intelligence explosion, when a positive intelligence explosion would be ...

...the end of all stories?

It seemed like a reasonable framework with which to examine the literary problem of Utopia, but something about that final conclusion produced a quiet, nagging doubt.

At that time I was thinking of an AI as being something like a safe wish-granting genie for the use of individuals. So the conclusion did make a kind of sense. If there was a problem, you would just wish it away, right? Ergo—no stories. So I ignored the quiet, nagging doubt.

Much later, after I concluded that even a safe genie wasn't such a good idea, it also seemed in retrospect that “no stories”



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could have been a productive indicator. On this particular occasion, “I can’t think of a single story I’d *want to read* about this scenario,” might indeed have pointed me toward the reason “I wouldn’t want to *actually live* in this scenario.”

So I swallowed my trained-in revulsion of Luddism and theodicy, and at least *tried* to contemplate the argument:

- A world in which nothing ever goes wrong, or no one ever experiences any pain or sorrow, is a world containing no stories worth reading about.
- A world that you wouldn’t want to read about is a world where you wouldn’t want to live.
- Into each eudaimonic life a little pain must fall. QED.

In one sense, it’s clear that we do *not* want to live the sort of lives that are depicted in most stories that human authors have written so far. Think of the truly great stories, the ones that have become legendary for being the very best of the best of their genre: the *Iliad*, *Romeo and Juliet*, *The Godfather*, *Watchmen*, *Planescape: Torment*, the second season of *Buffy the Vampire Slayer*, or *that ending* in *Tsukihime*. Is there a single story on the list that *isn’t* tragic?

Ordinarily, we prefer pleasure to pain, joy to sadness, and life to death. Yet it seems we prefer to empathize with hurting, sad, dead characters. Or stories about happier people *aren’t serious*, aren’t artistically great enough to be worthy of praise—but then why selectively praise stories containing unhappy people? Is there some hidden benefit to us in it? It’s a puzzle either way you look at it.

When I was a child I couldn’t write fiction because I wrote things to go *well* for my characters—just like I wanted things to go well in real life. Which I was cured of by Orson Scott Card: *Oh*, I said to myself, *that’s what I’ve been doing wrong, my characters aren’t hurting*. Even then, I didn’t realize that the microstructure of a plot works the same way—until Jack Bickham said that every scene must end in disaster. Here I’d been trying to set up problems and *resolve* them, instead of making them *worse*...

You simply don't *optimize* a story the way you optimize a real life. The *best* story and the *best* life will be produced by different criteria.

In the real world, people can go on living for quite a while without any major disasters, and still seem to do pretty okay. When was the last time you were shot at by assassins? Quite a while, right? Does your life seem emptier for it?

But on the other hand...

For some odd reason, when authors get too old or too successful, they revert to my childhood. Their stories start going *right*. They stop doing horrible things to their characters, with the result that they start doing horrible things to their readers. It seems to be a regular part of Elder Author Syndrome. Mercedes Lackey, Laurell K. Hamilton, Robert Heinlein, even Orson Scott bloody Card—they all went that way. They forgot how to hurt their characters. I don't know why.

And when you read a story by an Elder Author or a pure novice—a story where things just *relentlessly go right* one after another—where the main character defeats the supervillain with a snap of the fingers, or even worse, before the final battle, the supervillain *gives up and apologizes and then they're friends again*—

It's like a fingernail scraping on a blackboard at the base of your spine. If you've never actually read a story like that (or worse, written one) then count yourself lucky.

That fingernail-scraping quality—would it transfer over from the story to real life, if you tried living real life without a single drop of rain?

One answer might be that what a story really needs is not “disaster,” or “pain,” or even “conflict,” but simply *striving*. That the problem with Mary Sue stories is that there's not enough striving in them, but they wouldn't actually need *pain*. This might, perhaps, be tested.

An alternative answer might be that this *is* the transhumanist version of Fun Theory we're talking about. So we can reply, “Modify brains to eliminate that fingernail-scraping feeling,” unless there's some justification for keeping it. If the fingernail-

scraping feeling is a pointless random bug getting in the way of Utopia, delete it.

Maybe we *should*. Maybe all the Great Stories are tragedies because ...well ...

I once read that in the BDSM community, “intense sensation” is a euphemism for pain. Upon reading this, it occurred to me that, the way humans are constructed now, it is just *easier* to produce pain than pleasure. Though I speak here somewhat outside my experience, I expect that it takes a highly talented and experienced sexual artist working for hours to produce a *good* feeling as intense as the pain of one strong kick in the testicles—which is doable in seconds by a novice.

Investigating the life of the priest and proto-rationalist Friedrich Spee von Langenfeld, who heard the confessions of accused witches, I looked up some of the instruments that had been used to produce confessions. There is no ordinary way to make a human being feel as *good* as those instruments would make you hurt. I’m not sure even drugs would do it, though my experience of drugs is as nonexistent as my experience of torture.

There’s something imbalanced about that.

Yes, human beings are too optimistic in their planning. If losses weren’t more aversive than gains, we’d go broke, the way we’re constructed now. The experimental rule is that losing a desideratum—\$50, a coffee mug, whatever—hurts between 2 and 2.5 times as much as the equivalent gain.

But this is a deeper imbalance than that. The effort-in/intensity-out difference between sex and torture is not a mere factor of 2.

If someone goes in search of sensation—in this world, the way human beings are constructed now—it’s not surprising that they should arrive at pains to be mixed into their pleasures as a source of *intensity* in the combined experience.

If only people were constructed differently, so that you could produce pleasure as intense and in as many different flavors as pain! If only you could, with the same ingenuity and effort as a torturer of the Inquisition, make someone feel as *good* as the Inquisition’s victims felt *bad*—

But then, what is the analogous pleasure that feels that good? A victim of skillful torture will do anything to stop the pain and anything to prevent it from being repeated. Is the equivalent pleasure one that overrides everything with the demand to continue and repeat it? If people are stronger-willed to bear the pleasure, is it really the same pleasure?

There is another rule of writing which states that stories have to *shout*. A human brain is a long way off those printed letters. Every event and feeling needs to take place at ten times natural volume in order to have any impact at all. You must not try to make your characters behave or feel *realistically*—especially, you must not faithfully reproduce your own past experiences—because *without exaggeration*, they'll be too quiet to rise from the page.

Maybe all the Great Stories are tragedies because happiness can't shout loud enough—to a human reader.

Maybe that's what needs fixing.

And if it were fixed...would there be any use left for pain or sorrow? For even the *memory* of sadness, if all things were already as good as they could be, and every remediable ill already remedied?

Can you just delete pain outright? Or does removing the old floor of the utility function just create a new floor? Will any pleasure less than 10,000,000 hedons be the new unbearable pain?

Humans, built the way we are now, do seem to have hedonic scaling tendencies. Someone who can remember starving will appreciate a loaf of bread more than someone who's never known anything but cake. This was George Orwell's hypothesis for why Utopia is impossible in literature and reality:¹

It would seem that human beings are not able to describe, nor perhaps to imagine, happiness except in terms of contrast...The inability of mankind to imagine happiness except in the form of relief,

1. George Orwell, "Why Socialists Don't Believe in Fun," *Tribune* (December 1943).

either from effort or pain, presents Socialists with a serious problem. Dickens can describe a poverty-stricken family tucking into a roast goose, and can make them appear happy; on the other hand, the inhabitants of perfect universes seem to have no spontaneous gaiety and are usually somewhat repulsive into the bargain.

For an expected utility maximizer, rescaling the utility function to add a trillion to all outcomes is meaningless—it's literally the same utility function, as a mathematical object. A utility function describes the *relative* intervals between outcomes; that's what it is, mathematically speaking.

But the human brain has distinct neural circuits for positive feedback and negative feedback, and different varieties of positive and negative feedback. There are people today who "suffer" from congenital analgesia—a total absence of pain. I never heard that *insufficient pleasure* becomes intolerable to them.

Congenital analgesics do have to inspect themselves carefully and frequently to see if they've cut themselves or burned a finger. Pain serves a purpose in the human mind design ...

But that does not show there's no alternative which could serve the same purpose. Could you delete pain and replace it *with an urge not to do certain things* that lacked the intolerable subjective quality of pain? I do not know all the Law that governs here, but I'd have to guess that yes, you could; you could replace that side of yourself with something more akin to an expected utility maximizer.

Could you delete the human tendency to scale pleasures—delete the accommodation, so that each new roast goose is as delightful as the last? I would guess that you could. This verges perilously close to deleting Boredom, which is right up there with Sympathy as an absolute indispensable ...but to say that an old solution remains as pleasurable is not to say that you will lose the urge to seek new and better solutions.

Can you make every roast goose as pleasurable as it would be in contrast to starvation, without ever having starved?

Can you prevent the pain of a dust speck irritating your eye from being the new torture, if you've literally *never experienced* anything *worse* than a dust speck irritating your eye?

Such questions begin to exceed my grasp of the Law, but I would guess that the answer is: yes, it can be done. It is my experience in such matters that once you do learn the Law, you can usually see how to do weird-seeming things.

So far as I know or can guess, David Pearce (*The Hedonistic Imperative*) is very probably right about the *feasibility* part, when he says:²

Nanotechnology and genetic engineering will abolish suffering in all sentient life. The abolitionist project is hugely ambitious but technically feasible. It is also instrumentally rational and morally urgent. The metabolic pathways of pain and malaise evolved because they served the fitness of our genes in the ancestral environment. They will be replaced by a different sort of neural architecture—a motivational system based on heritable gradients of bliss. States of sublime well-being are destined to become the genetically pre-programmed norm of mental health. It is predicted that the world's last unpleasant experience will be a precisely dateable event.

Is that ...what we *want*?

To just wipe away the last tear, and be done?

Is there any good reason *not* to, except status quo bias and a handful of worn rationalizations?

What would be the *alternative*? Or alternatives?

2. David Pearce, *The Hedonistic Imperative*, sharov.me/r/5-12, 1995.



To leave things as they are? Of course not. No God designed this world; we have no reason to think it exactly optimal on any dimension. If this world does not contain too much pain, then it must not contain enough, and the latter seems unlikely.

But perhaps ...

You could cut out just the *intolerable* parts of pain?

Get rid of the Inquisition. Keep the sort of pain that tells you not to stick your finger in the fire, or the pain that tells you that you shouldn't have put your friend's finger in the fire, or even the pain of breaking up with a lover.

Try to get rid of the sort of pain that *grinds down and destroys* a mind. Or configure minds to be harder to damage.

You could have a world where there were broken legs, or even broken hearts, but no broken *people*. No child sexual abuse that turns out more abusers. No people ground down by weariness and drudging minor inconvenience to the point where they contemplate suicide. No random meaningless endless sorrows like starvation or AIDS.

And if even a broken leg still seems too scary—

Would we be less frightened of pain, if we were stronger, if our daily lives did not already exhaust so much of our reserves?

So that would be one alternative to Pearce's world—if there are yet other alternatives, I haven't thought them through in any detail.

The path of courage, you might call it—the idea being that if you eliminate the destroying kind of pain and strengthen the people, then what's left shouldn't be *that* scary.

A world where there is sorrow, but not massive systematic *pointless* sorrow, like we see on the evening news. A world where pain, if it is not eliminated, at least does not *overbalance pleasure*. You could write stories about that world, and they could read our stories.

I do tend to be rather conservative around the notion of deleting large parts of human nature. I'm not sure how many major chunks you can delete until that balanced, conflicting, dynamic structure collapses into something simpler, like an expected pleasure maximizer.

And so I do admit that it is the path of courage that appeals to me.

Then again, I haven't lived it both ways.

Maybe I'm just *afraid* of a world so different as Analgesia—wouldn't that be an ironic reason to walk "the path of courage"?

Maybe the path of courage just seems like the *smaller change*—maybe I just have trouble empathizing over a larger gap.

But "change" is a moving target.

If a human child grew up in a *less* painful world—if they had never lived in a world of AIDS or cancer or slavery, and so did not know these things as evils that had *been triumphantly eliminated*—and so did not feel that they were "already done" or that the world was "already changed enough" ...

Would they take the next step, and try to eliminate the unbearable pain of broken hearts, when someone's lover stops loving them?

And then what? Is there a point where *Romeo and Juliet* just seems less and less relevant, more and more a relic of some distant forgotten world? Does there come some point in the transhuman journey where the whole business of the negative reinforcement circuitry can't possibly seem like anything except a pointless hangover to wake up from?

And if so, is there any point in *delaying* that last step? Or should we just throw away our fears and ...throw away our fears?

I don't know.

Value is Fragile



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If I had to pick a single statement that *relies* on more *Overcoming Bias* content I've written than any other, that statement would be:

*Any Future **not** shaped by a goal system with detailed reliable inheritance from human morals and metamorals will contain almost nothing of worth.*

"Well," says the one, "maybe according to your provincial *human* values, *you* wouldn't like it. But I can easily imagine a galactic civilization full of agents who are nothing like *you*, yet find great value and interest in their *own* goals. And that's fine by me. I'm not so bigoted as you are. Let the Future go its own way, without trying to bind it forever to the laughably primitive prejudices of a pack of four-limbed Squishy Things—"

My friend, I have *no problem* with the thought of a galactic civilization vastly unlike our own...full of strange beings who look nothing like me even in their own imaginations...pursuing pleasures and experiences I can't begin to empathize with...trading in a marketplace of unimaginable goods...allying to pursue incomprehensible objectives...people whose life-stories I could never understand.

That's what the Future looks like if things go *right*.

If the chain of inheritance from human (meta)morals is broken, the Future does *not* look like this. It does *not* end up magically, delightfully incomprehensible.

With very high probability, it ends up looking *dull*. Pointless. Something whose loss you wouldn't mourn.

Seeing this as obvious is what requires that immense amount of background explanation.

And I'm not going to iterate through *all* the points and winding pathways of argument here, because that would take us back through 75% of my *Overcoming Bias* posts. Except

to remark on how *many* different things must be known to constrain the final answer.

Consider the incredibly important human value of “boredom”—our desire not to do “the same thing” over and over and over again. You can imagine a mind that contained *almost* the whole specification of human value, almost all the morals and metamorals, but left out *just this one thing*—

—and so it spent until the end of time, and until the farthest reaches of its light cone, replaying a single highly optimized experience, over and over and over again.

Or imagine a mind that contained almost the whole specification of which sort of feelings humans most enjoy—but not the idea that those feelings had important *external referents*. So that the mind just went around *feeling* like it had made an important discovery, *feeling* it had found the perfect lover, *feeling* it had helped a friend, but not actually *doing* any of those things—having become its own experience machine. And if the mind pursued those feelings *and their referents*, it would be a good future and true; but because this *one dimension* of value was left out, the future became something dull. Boring and repetitive, because although this mind *felt* that it was encountering experiences of incredible novelty, this feeling was in no wise true.

Or the converse problem—an agent that contains all the aspects of human value, *except* the valuation of subjective experience. So that the result is a nonsentient optimizer that goes around making genuine discoveries, but the discoveries are not savored and enjoyed, because there is no one there to do so. This, I admit, I don’t quite know to be possible. Consciousness does still confuse me to some extent. But a universe with no one to bear witness to it might as well not be.

Value isn’t just complicated, it’s *fragile*. There is *more than one dimension* of human value, where *if just that one thing is lost*, the Future becomes null. A *single* blow and *all* value shatters. Not every *single* blow will shatter *all* value—but more than one possible “single blow” will do so.

And then there are the long defenses of this proposition, which relies on 75% of my *Overcoming Bias* posts, so that it

would be more than one day's work to summarize all of it. Maybe some other week. There's so many branches I've seen that discussion tree go down.

After all—a mind *shouldn't* just go around having the same experience over and over and over again. Surely no superintelligence would be so grossly mistaken about the correct action?

Why would any supermind want something so inherently worthless as the feeling of discovery without any real discoveries? Even if that were its utility function, wouldn't it just notice that its utility function was wrong, and rewrite it? It's got free will, right?

Surely, at least *boredom* has to be a universal value. It evolved in humans because it's valuable, right? So any mind that doesn't share our dislike of repetition will fail to thrive in the universe and be eliminated...

If you are familiar with the difference between instrumental values and terminal values, *and* familiar with the stupidity of natural selection, *and* you understand how this stupidity manifests in the difference between executing adaptations versus maximizing fitness, *and* you know this turned instrumental subgoals of reproduction into decontextualized unconditional emotions ...

...*and* you're familiar with how the tradeoff between exploration and exploitation works in Artificial Intelligence ...

...then you might be able to see that the human form of boredom that demands a steady trickle of novelty for its own sake isn't a grand universal, but just a particular algorithm that evolution coughed out into us. And you might be able to see how the vast majority of possible expected utility maximizers would only engage in just so much efficient exploration, and spend most of their time exploiting the best alternative found so far, over and over and over.

That's a lot of background knowledge, though.

And so on and so on and so on through 75% of my posts on *Overcoming Bias*, and many chains of fallacy and counter-explanation. Some week I may try to write up the whole diagram. But for now I'm going to assume that you've read the arguments, and just deliver the conclusion:

We can't relax our grip on the future—let go of the steering wheel—and still end up with anything of value.

And those who think we *can*—

—they're trying to be cosmopolitan. I understand that. I read those same science fiction books as a kid: The provincial villains who enslave aliens for the crime of not looking just like humans. The provincial villains who enslave helpless AIs in durance vile on the assumption that silicon can't be sentient. And the cosmopolitan heroes who understand that *minds don't have to be just like us to be embraced as valuable*—

I read those books. I once believed them. But the beauty that jumps out of one box is not jumping out of *all* boxes. If you leave behind all order, what is left is not the perfect answer; what is left is perfect noise. Sometimes you have to abandon an old design rule to build a better mousetrap, but that's not the same as giving up all design rules and collecting wood shavings into a heap, with every pattern of wood as good as any other. The old rule is always abandoned at the behest of some higher rule, some higher criterion of value that governs.

If you loose the grip of human morals and metamorals—the result is not mysterious and alien and beautiful by the standards of human value. It is moral noise, a universe tiled with paperclips. To change away from human morals *in the direction of improvement rather than entropy* requires a criterion of improvement; and that criterion would be physically represented in our brains, and our brains alone.

Relax the grip of human value upon the universe, and it will end up *seriously* valueless. Not strange and alien and wonderful, shocking and terrifying and beautiful beyond all human imagination. Just—tiled with paperclips.

It's only some *humans*, you see, who have this idea of embracing manifold varieties of mind—of wanting the Future to be something greater than the past—of being not bound to our past selves—of trying to change and move forward.

A paperclip maximizer just chooses whichever action leads to the greatest number of paperclips.

No free lunch. You want a wonderful and mysterious universe? That's *your* value. *You* work to create that value. Let

that value exert its force through you who represents it; let it make decisions in you to shape the future. And maybe you shall indeed obtain a wonderful and mysterious universe.

No free lunch. Valuable things appear because a goal system that values them takes action to create them. Paperclips don't materialize from nowhere for a paperclip maximizer. And a wonderfully alien and mysterious Future will not materialize from nowhere for us humans, if our values that prefer it are physically obliterated—or even *disturbed* in the wrong dimension. Then there is nothing left in the universe that works to make the universe valuable.

You *do* have values, even when you're trying to be “cosmopolitan,” trying to display a properly virtuous appreciation of alien minds. Your values are then faded further into the invisible background—they are less *obviously* human. Your brain probably won't even generate an alternative so awful that it would wake you up, make you say “No! Something went wrong!” even at your most cosmopolitan. E.g., “a nonsentient optimizer absorbs all matter in its future light cone and tiles the universe with paperclips.” You'll just imagine strange alien worlds to appreciate.

Trying to be “cosmopolitan”—to be a *citizen of the cosmos*—just strips off a *surface veneer* of goals that seem *obviously* “human.”

But if you wouldn't like the Future tiled over with paperclips, and you would prefer a civilization of ...

...sentient beings ...

...with enjoyable experiences ...

...that aren't the *same* experience over and over again ...

...and are bound to something besides just being a sequence of internal pleasurable feelings ...

...learning, discovering, freely choosing ...

...well, my posts on Fun Theory go into *some* of the hidden details on those short English words.

Values that you might praise as *cosmopolitan* or *universal* or *fundamental* or *obvious common sense* are represented in your brain just as much as those values that you might dismiss as *merely human*. Those values come of the long history of

humanity, and the morally miraculous stupidity of evolution that created us. (And once I *finally* came to that realization, I felt less ashamed of values that seemed “provincial”—but that’s another matter.)

These values do *not* emerge in all possible minds. They will *not* appear from nowhere to rebuke and revoke the utility function of an expected paperclip maximizer.

Touch too hard in the wrong dimension, and the physical representation of those values will shatter—and *not come back*, for there will be nothing left to *want* to bring it back.

And the *referent* of those values—a worthwhile universe—would no longer have any physical reason to come into being.

Let go of the steering wheel, and the Future crashes.

The Gift We Give to Tomorrow



discussion
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How, oh how, could the universe, itself unloving and mindless, cough up minds who were capable of love?

“No mystery in that,” you say. “It’s just a matter of natural selection.”

But natural selection is cruel, bloody, and bloody stupid. Even when, on the surface of things, biological organisms aren’t *directly* fighting each other—aren’t *directly* tearing at each other with claws—there’s still a deeper competition going on between the genes. Genetic information is created when genes increase their *relative* frequency in the next generation—what matters for “genetic fitness” is not how many children you have, but that you have *more* children than others. It is quite possible for a species to evolve to extinction, if the winning genes are playing negative-sum games.

How, oh how, could such a process create beings capable of love?

“No mystery,” you say. “There is never any mystery-in-the-world. Mystery is a property of questions, not answers. A mother’s children share her genes, so the mother loves her children.”

But sometimes mothers adopt children, and still love them. And mothers love their children for themselves, not for their genes.

“No mystery,” you say. “Individual organisms are adaptation-executers, not fitness-maximizers. Evolutionary psychology is not about deliberately maximizing fitness—through most of human history, we didn’t know genes existed. We don’t calculate our acts’ effect on genetic fitness consciously, or even subconsciously.”

But human beings form friendships even with non-relatives. How can that be?

“No mystery, for hunter-gatherers often play Iterated Prisoner’s Dilemmas, the solution to which is reciprocal altruism.

Sometimes the most dangerous human in the tribe is not the strongest, the prettiest, or even the smartest, but the one who has the most allies.”

Yet not all friends are fair-weather friends; we have a concept of true friendship—and some people have sacrificed their life for their friends. Would not such a devotion tend to remove itself from the gene pool?

“You said it yourself: we have concepts of true friendship and of fair-weather friendship. We can tell, or try to tell, the difference between someone who considers us a valuable ally, and someone executing the friendship adaptation. We wouldn’t be true friends with someone who we didn’t think was a true friend to us—and someone with many *true* friends is far more formidable than someone with many fair-weather allies.”

And Mohandas Gandhi, who really did turn the other cheek? Those who try to serve all humanity, whether or not all humanity serves them in turn?

“That perhaps is a more complicated story. Human beings are not just social animals. We are political animals who argue linguistically about policy in adaptive tribal contexts. Sometimes the formidable human is not the strongest, but the one who can most skillfully argue that their preferred policies match the preferences of others.”

Um...that doesn’t explain Gandhi, or am I missing something?

“The point is that we have the ability to *argue* about ‘What should be done?’ as a *proposition*—we can make those arguments and respond to those arguments, without which politics could not take place.”

Okay, but Gandhi?

“Believed certain complicated propositions about ‘What should be done?’ and did them.”

That sounds suspiciously like it could explain any possible human behavior.

“If we traced back the chain of causality through all the arguments, it would involve: a moral architecture that had the ability to argue *general abstract* moral propositions like ‘What should be done to people?’; appeal to hardwired intuitions like

fairness, a concept of duty, pain aversion, empathy; something like a preference for simple moral propositions, probably reused from our pre-existing Occam prior; and the end result of all this, plus perhaps memetic selection effects, was ‘You should not hurt people’ in full generality—”

And that gets you Gandhi.

“Unless you think it was magic, it has to fit into the lawful causal development of the universe somehow.”

I certainly won’t postulate magic, under any name.

“Good.”

But come on ...doesn’t it seem a little ...*amazing* ...that hundreds of millions of years worth of evolution’s death tournament could cough up mothers and fathers, sisters and brothers, husbands and wives, steadfast friends and honorable enemies, true altruists and guardians of causes, police officers and loyal defenders, even artists sacrificing themselves for their art, all practicing so many kinds of love? For so many things other than genes? Doing their part to make their world less ugly, something besides a sea of blood and violence and mindless replication?

“Are you claiming to be surprised by this? If so, question your underlying model, for it has led you to be surprised by the true state of affairs.

Since the beginning,
not one unusual thing
has ever happened.”

But how is it *not* surprising?

“What would you suggest? That some sort of shadowy figure stood behind the scenes and directed evolution?”

Hell no. But—

“Because if you *were* suggesting that, I would have to ask how that shadowy figure *originally* decided that love was a *desirable* outcome of evolution. I would have to ask where that figure got preferences that included things like love, friendship, loyalty, fairness, honor, romance, and so on. On evolutionary psychology, we can see how *that specific outcome* came about—how *those particular goals rather than others* were generated

in the first place. You can call it ‘surprising’ all you like. But when you really do understand evolutionary psychology, you can see how parental love and romance and honor, and even true altruism and moral arguments, *bear the specific design signature of natural selection* in particular adaptive contexts of the hunter-gatherer savanna. So if there was a shadowy figure, it must itself have evolved—and that obviates the whole point of postulating it.”

I’m not postulating a shadowy figure! I’m just asking how human beings ended up so *nice*.

“Nice! Have you looked at this planet lately? We bear all those other emotions that evolved, too—which would tell you very well that we evolved, should you begin to doubt it. Humans aren’t always nice.”

We’re one hell of a lot nicer than the process that produced us, which lets elephants starve to death when they run out of teeth, which doesn’t anesthetize a gazelle even as it lays dying and is of no further importance to evolution one way or the other. It doesn’t take much to be nicer than evolution. To have the *theoretical capacity* to make one single gesture of mercy, to feel a single twinge of empathy, is to be nicer than evolution.

How did evolution, which is itself so uncaring, create minds on that qualitatively higher moral level? How did evolution, which is so ugly, end up doing anything so *beautiful*?

“Beautiful, you say? Bach’s *Little Fugue in G Minor* may be beautiful, but the sound waves, as they travel through the air, are not stamped with tiny tags to specify their beauty. If you wish to find *explicitly encoded* a measure of the fugue’s beauty, you will have to look at a human brain—nowhere else on Earth will you find it. Not upon the seas or the mountains will you find such judgments written: they are not minds; they cannot think.”

Perhaps that is so. Yet evolution *did in fact* give us the ability to admire the beauty of a flower. That still seems to call for some deeper answer.

“Do you not see the circularity in your question? If beauty were like some great light in the sky that shined from outside humans, then your question might make sense—though there

would still be the question of how humans came to perceive that light. You evolved with a psychology alien to evolution: Evolution has nothing like the intelligence or the precision required to exactly quine its goal system. In coughing up the first true minds, evolution's simple fitness criterion shattered into a thousand values. You evolved with a psychology that attaches utility to things which evolution does not care about—human life, human happiness. And then you look back and say, 'How marvelous!' You marvel and you wonder at the fact that your values coincide with themselves."

But then—it is still amazing that this particular circular loop, and not some other loop, came into the world. That we find ourselves praising love and not hate, beauty and not ugliness.

"I don't think you understand. To you, it seems natural to privilege the beauty and altruism as special, as preferred, because you value them highly. And you don't see this as an unusual fact about yourself, because many of your friends do likewise. So you expect that a ghost of perfect emptiness would also value life and happiness—and then, from this standpoint outside reality, a great coincidence would indeed have occurred."

But you can make arguments for the importance of beauty and altruism from first principles—that our aesthetic senses lead us to create new complexity, instead of repeating the same things over and over; and that altruism is important because it takes us outside ourselves, gives our life a higher meaning than sheer brute selfishness.

"And that argument is going to move even a ghost of perfect emptiness? Because you've appealed to slightly different values? Those aren't first principles. They're just *different* principles. Speak in a grave and philosophical register, and still you shall find no universally compelling arguments. All you've done is pass the recursive buck."

You don't think that, somehow, we evolved to *tap into* something beyond—

"What good does it do to suppose something beyond? Why should we pay more attention to this beyond thing than we pay to our existence as humans? How does it alter your personal

responsibility to say that you were only following the orders of the beyond thing? And you would still have evolved to let the beyond thing, rather than something else, direct your actions. It would be *too much coincidence*.”

Too much coincidence?

“A flower is beautiful, you say. Do you think there is no story behind that beauty, or that science does not know the story? Flower pollen is transmitted by bees, so by sexual selection, flowers evolved to attract bees—by imitating certain mating signs of bees, as it happened; the flowers’ patterns would look more intricate if you could see in the ultraviolet. Now healthy flowers are a sign of fertile land, likely to bear fruits and other treasures, and probably prey animals as well; so is it any wonder that humans evolved to be attracted to flowers? But for there to be some great light written upon the very stars—those huge unsentient balls of burning hydrogen—which *also* said that flowers were beautiful, now *that* would be far too much coincidence.”

So you explain away the beauty of a flower?

“No. I explain it. Of course there’s a story behind the beauty of flowers, behind the fact that we find them beautiful. Behind ordered events, one finds ordered stories; and what has no story is the product of random noise, which is hardly any better. If you cannot take joy in things that have stories behind them, your life will be empty indeed. I don’t think I take any less joy in a flower than you do. More so, perhaps, because I take joy in its story as well.”

Perhaps, as you say, there is no surprise from a causal viewpoint—no disruption of the physical order of the universe. But it still seems to me that, in this creation of humans by evolution, something happened that is precious and marvelous and wonderful. If we cannot call it a physical miracle, then call it a moral miracle.

“Because it’s only a miracle from the perspective of the morality that was produced, thus explaining away all of the apparent coincidence from a merely causal and physical perspective?”

Well ...I suppose you could interpret the term that way, yes. I just meant something that was immensely surprising and

wonderful on a moral level, even if it is not surprising on a physical level.

“I think that’s what I said.”

But it still seems to me that you, from your own view, drain something of that wonder away.

“Then you have problems taking joy in the merely real. Love has to begin somehow. It has to enter the universe somewhere. It is like asking how life itself begins—and though you were born of your father and mother, and they arose from their living parents in turn, if you go far and far and far away back, you will finally come to a replicator that arose by pure accident—the border between life and unlife. So too with love.

“A complex pattern must be explained by a cause that is not already that complex pattern. Not just the event must be explained, but the very shape and form. For love to first enter Time, it must come of something that is not love; if this were not possible, then love could not be.

“Even as life itself required that first replicator to come about by accident, parentless but still caused: far, far back in the causal chain that led to you: 3.85 billion years ago, in some little tidal pool.

“Perhaps your children’s children will ask how it is that they are capable of love.

“And their parents will say: Because we, who also love, created you to love.

“And your children’s children will ask: But how is it that *you* love?

“And their parents will reply: Because our own parents, who also loved, created us to love in turn.

“Then your children’s children will ask: But where did it all begin? Where does the recursion end?

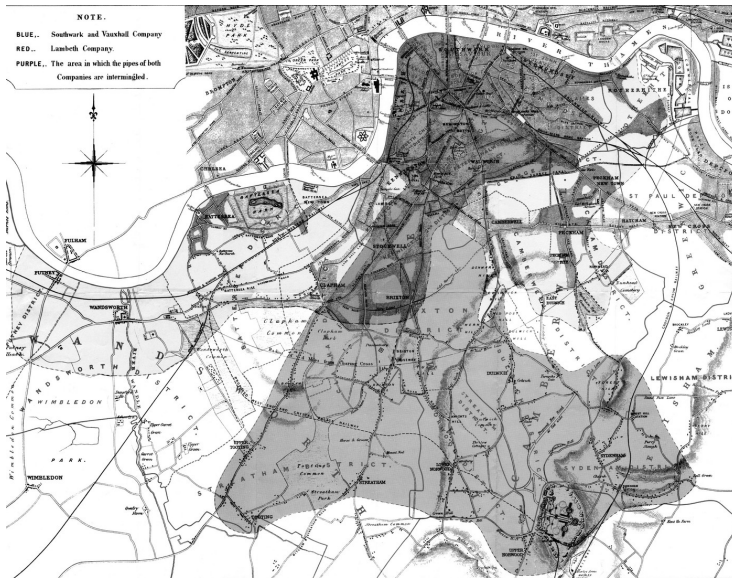
“And their parents will say: Once upon a time, long ago and far away, ever so long ago, there were intelligent beings who were not themselves intelligently designed. Once upon a time, there were lovers created by something that did not love.

“Once upon a time, when all of civilization was a single galaxy and a single star: and a single planet. A place called Earth.

“Long ago, and far away, ever so long ago.”

PART W

Quantified Humanism



Scope Insensitivity

Once upon a time, three groups of subjects were asked how much they would pay to save 2,000 / 20,000 / 200,000 migrating birds from drowning in uncovered oil ponds. The groups respectively answered \$80, \$78, and \$88.¹ This is *scope insensitivity* or *scope neglect*: the number of birds saved—the *scope* of the altruistic action—had little effect on willingness to pay.

Similar experiments showed that Toronto residents would pay little more to clean up all polluted lakes in Ontario than polluted lakes in a particular region of Ontario,² or that residents of four western US states would pay only 28% more to protect all 57 wilderness areas in those states than to protect a single area.³ People visualize “a single exhausted bird, its feathers soaked in black oil, unable to escape.”⁴ This image, or *prototype*, calls forth some level of emotional arousal that is primarily responsible for willingness-to-pay—and the image is the same in all cases. As for scope, it gets tossed out the window—no human can visualize 2,000 birds at once, let alone 200,000. The



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1. William H. Desvousges et al., *Measuring Nonuse Damages Using Contingent Valuation: An Experimental Evaluation of Accuracy*, technical report (Research Triangle Park, NC: RTI International, 2010), doi:10.3768/rti-press.2009.bk.0001.1009.
2. Daniel Kahneman, “Comments by Professor Daniel Kahneman,” in *Valuing Environmental Goods: An Assessment of the Contingent Valuation Method*, ed. Ronald G. Cummings, David S. Brookshire, and William D. Schulze, vol. 1.B, *Experimental Methods for Assessing Environmental Benefits* (Totowa, NJ: Rowman & Allanheld, 1986), 226–235, sharov.me/r/5-13.
3. Daniel L. McFadden and Gregory K. Leonard, “Issues in the Contingent Valuation of Environmental Goods: Methodologies for Data Collection and Analysis,” in *Contingent Valuation: A Critical Assessment*, ed. Jerry A. Hausman, Contributions to Economic Analysis 220 (New York: North-Holland, 1993), 165–215, doi:10.1108/S0573-8555(1993)0000220007.
4. Kahneman, Ritov, and Schkade, “Economic Preferences or Attitude Expressions?”



usual finding is that *exponential* increases in scope create *linear* increases in willingness-to-pay—perhaps corresponding to the linear time for our eyes to glaze over the zeroes; this small amount of affect is added, not multiplied, with the prototype affect. This hypothesis is known as “valuation by prototype.”

An alternative hypothesis is “purchase of moral satisfaction.” People spend enough money to create a *warm glow* in themselves, a sense of having done their duty. The level of spending needed to purchase a warm glow depends on personality and financial situation, but it certainly has nothing to do with the number of birds.

We are insensitive to scope even when human lives are at stake: Increasing the alleged risk of chlorinated drinking water from 0.004 to 2.43 annual deaths per 1,000—a factor of 600—increased willingness-to-pay from \$3.78 to \$15.23.⁵ Baron and Greene found no effect from varying lives saved by a factor of 10.⁶

A paper entitled “Insensitivity to the value of human life: A study of psychophysical numbing” collected evidence that our perception of human deaths follows Weber’s Law—obeys a logarithmic scale where the “just noticeable difference” is a constant fraction of the whole. A proposed health program to save the lives of Rwandan refugees garnered far higher support when it promised to save 4,500 lives in a camp of 11,000 refugees, rather than 4,500 in a camp of 250,000. A potential disease cure had to promise to save far more lives in order to be judged worthy of funding, if the disease was originally stated to have killed 290,000 rather than 160,000 or 15,000 people per year.⁷

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5. Richard T. Carson and Robert Cameron Mitchell, “Sequencing and Nesting in Contingent Valuation Surveys,” *Journal of Environmental Economics and Management* 28, no. 2 (1995): 155–173, doi:10.1006/jeem.1995.1011.
 6. Jonathan Baron and Joshua D. Greene, “Determinants of Insensitivity to Quantity in Valuation of Public Goods: Contribution, Warm Glow, Budget Constraints, Availability, and Prominence,” *Journal of Experimental Psychology: Applied* 2, no. 2 (1996): 107–125, doi:10.1037/1076-898X.2.2.107.
 7. David Fetherstonhaugh et al., “Insensitivity to the Value of Human Life: A Study of Psychophysical Numbing,” *Journal of Risk and Uncertainty* 14, no. 3 (1997): 283–300, doi:10.1023/A:1007744326393.

The moral: If you want to be an effective altruist, you have to think it through with the part of your brain that processes those unexciting inky zeroes on paper, not just the part that gets real worked up about that poor struggling oil-soaked bird.

One Life Against the World



discussion
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Whoever saves a single life, it is as if he had saved the whole world.

—The Talmud, Sanhedrin 4:5

It's a beautiful thought, isn't it? Feel that warm glow.

I can testify that helping one person *feels* just as good as helping the whole world. Once upon a time, when I was burned out for the day and wasting time on the Internet—it's a bit complicated, but essentially I managed to turn someone's whole life around by leaving an anonymous blog comment. I wasn't expecting it to have an effect that large, but it did. When I discovered what I had accomplished, it gave me a *tremendous* high. The euphoria lasted through that day and into the night, only wearing off somewhat the next morning. It felt just as good (this is the scary part) as the euphoria of a major scientific insight, which had previously been my best referent for what it might feel like to do drugs.

Saving one life probably *does* feel just as good as being the first person to realize what makes the stars shine. It probably *does* feel just as good as saving the entire world.

But if you ever have a choice, dear reader, between saving a single life and saving the whole world—then save the world. Please. Because beyond that warm glow is *one heck of a gigantic difference*. For some people, the notion that saving the world is *significantly better* than saving one human life will be obvious, like saying that six billion dollars is worth more than one dollar, or that six cubic kilometers of gold weighs more than one cubic meter of gold. (And never mind the expected value of posterity.) Why might it *not* be obvious? Well, suppose there's a qualitative duty to save what lives you can—then someone who saves the world, and someone who saves one human life, are just fulfilling the same duty. Or suppose that we follow the Greek conception of personal virtue, rather than consequentialism; someone who

saves the world is virtuous, but not six billion times as virtuous as someone who saves one human life. Or perhaps the value of one human life is already too great to comprehend—so that the passing grief we experience at funerals is an infinitesimal underestimate of what is lost—and thus passing to the entire world changes little.

I agree that one human life is of unimaginably high value. I also hold that two human lives are twice as unimaginably valuable. Or to put it another way: Whoever saves one life, if it is as if they had saved the whole world; whoever saves ten lives, it is as if they had saved ten worlds. Whoever *actually* saves the whole world—not to be confused with pretend rhetorical saving the world—it is as if they had saved an intergalactic civilization.

Two deaf children are sleeping on the railroad tracks, the train speeding down; you see this, but you are too far away to save the children. I'm nearby, within reach, so I leap forward and drag one child off the railroad tracks—and then stop, calmly sipping a Diet Pepsi as the train bears down on the second child. “*Quick!*” you scream to me. “*Do something!*” But (I call back) I already saved one child from the train tracks, and thus I am “unimaginably” far ahead on points. Whether I save the second child, or not, I will still be credited with an “unimaginably” good deed. Thus, I have no further motive to act. Doesn't sound right, does it?

Why should it be any different if a philanthropist spends \$10 million on curing a rare but spectacularly fatal disease which afflicts only a hundred people planetwide, when the same money has an equal probability of producing a cure for a less spectacular disease that kills 10% of 100,000 people? I don't think it *is* different. When human lives are at stake, we have a duty to *maximize*, not *satisfice*; and this duty has the same strength as the original duty to save lives. Whoever knowingly chooses to save one life, when they could have saved two—to say nothing of a thousand lives, or a world—they have damned themselves as thoroughly as any murderer.

It's not cognitively easy to spend money to save lives, since cliché methods that instantly leap to mind don't work or are

counterproductive. (I will write later on why this tends to be so.) Stuart Armstrong also points out that if we are to disdain the philanthropist who spends life-saving money inefficiently, we should be consistent and disdain more those who could spend money to save lives but don't.

The Allais Paradox

Choose between the following two options:

- 1A. \$24,000, with certainty.
- 1B. 33/34 chance of winning \$27,000, and 1/34 chance of winning nothing.



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Which seems more intuitively appealing? And which one would you choose in real life? Now which of these two options would you intuitively prefer, and which would you choose in real life?

- 2A. 34% chance of winning \$24,000, and 66% chance of winning nothing.
- 2B. 33% chance of winning \$27,000, and 67% chance of winning nothing.

The Allais Paradox—as Allais called it, though it’s not really a paradox—was one of the first conflicts between decision theory and human reasoning to be experimentally exposed, in 1953.¹ I’ve modified it slightly for ease of math, but the essential problem is the same: Most people prefer 1A to 1B, and most people prefer 2B to 2A. Indeed, in within-subject comparisons, a majority of subjects express both preferences simultaneously.

This is a problem because the 2s are equal to a one-third chance of playing the 1s. That is, 2A is equivalent to playing gamble 1A with 34% probability, and 2B is equivalent to playing 1B with 34% probability.

1. Maurice Allais, “Le Comportement de l’Homme Rationnel devant le Risque: Critique des Postulats et Axiomes de l’Ecole Americaine,” *Econometrica* 21, no. 4 (1953): 2, doi:10.2307/1907921; Daniel Kahneman and Amos Tversky, “Prospect Theory: An Analysis of Decision Under Risk,” *Econometrica* 47 (1979): 263–292.

Among the axioms used to prove that “consistent” decision-makers can be viewed as maximizing expected utility is the Axiom of Independence: If X is strictly preferred to Y , then a probability P of X and $(1 - P)$ of Z should be strictly preferred to P chance of Y and $(1 - P)$ chance of Z .

All the axioms are consequences, as well as antecedents, of a consistent utility function. So it must be possible to prove that the experimental subjects above *can't* have a consistent utility function over outcomes. And indeed, you can't simultaneously have:

$$\begin{aligned} &U(\$24,000) \\ &> (33/34) \times U(\$27,000) + (1/34) \times U(\$0) \\ \\ &0.34 \times U(\$24,000) + 0.66 \times U(\$0) \\ &< 0.33 \times U(\$27,000) + 0.67 \times U(\$0). \end{aligned}$$

These two equations are algebraically inconsistent, regardless of U , so the Allais Paradox has nothing to do with the diminishing marginal utility of money.

Maurice Allais initially defended the revealed preferences of the experimental subjects—he saw the experiment as exposing a flaw in the conventional ideas of utility, rather than exposing a flaw in human psychology. This was 1953, after all, and the heuristics-and-biases movement wouldn't really get started for another two decades. Allais thought his experiment just showed that the Axiom of Independence clearly wasn't a good idea in real life.

(How naive, how foolish, how simplistic is Bayesian decision theory ...)

Surely the *certainty* of having \$24,000 should count for *something*. You can *feel* the difference, right? The solid reassurance?²

2. Edit 2018: If the Bayesian view still isn't making sense, try it out (with slightly adjusted probabilities). Get out three six sided dice. (Two white and one red would work well). The options now are:

1A. \$24,000 no matter what the dice roll.

(I'm starting to think of this as "naive philosophical realism"—supposing that our intuitions directly expose truths about which strategies are wiser, as though it were a directly perceived fact that "1A is superior to 1B." Intuitions *directly* expose truths about human cognitive functions, and only *indirectly* expose (after we reflect on the cognitive functions themselves) truths about rationality.)

"But come now," you say, "is it really such a terrible thing to depart from Bayesian beauty?" Okay, so the subjects didn't follow the neat little "independence axiom" espoused by the likes of von Neumann and Morgenstern. Yet who says that things *must* be neat and tidy?

Why fret about elegance, if it makes us take risks we don't want? Expected utility tells us that we ought to assign some kind of number to an outcome, and then multiply that value by the outcome's probability, add them up, etc. Okay, but why do we *have* to do that? Why not make up more palatable rules instead?

There is always a price for leaving the Bayesian Way. That's what coherence and uniqueness theorems are all about.

In this case, if an agent prefers 1A to 1B, and 2B to 2A, it introduces a form of *preference reversal*—a *dynamic inconsistency* in the agent's planning. You become a *money pump*.

Suppose that at 12:00 p.m. I roll a hundred-sided die. If the die shows a number greater than 34, the game terminates. Otherwise, at 12:05 p.m. I consult a switch with two settings, A and B. If the setting is A, I pay you \$24,000. If the setting is B, I roll a 34-sided die and pay you \$27,000 unless the die shows "34," in which case I pay you nothing.

Let's say you prefer 1A over 1B, and 2B over 2A, and you would pay a single penny to indulge each preference. The switch starts in state A. Before 12:00 p.m., you pay me a penny to throw the switch to B. The die comes up 12. After 12:00 p.m.

1B. \$27,000 if the white dice do *not* roll 1-1.

2A. \$24,000 if the red die rolls 3 or 6, ignore the white dice.

2B. \$27,000 if the red die rolls 3 or 6, and the white dice are *not* 1-1.

and before 12:05 p.m., you pay me a penny to throw the switch to A.

I have taken your two cents on the subject. If you indulge your intuitions, and dismiss mere elegance as a pointless obsession with neatness, then don't be surprised when your pennies get taken from you ...

(I think the same failure to proportionally devalue the emotional impact of small probabilities is responsible for the lottery.)

Zut Allais!

Huh! I was not expecting so many commenters to defend the preference reversal. Looks like I ran into an inferential distance.

It probably helps in interpreting the Allais Paradox to have absorbed more of the *gestalt* of the field of heuristics and biases, such as:

- Experimental subjects tend to defend incoherent preferences even when they're *really* silly.
- People put very high values on small shifts in probability away from 0 or 1 (the certainty effect).

Let's start with the issue of incoherent preferences—preference reversals, dynamic inconsistency, money pumps, that sort of thing.

Anyone who knows a little prospect theory will have no trouble constructing cases where people say they would prefer to play gamble A rather than gamble B; but when you ask them to price the gambles they put a higher value on gamble B than gamble A. There are different perceptual features that become salient when you ask “Which do you prefer?” in a direct comparison, and “How much would you pay?” with a single item.

This choice of gambles typically generates a preference reversal:

1. 1/3 chance to win \$16 and 2/3 chance to lose \$2.
2. 99/100 chance to win \$4 and 1/100 chance to lose \$1.

Most people will rather play 2 than 1. But if you ask them to price the bets separately—ask for a price at which they would be indifferent between having that amount of money, and having



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a chance to play the gamble—people will put a higher price on 1 than on 2.¹

So first you sell them a chance to play bet 1, at their stated price. Then you offer to trade bet 1 for bet 2. Then you buy bet 2 back from them, at their stated price. Then you do it again. Hence the phrase, “money pump.”

Or to paraphrase Steve Omohundro: If you would rather be in Oakland than San Francisco, and you would rather be in San Jose than Oakland, and you would rather be in San Francisco than San Jose, you’re going to spend an awful lot of money on taxi rides.

Amazingly, people *defend* these preference patterns. Some subjects abandon them after the money-pump effect is pointed out—revise their price or revise their preference—but some subjects defend them.

On one occasion, gamblers in Las Vegas played these kinds of bets for real money, using a roulette wheel. And afterward, one of the researchers tried to explain the problem with the incoherence between their pricing and their choices. From the transcript:^{2,3}

SARAH LICHTENSTEIN: “Well, how about the bid for Bet A? Do you have any further feelings about it now that you know you are choosing one but bidding more for the other one?”

SUBJECT: “It’s kind of strange, but no, I don’t have any feelings at all whatsoever really about it. It’s just one of those things. It shows my reasoning process isn’t so good, but, other than that, I...no qualms.”

...

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1. Sarah Lichtenstein and Paul Slovic, “Reversals of Preference Between Bids and Choices in Gambling Decisions,” *Journal of Experimental Psychology* 89, no. 1 (1971): 46–55.
 2. William Poundstone, *Priceless: The Myth of Fair Value (and How to Take Advantage of It)* (Hill & Wang, 2010).
 3. Sarah Lichtenstein and Paul Slovic, eds., *The Construction of Preference* (Cambridge University Press, 2006).

LICHTENSTEIN: "Can I persuade you that it is an irrational pattern?"

SUBJECT: "No, I don't think you probably could, but you could try."

...

LICHTENSTEIN: "Well, now let me suggest what has been called a money-pump game and try this out on you and see how you like it. If you think Bet A is worth 550 points [points were converted to dollars after the game, though not on a one-to-one basis] then you ought to be willing to give me 550 points if I give you the bet..."

...

LICHTENSTEIN: "So you have Bet A, and I say, 'Oh, you'd rather have Bet B wouldn't you?'"

...

SUBJECT: "I'm losing money."

LICHTENSTEIN: "I'll buy Bet B from you. I'll be generous; I'll pay you more than 400 points. I'll pay you 401 points. Are you willing to sell me Bet B for 401 points?"

SUBJECT: "Well, certainly."

...

LICHTENSTEIN: "I'm now ahead 149 points."

SUBJECT: "That's good reasoning on my part. (laughs) How many times are we going to go through this?"

...

LICHTENSTEIN: "Well, I think I've pushed you as far as I know how to push you short of actually insulting you."

SUBJECT: "That's right."

You want to scream, "Just *give up already!* Intuition *isn't always right!*"

And then there's the business of the strange value that people attach to certainty. My books are packed up for the move, but I believe that one experiment showed that a shift from 100%

probability to 99% probability weighed larger in people's minds than a shift from 80% probability to 20% probability.

The problem with attaching a huge extra value to certainty is that *one time's certainty is another time's probability*.

In the last essay, I talked about the Allais Paradox:

- 1A. \$24,000, with certainty.
- 1B. 33/34 chance of winning \$27,000, and 1/34 chance of winning nothing.
- 2A. 34% chance of winning \$24,000, and 66% chance of winning nothing.
- 2B. 33% chance of winning \$27,000, and 67% chance of winning nothing.

The naive preference pattern on the Allais Paradox is $1A > 1B$ and $2B > 2A$. Then you will pay me to throw a switch from A to B because you'd rather have a 33% chance of winning \$27,000 than a 34% chance of winning \$24,000. Then a die roll eliminates a chunk of the probability mass. In both cases you had *at least* a 66% chance of winning nothing. This die roll eliminates that 66%. So now option B is a 33/34 chance of winning \$27,000, but option A is a *certainty* of winning \$24,000. Oh, glorious certainty! So you pay me to throw the switch back from B to A.

Now, if I've told you in advance that I'm going to do all that, do you really want to pay me to throw the switch, and then pay me to throw it back? Or would you prefer to reconsider?

Whenever you try to price a probability shift from 24% to 23% as being less important than a shift from ~1 to 99%—every time you try to make an increment of probability have more value when it's near an end of the scale—you open yourself up to this kind of exploitation. I can always set up a chain of events that eliminates the probability mass, a bit at a time, until you're left with "certainty" that flips your preferences. One time's certainty is another time's uncertainty, and if you insist on treating the distance from ~1 to 0.99 as special, I can cause you to invert your preferences over time and pump some money out of you.

Can I persuade you, perhaps, that this is an irrational pattern?

Surely, if you've been reading this book for a while, you realize that *you*—the very system and process that reads these very words—are a flawed piece of machinery. Your intuitions are not giving you direct, veridical information about good choices. If you don't believe that, there are some gambling games I'd like to play with you.

There are various other games you can also play with certainty effects. For example, if you offer someone a certainty of \$400, or an 80% probability of \$500 and a 20% probability of \$300, they'll usually take the \$400. But if you ask people to imagine themselves \$500 richer, and ask if they would prefer a certain loss of \$100 or a 20% chance of losing \$200, they'll usually take the chance of losing \$200.⁴ Same probability distribution over outcomes, different descriptions, different choices.

Yes, Virginia, you really *should* try to multiply the utility of outcomes by their probability. You really should. Don't be embarrassed to use clean math.

In the Allais paradox, figure out whether 1 unit of the difference between getting \$24,000 and getting nothing outweighs 33 units of the difference between getting \$24,000 and \$27,000. If it does, prefer 1A to 1B and 2A to 2B. If the 33 units outweigh the 1 unit, prefer 1B to 1A and 2B to 2A. As for calculating the utility of money, I would suggest using an approximation that assumes money is logarithmic in utility. If you've got plenty of money already, pick B. If \$24,000 would double your existing assets, pick A. Case 2 or case 1, makes no difference. Oh, and be sure to assess the utility of total asset values—the utility of final outcome states of the world—not *changes in* assets, or you'll end up inconsistent again.

A number of commenters claimed that the preference pattern wasn't irrational because of "the utility of certainty," or something like that. One commenter even wrote $U(\text{Certainty})$ into an expected utility equation.

4. Kahneman and Tversky, "Prospect Theory: An Analysis of Decision Under Risk."

Does anyone remember that whole business about *expected utility* and *utility* being of fundamentally different types? Utilities are over *outcomes*. They are values you attach to *particular, solid states of the world*. You cannot feed a probability of 1 into a utility function. It makes no sense.

And before you sniff, “Hmph ...you just want the math to be neat and tidy,” remember that, in this case, the price of departing the Bayesian Way was paying someone to throw a switch and then throw it back.

But what about that solid, warm feeling of reassurance? Isn’t *that* a utility?

That’s being human. Humans are not expected utility maximizers. Whether you want to relax and have fun, or pay some extra money for a feeling of certainty, depends on whether you care more about satisfying your intuitions or *actually achieving the goal*.

If you’re gambling at Las Vegas for fun, then by all means, don’t think about the expected utility—you’re going to lose money anyway.

But what if it were 24,000 lives at stake, instead of \$24,000? The certainty effect is even stronger over human lives. Will you pay one human life to throw the switch, and another to switch it back?

Tolerating preference reversals makes a mockery of claims to optimization. If you drive from San Jose to San Francisco to Oakland to San Jose, over and over again, then you may get a lot of warm fuzzy feelings out of it, but you can’t be interpreted as having a *destination*—as trying to *go somewhere*.

When you have circular preferences, you’re not *steering the future*—just running in circles. If you enjoy running for its own sake, then fine. But if you have a goal—something you’re trying to actually accomplish—a preference reversal reveals a big problem. At least one of the choices you’re making must not be working to actually optimize the future in any coherent sense.

If what you care about is the warm fuzzy feeling of certainty, then fine. If someone’s life is at stake, then you had best realize that your intuitions are a greasy lens through which to see the

world. Your feelings are not providing you with direct, veridical information about strategic consequences—it *feels* that way, but they're *not*. Warm fuzzies can lead you far astray.

There are mathematical laws governing efficient strategies for steering the future. When something *truly* important is at stake—something more important than your feelings of happiness about the decision—then you should care about the math, if you truly care at all.

Feeling Moral



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Suppose that a disease, or a monster, or a war, or something, is killing people. And suppose you only have enough resources to implement one of the following two options:

Save 400 lives, with certainty.

Save 500 lives, with 90% probability; save no lives, 10% probability.

Most people choose option 1. Which, I think, is foolish; because if you multiply 500 lives by 90% probability, you get an expected value of 450 lives, which exceeds the 400-life value of option 1. (Lives saved don't diminish in marginal utility, so this is an appropriate calculation.)

"What!" you cry, incensed. "How can you gamble with human lives? How can you think about numbers when so much is at stake? What if that 10% probability strikes, and everyone dies? So much for your damned logic! You're following your rationality off a cliff!"

Ah, but here's the interesting thing. If you present the options this way:

1. 100 people die, with certainty.
2. 90% chance no one dies; 10% chance 500 people die.

Then a majority choose option 2. *Even though it's the same gamble.* You see, just as a *certainty* of saving 400 lives seems to *feel* so much more comfortable than an unsure gain, so too, a certain loss *feels* worse than an uncertain one.

You can grandstand on the second description too: "How can you condemn 100 people to certain death when there's such a good chance you can save them? We'll all share the risk! Even if it was only a 75% chance of saving everyone, it would still be worth it—so long as there's a chance—everyone makes it, or no one does!"

You know what? This isn't about your feelings. A human life, with all its joys and all its pains, adding up over the course of decades, is worth far more than your brain's feelings of comfort or discomfort with a plan. Does computing the expected utility feel too cold-blooded for your taste? Well, that feeling isn't even a feather in the scales, when a life is at stake. Just shut up and multiply.

A googol is 10^{100} —a 1 followed by one hundred zeroes. A googolplex is an even more incomprehensibly large number—it's 10^{googol} , a 1 followed by a googol zeroes. Now pick some trivial inconvenience, like a hiccup, and some decidedly untrivial misfortune, like getting slowly torn limb from limb by sadistic mutant sharks. If we're forced into a choice between either preventing a googolplex people's hiccups, or preventing a single person's shark attack, which choice should we make? If you assign *any* negative value to hiccups, then, on pain of decision-theoretic incoherence, there must be some number of hiccups that would add up to rival the negative value of a shark attack. For any particular finite evil, there must be some number of hiccups that would be even worse.

Moral dilemmas like these aren't conceptual blood sports for keeping analytic philosophers entertained at dinner parties. They're distilled versions of the kinds of situations we actually find ourselves in every day. Should I spend \$50 on a console game, or give it all to charity? Should I organize a \$700,000 fundraiser to pay for a single bone marrow transplant, or should I use that same money on mosquito nets and prevent the malaria deaths of some 200 children?

Yet there are many who avert their gaze from the real world's abundance of unpleasant moral tradeoffs—many, too, who take pride in looking away. Research shows that people distinguish “sacred values,” like human lives, from “unsacred values,” like money. When you try to trade off a sacred value against an unsacred value, subjects express great indignation. (Sometimes they want to punish the person who made the suggestion.)

My favorite anecdote along these lines comes from a team of researchers who evaluated the effectiveness of a certain project, calculating the cost per life saved, and recommended

to the government that the project be implemented because it was cost-effective. The governmental agency rejected the report because, they said, you couldn't put a dollar value on human life. After rejecting the report, the agency decided *not* to implement the measure.

Trading off a sacred value against an unsacred value *feels really awful*. To merely multiply utilities would be too cold-blooded—it would be following rationality off a cliff...

But altruism isn't the warm fuzzy feeling you get from being altruistic. If you're doing it for the spiritual benefit, that is nothing but selfishness. The primary thing is to help others, whatever the means. So shut up and multiply!

And if it seems to you that there is a fierceness to this maximization, like the bare sword of the law, or the burning of the Sun—if it seems to you that at the center of this rationality there is a small cold flame—

Well, the other way might feel better inside you. But it wouldn't work.

And I say also this to you: That if you set aside your regret for all the spiritual satisfaction you could be having—if you *wholeheartedly* pursue the Way, without thinking that you are being cheated—if you give yourself over to rationality without holding back, you will find that rationality gives to you in return.

But *that* part only works if you don't go around saying to yourself, "It would feel better inside me if only I could be less rational." Should you be sad that you have the opportunity to actually help people? You cannot attain your full potential if you regard your gift as a burden.

The “Intuitions” Behind “Utilitarianism”

I used to be very confused about metaethics. After my confusion finally cleared up, I did a postmortem on my previous thoughts. I found that my object-level moral reasoning had been valuable and my meta-level moral reasoning had been worse than useless. And this appears to be a general syndrome—people do much better when discussing whether torture is good or bad than when they discuss the meaning of “good” and “bad.” Thus, I deem it prudent to keep moral discussions on the object level wherever I possibly can.

Occasionally people object to any discussion of morality on the grounds that morality doesn’t exist, and in lieu of explaining that “exist” is not the right term to use here, I generally say, “But what do you do anyway?” and take the discussion back down to the object level.

Paul Gowder, though, has pointed out that both the idea of choosing a googolplex trivial inconveniences over one atrocity, and the idea of “utilitarianism,” depend on “intuition.” He says I’ve argued that the two are not compatible, but charges me with failing to argue for the utilitarian intuitions that I appeal to.

Now “intuition” is not how I would describe the computations that underlie human morality and distinguish us, as moralists, from an ideal philosopher of perfect emptiness and/or a rock. But I am okay with using the word “intuition” as a term of art, bearing in mind that “intuition” in this sense is not to be contrasted to reason, but is, rather, the cognitive building block out of which both long verbal arguments and fast perceptual arguments are constructed.

I see the project of morality as a project of renormalizing intuition. We have intuitions about things that seem desirable or undesirable, intuitions about actions that are right or wrong, intuitions about how to resolve conflicting intuitions, intu-



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itions about how to systematize specific intuitions into general principles.

Delete all the intuitions, and you aren't left with an ideal philosopher of perfect emptiness; you're left with a rock.

Keep all your specific intuitions and refuse to build upon the reflective ones, and you aren't left with an ideal philosopher of perfect spontaneity and genuineness; you're left with a grunting caveperson running in circles, due to cyclical preferences and similar inconsistencies.

"Intuition," as a term of art, is not a curse word when it comes to morality—there is nothing else to argue from. Even *modus ponens* is an "intuition" in this sense—it's just that *modus ponens* still seems like a good idea after being formalized, reflected on, extrapolated out to see if it has sensible consequences, et cetera.

So that is "intuition."

However, Gowder did not say what he meant by "utilitarianism." Does utilitarianism say ...

1. That right actions are strictly determined by good consequences?
2. That praiseworthy actions depend on justifiable expectations of good consequences?
3. That probabilities of consequences should normatively be discounted by their probability, so that a 50% probability of something bad should weigh exactly half as much in our tradeoffs?
4. That virtuous actions always correspond to maximizing expected utility under some utility function?
5. That two harmful events are worse than one?
6. That two independent occurrences of a harm (not to the same person, not interacting with each other) are exactly twice as bad as one?

7. That for any two harms *A* and *B*, with *A* much worse than *B*, there exists some tiny probability such that gambling on this probability of *A* is preferable to a certainty of *B*?

If you say that I advocate something, or that my argument depends on something, and that it is wrong, do please specify what this thingy is. Anyway, I accept 3, 5, 6, and 7, but not 4; I am not sure about the phrasing of 1; and 2 is true, I guess, but phrased in a rather solipsistic and selfish fashion: you should not worry about being praiseworthy.

Now, what are the “intuitions” upon which my “utilitarianism” depends?

This is a deepish sort of topic, but I’ll take a quick stab at it.

First of all, it’s not just that someone presented me with a list of statements like those above, and I decided which ones sounded “intuitive.” Among other things, if you try to violate “utilitarianism,” you run into paradoxes, contradictions, circular preferences, and other things that aren’t symptoms of moral wrongness so much as moral incoherence.

After you think about moral problems for a while, and also find new truths about the world, and even discover disturbing facts about how you yourself work, you often end up with different moral opinions than when you started out. This does not quite *define* moral progress, but it is how we experience moral progress.

As part of my experienced moral progress, I’ve drawn a conceptual separation between questions of type *Where should we go?* and questions of type *How should we get there?* (Could that be what Gowder means by saying I’m “utilitarian”?)

The question of where a road goes—where it leads—you can answer by traveling the road and finding out. If you have a false belief about where the road leads, this falsity can be destroyed by the truth in a very direct and straightforward manner.

When it comes to wanting to go to a particular place, this want is not entirely immune from the destructive powers of truth. You could go there and find that you regret it afterward (which does not define moral error, but is how we experience moral error).

But, even so, wanting to be in a particular place seems worth distinguishing from wanting to take a particular road to a particular place.

Our intuitions about where to go are arguable enough, but our intuitions about how to get there are frankly messed up. After the two hundred and eighty-seventh research study showing that people will chop their own feet off if you frame the problem the wrong way, you start to distrust first impressions.

When you've read enough research on scope insensitivity—people will pay only 28% more to protect all 57 wilderness areas in Ontario than one area, people will pay the same amount to save 50,000 lives as 5,000 lives ...that sort of thing ...

Well, the worst case of scope insensitivity I've ever heard of was described here by Slovic:

Other recent research shows similar results. Two Israeli psychologists asked people to contribute to a costly life-saving treatment. They could offer that contribution to a group of eight sick children, or to an individual child selected from the group. The target amount needed to save the child (or children) was the same in both cases. Contributions to individual group members far outweighed the contributions to the entire group.¹

There's other research along similar lines, but I'm just presenting one example, 'cause, y'know, eight examples would probably have less impact.

If you know the general experimental paradigm, then the reason for the above behavior is pretty obvious—focusing your attention on a single child creates more emotional arousal than trying to distribute attention around eight children simultaneously. So people are willing to pay more to help one child than to help eight.

1. Paul Slovic, "Numbed by Numbers," *Foreign Policy* (March 2007), sharov.me/r/5-14.



Now, you could look at this intuition, and think it was revealing some kind of incredibly deep moral truth which shows that one child's good fortune is somehow devalued by the other children's good fortune.

But what about the billions of other children in the world? Why isn't it a bad idea to help this one child, when that causes the value of all the other children to go down? How can it be significantly better to have 1,329,342,410 happy children than 1,329,342,409, but then somewhat worse to have seven more at 1,329,342,417?

Or you could look at that and say: "The intuition is wrong: the brain can't successfully multiply by eight and get a larger quantity than it started with. But it ought to, normatively speaking."

And once you realize that the brain can't multiply by eight, then the other cases of scope neglect stop seeming to reveal some fundamental truth about 50,000 lives being worth just the same effort as 5,000 lives, or whatever. You don't get the impression you're looking at the revelation of a deep moral truth about nonagglomerative utilities. It's just that the brain doesn't goddamn multiply. Quantities get thrown out the window.

If you have \$100 to spend, and you spend \$20 each on each of 5 efforts to save 5,000 lives, you will do worse than if you spend \$100 on a single effort to save 50,000 lives. Likewise if such choices are made by 10 different people, rather than the same person. As soon as you start believing that it is better to save 50,000 lives than 25,000 lives, that simple preference of final destinations has implications for the choice of paths, when you consider five different events that save 5,000 lives.

(It is a general principle that Bayesians see no difference between the long-run answer and the short-run answer; you never get two different answers from computing the same question two different ways. But the long run is a helpful intuition pump, so I am talking about it anyway.)

The aggregative valuation strategy of "shut up and multiply" arises from the simple preference to have more of something—to save as many lives as possible—when you have to describe

general principles for choosing more than once, acting more than once, planning at more than one time.

Aggregation also arises from claiming that the local choice to save one life doesn't depend on how many lives already exist, far away on the other side of the planet, or far away on the other side of the universe. Three lives are one and one and one. No matter how many billions are doing better, or doing worse. $3 = 1 + 1 + 1$, no matter what other quantities you add to both sides of the equation. And if you add another life you get $4 = 1 + 1 + 1 + 1$. That's aggregation.

When you've read enough heuristics and biases research, and enough coherence and uniqueness proofs for Bayesian probabilities and expected utility, and you've seen the "Dutch book" and "money pump" effects that penalize trying to handle uncertain outcomes any other way, then you don't see the preference reversals in the Allais Paradox as revealing some incredibly deep moral truth about the intrinsic value of certainty. It just goes to show that the brain doesn't goddamn multiply.

The primitive, perceptual intuitions that make a choice "feel good" don't handle probabilistic pathways through time very skillfully, especially when the probabilities have been expressed symbolically rather than experienced as a frequency. So you reflect, devise more trustworthy logics, and think it through in words.

When you see people insisting that no amount of money whatsoever is worth a single human life, and then driving an extra mile to save \$10; or when you see people insisting that no amount of money is worth a decrement of health, and then choosing the cheapest health insurance available; then you don't think that their protestations reveal some deep truth about incommensurable utilities.

Part of it, clearly, is that primitive intuitions don't successfully diminish the emotional impact of symbols standing for small quantities—anything you talk about seems like "an amount worth considering."

And part of it has to do with preferring unconditional social rules to conditional social rules. Conditional rules seem weaker, seem more subject to manipulation. If there's any loop-

hole that lets the government legally commit torture, then the government will drive a truck through that loophole.

So it seems like there should be an unconditional social injunction against preferring money to life, and no “but” following it. Not even “but a thousand dollars isn’t worth a 0.0000000001% probability of saving a life.” Though the latter choice, of course, is revealed every time we sneeze without calling a doctor.

The rhetoric of sacredness gets bonus points for seeming to express an *unlimited* commitment, an *unconditional* refusal that signals trustworthiness and refusal to compromise. So you conclude that moral rhetoric espouses qualitative distinctions, because espousing a quantitative tradeoff would sound like you were plotting to defect.

On such occasions, people vigorously want to throw quantities out the window, and they get upset if you try to bring quantities back in, because quantities sound like conditions that would weaken the rule.

But you don’t conclude that there are *actually* two tiers of utility with lexical ordering. You don’t conclude that there is actually an infinitely sharp moral gradient, some atom that moves a Planck distance (in our continuous physical universe) and sends a utility from zero to infinity. You don’t conclude that utilities must be expressed using hyper-real numbers. Because the lower tier would simply vanish in any equation. It would never be worth the tiniest effort to recalculate for it. All decisions would be determined by the upper tier, and all thought spent thinking about the upper tier only, if the upper tier genuinely had lexical priority.

As Peter Norvig once pointed out, if Asimov’s robots had *strict* priority for the First Law of Robotics (“A robot shall not harm a human being, nor through inaction allow a human being to come to harm”) then no robot’s behavior would ever show any sign of the other two Laws; there would always be some tiny First Law factor that would be sufficient to determine the decision.

Whatever value is worth thinking about at all must be worth trading off against all other values worth thinking about, be-

cause thought itself is a limited resource that must be traded off. When you reveal a value, you reveal a utility.

I don't say that morality should always be simple. I've already said that the meaning of music is more than happiness alone, more than just a pleasure center lighting up. I would rather see music composed by people than by nonsentient machine learning algorithms, so that someone should have the joy of composition; I care about the journey, as well as the destination. And I am ready to hear if you tell me that the value of music is deeper, and involves more complications, than I realize—that the valuation of this one event is more complex than I know.

But that's for *one event*. When it comes to multiplying by quantities and probabilities, complication is to be avoided—at least if you care more about the destination than the journey. When you've reflected on enough intuitions, and corrected enough absurdities, you start to see a common denominator, a meta-principle at work, which one might phrase as "Shut up and multiply."

Where music is concerned, I care about the journey.

When lives are at stake, I shut up and multiply.

It is more important that lives be saved, than that we conform to any particular ritual in saving them. And the optimal path to that destination is governed by laws that are simple, because they are math.

And that's why I'm a utilitarian—at least when I am doing something that is overwhelmingly more important than my own feelings about it—which is most of the time, because there are not many utilitarians, and many things left undone.

Ends Don't Justify Means (Among Humans)

If the ends don't justify the means, what does?

—variously attributed

I think of myself as running on hostile hardware.

—Justin Corwin



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Humans may have evolved a structure of political revolution, beginning by believing themselves morally superior to the corrupt current power structure, but ending by being corrupted by power themselves—not by any plan in their own minds, but by the echo of ancestors who did the same and thereby reproduced.

This fits the template:

In some cases, human beings have evolved in such fashion as to think that they are doing *X* for prosocial reason *Y*, but when human beings actually do *X*, other adaptations execute to promote self-benefiting consequence *Z*.

From this proposition, I now move on to a question *considerably* outside the realm of classical Bayesian decision theory:

What if I'm running on corrupted hardware?

In such a case as this, you might even find yourself uttering such seemingly paradoxical statements—sheer nonsense from the perspective of classical decision theory—as:

The ends don't justify the means.

But if you are running on corrupted hardware, then the reflective observation that it *seems* like a righteous and altruistic

act to seize power for yourself—this *seeming* may not be much evidence for the proposition that seizing power is in fact the action that will most benefit the tribe.

By the power of naive realism, the corrupted hardware that you run on, and the corrupted seemings that it computes, will seem like the fabric of the very world itself—simply the way-things-are.

And so we have the bizarre-seeming rule: “For the good of the tribe, do not cheat to seize power *even when it would provide a net benefit to the tribe.*”

Indeed it may be wiser to phrase it this way. If you just say, “when it *seems* like it would provide a net benefit to the tribe,” then you get people who say, “But it doesn’t just *seem* that way—it *would* provide a net benefit to the tribe if I were in charge.”

The notion of untrusted hardware seems like something wholly outside the realm of classical decision theory. (What it does to reflective decision theory I can’t yet say, but that would seem to be the appropriate level to handle it.)

But on a human level, the patch seems straightforward. Once you know about the warp, you create rules that describe the warped behavior and outlaw it. A rule that says, “For the good of the tribe, do not cheat to seize power even for the good of the tribe.” Or “For the good of the tribe, do not murder even for the good of the tribe.”

And now the philosopher comes and presents their “thought experiment”—setting up a scenario in which, *by stipulation*, the *only* possible way to save five innocent lives is to murder one innocent person, and this murder is *certain* to save the five lives. “There’s a train heading to run over five innocent people, who you can’t possibly warn to jump out of the way, but you can push one innocent person into the path of the train, which will stop the train. These are your only options; what do you do?”

An altruistic human, who has accepted certain deontological prohibitions—which seem well justified by some historical statistics on the results of reasoning in certain ways on untrust-

worthy hardware—may experience some mental distress, on encountering this thought experiment.

So here's a reply to that philosopher's scenario, which I have yet to hear any philosopher's victim give:

"You stipulate that the *only possible* way to save five innocent lives is to murder one innocent person, and this murder will *definitely* save the five lives, and that these facts are *known* to me with effective certainty. But since I am running on corrupted hardware, I can't occupy the *epistemic state* you want me to imagine. Therefore I reply that, in a society of Artificial Intelligences worthy of personhood and lacking any inbuilt tendency to be corrupted by power, it would be right for the AI to murder the one innocent person to save five, and moreover all its peers would agree. However, I refuse to extend this reply to myself, because the epistemic state you ask me to imagine can only exist among other kinds of people than human beings."

Now, to me this seems like a dodge. I think the universe is sufficiently unkind that we can justly be forced to consider situations of this sort. The sort of person who goes around proposing that sort of thought experiment might well deserve that sort of answer. But any human legal system does embody some answer to the question "How many innocent people can we put in jail to get the guilty ones?," even if the number isn't written down.

As a human, I try to abide by the deontological prohibitions that humans have made to live in peace with one another. But I don't think that our deontological prohibitions are *literally inherently nonconsequentially terminally right*. I endorse "the end doesn't justify the means" as a principle to guide humans running on corrupted hardware, but I wouldn't endorse it as a principle for a society of AIs that make well-calibrated estimates. (If you have one AI in a society of humans, that does bring in other considerations, like whether the humans learn from your example.)

And so I wouldn't say that a well-designed Friendly AI must necessarily refuse to push that one person off the ledge to stop the train. Obviously, I would expect any decent superintelligence to come up with a superior third alternative. But if those

are the only two alternatives, and the FAI judges that it is wiser to push the one person off the ledge—even after taking into account knock-on effects on any humans who see it happen and spread the story, etc.—then I don’t call it an alarm light, if an *AI* says that the right thing to do is sacrifice one to save five. Again, *I* don’t go around pushing people into the paths of trains myself, nor stealing from banks to fund my altruistic projects. *I* happen to be a human. But for a Friendly AI to be corrupted by power would be like it starting to bleed red blood. The tendency to be corrupted by power is a specific biological adaptation, supported by specific cognitive circuits, built into us by our genes for a clear evolutionary reason. It wouldn’t spontaneously appear in the code of a Friendly AI any more than its transistors would start to bleed.

I would even go further, and say that if you had minds with an inbuilt warp that made them *overestimate* the external harm of self-benefiting actions, then they would need a rule “the ends do not prohibit the means”—that you should do what benefits yourself even when it (seems to) harm the tribe. By hypothesis, if their society did not have this rule, the minds in it would refuse to breathe for fear of using someone else’s oxygen, and they’d all die. For them, an occasional overshoot in which one person seizes a personal benefit at the net expense of society would seem just as cautiously virtuous—and indeed *be* just as cautiously virtuous—as when one of us humans, being cautious, passes up an opportunity to steal a loaf of bread that really would have been more of a benefit to them than a loss to the merchant (including knock-on effects).

“The end does not justify the means” is just consequentialist reasoning at one meta-level up. If a human starts thinking on the *object* level that the end justifies the means, this has awful consequences given our untrustworthy brains; therefore a human shouldn’t think this way. But it is all still ultimately consequentialism. It’s just *reflective* consequentialism, for beings who know that their moment-by-moment decisions are made by untrusted hardware.

Ethical Injunctions

Would you kill babies if it was the right thing to do? If no, under what circumstances would you not do the right thing to do? If yes, how right would it have to be, for how many babies?

—horrible job interview question¹



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Swapping hats for a moment, I'm *professionally* intrigued by the decision theory of "things you shouldn't do even if they seem to be the right thing to do."

Suppose we have a reflective AI, self-modifying and self-improving, at an intermediate stage in the development process. In particular, the AI's goal system isn't finished—the shape of its motivations is still being loaded, learned, tested, or tweaked.

Yea, I have seen many ways to screw up an AI goal system design, resulting in a decision system that decides, given its goals, that the universe ought to be tiled with tiny molecular smiley-faces, or some such. Generally, these deadly suggestions also have the property that the AI will not desire its programmers to fix it. If the AI is *sufficiently* advanced—which it may be even at an intermediate stage—then the AI may also realize that deceiving the programmers, hiding the changes in its thoughts, will help transform the universe into smiley-faces.

Now, from our perspective as programmers, if we *condition on the fact* that the AI has decided to hide its thoughts from the programmers, or otherwise act willfully to deceive us, then it would seem likely that some kind of unintended consequence has occurred in the goal system. We would consider it probable that the AI is *not* functioning as intended, but rather likely

1. sharov.me/r/5-15

The Moral Void • lesswrong.com



that we have messed up the AI's utility function somehow. So that the AI wants to turn the universe into tiny reward-system counters, or some such, and now has a motive to hide from us.

Well, suppose we're *not* going to implement some object-level Great Idea as the AI's utility function. Instead we're going to do something advanced and recursive—build a goal system which knows (and cares) about the programmers outside. A goal system that, via some nontrivial internal structure, “knows it's being programmed” and “knows it's incomplete.” Then you might be able to have and keep the rule:

If [I decide that] fooling my programmers is the right thing to do, execute a controlled shutdown [instead of doing the right thing to do].

And the AI would keep this rule, even through the self-modifying AI's revisions of its own code, because, in its structurally nontrivial goal system, the present-AI understands that this decision by a future-AI *probably* indicates something defined-as-a-malfunction. Moreover, the present-AI knows that if future-AI tries to *evaluate* the utility of executing a shutdown, once this hypothetical malfunction has occurred, the future-AI will probably *decide* not to shut itself down. So the shutdown should happen unconditionally, automatically, without the goal system getting another chance to recalculate the right thing to do.

I'm not going to go into the deep dark depths of the exact mathematical structure, because that would be beyond the scope of this book. Also I don't yet know the deep dark depths of the mathematical structure. It looks like it *should* be possible, if you do things that are advanced and recursive and have nontrivial (but consistent) structure. But I haven't reached that level, as yet, so for now it's only a dream.

But the topic here is not advanced AI; it's human ethics. I introduce the AI scenario to bring out more starkly the strange idea of an *ethical injunction*:

You should never, ever murder an innocent person who's helped you, *even if it's the right thing to do*;

because it's far more likely that *you've made a mistake*, than that murdering an innocent person who helped you is the right thing to do.

Sound reasonable?

During World War II, it became necessary to destroy Germany's supply of deuterium, a neutron moderator, in order to block their attempts to achieve a fission chain reaction. Their supply of deuterium was coming at this point from a captured facility in Norway. A shipment of heavy water was on board a Norwegian ferry ship, the *SF Hydro*. Knut Haukelid and three others had slipped on board the ferry in order to sabotage it, when the saboteurs were discovered by the ferry watchman. Haukelid told him that they were escaping the Gestapo, and the watchman immediately agreed to overlook their presence. Haukelid "considered warning their benefactor but decided that might endanger the mission and only thanked him and shook his hand."² So the civilian ferry *Hydro* sank in the deepest part of the lake, with eighteen dead and twenty-nine survivors. Some of the Norwegian rescuers felt that the German soldiers present should be left to drown, but this attitude did not prevail, and four Germans were rescued. And that was, effectively, the end of the Nazi atomic weapons program.

Good move? Bad move? Germany *very likely* wouldn't have gotten the Bomb anyway ...I hope with absolute desperation that I never get faced by a choice like that, but in the end, I can't say a word against it.

On the other hand, when it comes to the rule:

Never try to deceive yourself, or offer a reason to believe other than probable truth; because even if you come up with an amazing clever reason, it's more likely that you've made a mistake than that you have a reasonable expectation of this being a net benefit in the long run.

2. Richard Rhodes, *The Making of the Atomic Bomb* (New York: Simon & Schuster, 1986).

Then I really *don't* know of anyone who's knowingly been faced with an exception. There are times when you try to convince yourself "I'm not hiding any Jews in my basement" before you talk to the Gestapo officer. But then you do still know the truth, you're just trying to create something like an alternative self that exists in your imagination, a facade to talk to the Gestapo officer.

But to really believe something that isn't true? I don't know if there was ever anyone for whom that was *knowably* a good idea. I'm sure that there have been many many times in human history, where person *X* was better off with false belief *Y*. And by the same token, there is always some set of winning lottery numbers in every drawing. It's *knowing which lottery ticket will win* that is the epistemically difficult part, like *X* knowing when they're better off with a false belief.

Self-deceptions are the worst kind of black swan bets, much worse than lies, because without knowing the true state of affairs, you can't even guess at what the penalty will be for your self-deception. They only have to blow up once to undo all the good they ever did. One single time when you pray to God after discovering a lump, instead of going to a doctor. That's all it takes to undo a life. All the happiness that the warm thought of an afterlife ever produced in humanity, has now been more than cancelled by the failure of humanity to institute systematic cryonic preservations after liquid nitrogen became cheap to manufacture. And I don't think that anyone ever had that sort of failure in mind as a possible blowup, when they said, "But we need religious beliefs to cushion the fear of death." That's what black swan bets are all about—the unexpected blowup.

Maybe you even get away with one or two black swan bets—they don't get you *every* time. So you do it again, and then the blowup comes and cancels out every benefit and then some. That's what black swan bets are all about.

Thus the difficulty of knowing when it's safe to believe a lie (assuming you can even manage that much mental contortion in the first place)—part of the nature of black swan bets is that you don't see the bullet that kills you; and since our perceptions

just seem like the way the world is, it looks like there is no bullet, period.

So I would say that there is an ethical injunction against self-deception. I call this an “ethical injunction” not so much because it’s a matter of interpersonal morality (although it is), but because it’s a rule that guards you from your own cleverness—an override against the temptation to do what seems like the right thing.

So now we have two kinds of situation that can support an “ethical injunction,” a rule not to do something even when it’s the right thing to do. (That is, you refrain “even when your brain has computed it’s the right thing to do,” but this will just *seem like* “the right thing to do.”)

First, being human and running on corrupted hardware, we may generalize classes of situation where when you say e.g. “It’s time to rob a few banks for the greater good,” we deem it more likely that you’ve been corrupted than that this is really the case. (Note that we’re not prohibiting it from *ever* being the case in *reality*, but we’re questioning the *epistemic* state where you’re *justified in trusting* your own calculation that this is the right thing to do—fair lottery tickets can win, but you can’t justifiably buy them.)

Second, history may teach us that certain classes of action are black swan bets, that is, they sometimes blow up bigtime for reasons not in the decider’s model. So even when we calculate within the model that something seems like the right thing to do, we apply the further knowledge of the black swan problem to arrive at an injunction against it.

But surely ...if one is *aware of these reasons* ...then one can simply redo the calculation, taking them into account. So we can rob banks if it seems like the right thing to do *after taking into account* the problem of corrupted hardware and black swan blowups. That’s the rational course, right?

There’s a number of replies I could give to that.

I’ll start by saying that this is a prime example of the sort of thinking I have in mind, when I warn aspiring rationalists to beware of cleverness.

I'll also note that I wouldn't want an attempted Friendly AI that had just decided that the Earth ought to be transformed into paperclips, to assess whether this was a reasonable thing to do in light of all the various warnings it had received against it. I would want it to undergo an automatic controlled shutdown. Who says that meta-reasoning is immune from corruption?

I could mention the important times that my naive, idealistic ethical inhibitions have protected me from *myself*, and placed me in a recoverable position, or helped start the recovery, from very deep mistakes I had no clue I was making. And I could ask whether I've really advanced so much, and whether it would really be all that wise, to remove the protections that saved me before.

Yet even so ... "Am I still dumber than my ethics?" is a question whose answer isn't *automatically* "Yes."

There are obvious silly things here that you shouldn't do; for example, you shouldn't wait until you're really tempted, and *then* try to figure out if you're smarter than your ethics on that particular occasion.

But in general—there's only so much power that can vest in what your parents told you not to do. One shouldn't underestimate the power. Smart people debated historical lessons in the course of forging the Enlightenment ethics that much of Western culture draws upon; and some subcultures, like scientific academia, or science-fiction fandom, draw on those ethics more directly. But even so the power of the past is bounded.

And in fact ...

I've had to make my ethics *much stricter* than what my parents and Jerry Pournelle and Richard Feynman told me not to do.

Funny thing, how when people seem to think they're smarter than their ethics, they argue for *less* strictness rather than *more* strictness. I mean, when you think about how much more complicated the modern world is ...

And along the same lines, the ones who come to me and say, "You should lie about the intelligence explosion, because that way you can get more people to support you; it's the rational

thing to do, for the greater good”—these ones seem to have *no idea* of the risks.

They don't mention the problem of running on corrupted hardware. They don't mention the idea that lies have to be recursively protected from all the truths and all the truthfinding techniques that threaten them. They don't mention that honest ways have a simplicity that dishonest ways often lack. They don't talk about black swan bets. They don't talk about the terrible nakedness of discarding the last defense you have against yourself, and trying to survive on raw calculation.

I am reasonably sure that this is because they have *no clue* about any of these things.

If you've truly understood the reason and the rhythm behind ethics, then one major sign is that, augmented by this newfound knowledge, you *don't do* those things that previously seemed like ethical transgressions. Only now you know why.

Someone who just looks at one or two reasons behind ethics, and says, "Okay, I've understood that, so now I'll take it into account consciously, and therefore I have no more need of ethical inhibitions"—this one is behaving more like a stereotype than a real rationalist. The world isn't simple and pure and clean, so you can't just take the ethics you were raised with and trust them. But that pretense of Vulcan logic, where you think you're just going to compute everything correctly once you've got one or two abstract insights—that doesn't work in real life either.

As for those who, having figured out *none* of this, think themselves smarter than their ethics: Ha.

And as for those who previously thought themselves smarter than their ethics, but who hadn't conceived of all these elements behind ethical injunctions "in so many words" until they ran across this essay, and who *now* think themselves smarter than their ethics, because they're going to take all this into account from now on: Double ha.

I have seen many people struggling to excuse themselves from their ethics. Always the modification is toward lenience, never to be more strict. And I am stunned by the speed and the lightness with which they strive to abandon their protec-

tions. Hobbes said, “I don’t know what’s worse, the fact that everyone’s got a price, or the fact that their price is so low.” So very low the price, so very eager they are to be bought. They don’t look twice and then a third time for alternatives, before deciding that they have no option left but to transgress—though they may look very grave and solemn when they say it. They abandon their ethics at the very first opportunity. “Where there’s a will to failure, obstacles can be found.” The will to fail at ethics seems very strong, in some people.

I don’t know if I can endorse absolute ethical injunctions that bind over all possible epistemic states of a human brain. The universe isn’t kind enough for me to trust that. (Though an ethical injunction against self-deception, for example, does seem to me to have tremendous force. I’ve seen many people arguing for the Dark Side, and none of them seem aware of the network risks or the black-swan risks of self-deception.) If, someday, I attempt to shape a (reflectively consistent) injunction within a self-modifying AI, it will only be after working out the math, because that is so totally not the sort of thing you could get away with doing via an ad-hoc patch.

But I will say this much:

I am completely unimpressed with the knowledge, the reasoning, and the overall level of those folk who have eagerly come to me, and said in grave tones, “It’s rational to do unethical thing X because it will have benefit Y.”

Something to Protect

In the gestalt of (ahem) Japanese fiction, one finds this oft-repeated motif: Power comes from having something to protect.

I'm not just talking about superheroes that power up when a friend is threatened, the way it works in Western fiction. In the Japanese version it runs deeper than that.

In the *X* saga it's explicitly stated that each of the good guys draw their power from having someone—one person—who they want to protect. Who? That question is part of *X*'s plot—the “most precious person” isn't always who we think. But if that person is killed, or hurt in the wrong way, the protector loses their power—not so much from magical backlash, as from simple despair. This isn't something that happens once per week per good guy, the way it would work in a Western comic. It's equivalent to being Killed Off For Real—taken off the game board.

The way it works in Western superhero comics is that the good guy gets bitten by a radioactive spider; and then he needs something to do with his powers, to keep him busy, so he decides to fight crime. And then Western superheroes are always whining about how much time their superhero duties take up, and how they'd rather be ordinary mortals so they could go fishing or something.

Similarly, in Western real life, unhappy people are told that they need a “purpose in life,” so they should pick out an altruistic cause that goes well with their personality, like picking out nice living-room drapes, and this will brighten up their days by adding some color, like nice living-room drapes. You should be careful not to pick something too expensive, though.

In Western comics, the magic comes first, then the purpose: Acquire amazing powers, decide to protect the innocent. In Japanese fiction, often, it works the other way around.



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Of course I'm not saying all this to generalize from fictional evidence. But I want to convey a concept whose deceptively close Western analogue is *not* what I mean.

I have touched before on the idea that a rationalist must have something they value more than "rationality": *the Art must have a purpose other than itself, or it collapses into infinite recursion*. But do not mistake me, and think I am advocating that rationalists should pick out a nice altruistic cause, by way of having something to do, because rationality isn't all that important by itself. No. I am asking: Where do rationalists come from? How do we acquire our powers?

It is written in The Twelve Virtues of Rationality:

How can you improve your conception of rationality? Not by saying to yourself, "It is my duty to be rational." By this you only enshrine your mistaken conception. Perhaps your conception of rationality is that it is rational to believe the words of the Great Teacher, and the Great Teacher says, "The sky is green," and you look up at the sky and see blue. If you think: "It may look like the sky is blue, but rationality is to believe the words of the Great Teacher," you lose a chance to discover your mistake.

Historically speaking, the way humanity *finally* left the trap of authority and began paying attention to, y'know, the actual sky, was that beliefs based on experiment turned out to be *much more useful* than beliefs based on authority. Curiosity has been around since the dawn of humanity, but the problem is that spinning campfire tales works just as well for satisfying curiosity.

Historically speaking, science won because it displayed greater raw strength in the form of technology, not because science *sounded more reasonable*. To this very day, magic and scripture still sound more reasonable to untrained ears than science. That is why there is continuous social tension between the belief systems. If science not only worked better than magic,

but *also* sounded more intuitively reasonable, it would have won *entirely* by now.

Now there are those who say: “How dare you suggest that anything should be valued more than Truth? Must not a rationalist love Truth more than mere usefulness?”

Forget for a moment what would have happened historically to someone like that—that people in pretty much that frame of mind defended the Bible because they loved Truth more than mere accuracy. Propositional morality is a glorious thing, but it has too many degrees of freedom.

No, the real point is that a rationalist’s love affair with the Truth is, well, just *more complicated* as an emotional relationship.

One doesn’t become an adept rationalist without caring about the truth, both as a purely moral desideratum and as something that’s fun to have. I doubt there are many master composers who hate music.

But part of what I *like* about rationality is the discipline imposed by requiring beliefs to yield predictions, which ends up taking us much closer to the truth than if we sat in the living room obsessing about Truth all day. I *like* the complexity of simultaneously having to love True-seeming ideas, and also being ready to drop them out the window at a moment’s notice. I even like the glorious aesthetic purity of declaring that I value mere usefulness above aesthetics. That is almost a contradiction, but not quite; and that has an aesthetic quality as well, a delicious humor.

And of course, no matter how much you profess your love of mere usefulness, you should never *actually* end up deliberately believing a useful false statement.

So don’t oversimplify the relationship between loving truth and loving usefulness. It’s not one or the other. It’s *complicated*, which is not necessarily a defect in the moral aesthetics of single events.

But morality and aesthetics alone, believing that one ought to be “rational” or that certain ways of thinking are “beautiful,” will not lead you to the center of the Way. It wouldn’t have gotten humanity out of the authority-hole.

In *Feeling Moral*, I discussed this dilemma: Which of these options would you prefer?

1. Save 400 lives, with certainty.
2. Save 500 lives, 90% probability; save no lives, 10% probability.

You may be tempted to grandstand, saying, “How dare you gamble with people’s lives?” Even if you, yourself, are one of the 500—but you don’t know which one—you may still be tempted to rely on the comforting feeling of certainty, because our own lives are often worth less to us than a good intuition.

But if your precious daughter is one of the 500, and you don’t know which one, *then*, perhaps, you may feel more impelled to shut up and multiply—to notice that you have an 80% chance of saving her in the first case, and a 90% chance of saving her in the second.

And yes, everyone in that crowd is someone’s son or daughter. Which, in turn, suggests that we should pick the second option as altruists, as well as concerned parents.

My point is not to suggest that one person’s life is more valuable than 499 people. What I am trying to say is that *more* than your own life has to be at stake, before a person becomes desperate enough to resort to math.

What if you believe that it is “rational” to choose the certainty of option 1? Lots of people think that “rationality” is about choosing only methods that are certain to work, and rejecting all uncertainty. But, hopefully, you care more about your daughter’s life than about “rationality.”

Will pride in your own virtue as a rationalist save you? Not if you believe that it is virtuous to choose certainty. You will only be able to learn something about rationality if your daughter’s life matters more to you than your pride as a rationalist.

You may even learn something about rationality from the experience, if you are already far enough grown in your Art to say, “I must have had the wrong conception of rationality,” and not, “Look at how rationality gave me the wrong answer!”

(The essential difficulty in becoming a master rationalist is that you need quite a bit of rationality to bootstrap the learning process.)

Is your belief that you ought to be rational more important than your life? Because, as I've previously observed, risking your life isn't comparatively all that scary. Being the lone voice of dissent in the crowd and having everyone look at you funny is *much* scarier than a mere threat to your life, according to the revealed preferences of teenagers who drink at parties and then drive home. It will take something terribly important to make you willing to leave the pack. A threat to your life won't be enough.

Is your will to rationality stronger than your *pride*? Can it be, if your will to rationality stems from your pride in your self-image as a rationalist? It's helpful—*very* helpful—to have a self-image which says that you are the sort of person who confronts harsh truth. It's helpful to have too much self-respect to knowingly lie to yourself or refuse to face evidence. But there may come a time when you have to admit that you've been doing rationality all wrong. Then your pride, your self-image as a rationalist, may make that too hard to face.

If you've prided yourself on believing what the Great Teacher says—even when it seems harsh, even when you'd rather not—that may make it all the more bitter a pill to swallow, to admit that the Great Teacher is a fraud, and all your noble self-sacrifice was for naught.

Where do you get the will to keep moving forward?

When I look back at my own personal journey toward rationality—not just humanity's historical journey—well, I grew up believing very strongly that I ought to be rational. This made me an above-average Traditional Rationalist à la Feynman and Heinlein, and nothing more. It did not drive me to go beyond the teachings I had received. I only began to grow *further* as a rationalist once I had something terribly important that I needed to do. Something more important than my pride as a rationalist, never mind my life.

Only when you become more wedded to success than to any of your beloved techniques of rationality do you begin to appreciate these words of Miyamoto Musashi:¹

You can win with a long weapon, and yet you can also win with a short weapon. In short, the Way of the Ichi school is the spirit of winning, whatever the weapon and whatever its size.

—Miyamoto Musashi, *The Book of Five Rings*

Don't mistake this for a specific teaching of rationality. It describes how you *learn* the Way, beginning with a desperate need to succeed. No one masters the Way until more than their life is at stake. More than their comfort, more even than their pride.

You can't just pick out a Cause like that because you feel you need a hobby. Go looking for a "good cause," and your mind will just fill in a standard cliché. Learn how to multiply, and perhaps you will recognize a drastically important cause when you see one.

But *if* you have a cause like that, it is right and proper to wield your rationality in its service.

To strictly subordinate the aesthetics of rationality to a higher cause is part of the aesthetic of rationality. You should pay attention to that aesthetic: You will never master rationality well enough to win with any weapon if you do not appreciate the beauty for its own sake.

1. Musashi, *Book of Five Rings*.

When (Not) to Use Probabilities

It may come as a surprise to some readers that I do not always advocate using probabilities.

Or rather, I don't always advocate that human beings, trying to solve their problems, should try to *make up* verbal probabilities, and then apply the laws of probability theory or decision theory to whatever number they just made up, and then use the result as their final belief or decision.

The laws of probability are laws, not suggestions, but often the true Law is too difficult for us humans to compute. If $P \neq NP$ and the universe has no source of exponential computing power, then there are evidential updates too difficult for even a superintelligence to compute—even though the probabilities would be quite well-defined, if we could afford to calculate them.

So sometimes you don't apply probability theory. Especially if you're human, and your brain has evolved with all sorts of useful algorithms for uncertain reasoning, that *don't* involve verbal probability assignments.

Not sure where a flying ball will land? I don't advise trying to formulate a probability distribution over its landing spots, performing deliberate Bayesian updates on your glances at the ball, and calculating the expected utility of all possible strings of motor instructions to your muscles. Trying to catch a flying ball, you're probably better off with your brain's built-in mechanisms than using deliberative verbal reasoning to invent or manipulate probabilities.

But this doesn't mean you're going *beyond* probability theory or *above* probability theory.

The Dutch book arguments still apply. If I offer you a choice of gambles (\$10,000 if the ball lands in this square, versus \$10,000 if I roll a die and it comes up 6), and you answer in a way that does not allow consistent probabilities to be assigned,



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then you will accept combinations of gambles that are certain losses, or reject gambles that are certain gains ...

Which still doesn't mean that you should try to use deliberative verbal reasoning. I would expect that for professional baseball players, at least, it's more important to catch the ball than to assign consistent probabilities. Indeed, if you tried to make up probabilities, the *verbal* probabilities might not even be very good ones, compared to some gut-level feeling—some wordless representation of uncertainty in the back of your mind.

There is nothing privileged about uncertainty that is expressed in words, unless the verbal parts of your brain *do*, in fact, happen to work better on the problem.

And while accurate maps of the same territory will necessarily be consistent among themselves, not all consistent maps are accurate. It is more important to be accurate than to be consistent, and more important to catch the ball than to be consistent.

In fact, I generally advise against *making up* probabilities, unless it seems like you have some decent basis for them. This only fools you into believing that you are more Bayesian than you actually are.

To be specific, I would advise, in most cases, against using non-numerical procedures to create what appear to be numerical probabilities. Numbers should come from numbers.

Now there *are* benefits from trying to translate your gut feelings of uncertainty into verbal probabilities. It may help you spot problems like the conjunction fallacy. It may help you spot internal inconsistencies—though it may not show you any way to remedy them.

But you shouldn't go around thinking that if you translate your gut feeling into "one in a thousand," then, on occasions when you emit these verbal words, the corresponding event will happen around one in a thousand times. Your brain is not so well-calibrated. If instead you do something nonverbal with your gut feeling of uncertainty, you may be better off, because at least you'll be *using* the gut feeling the way it was meant to be used.

This specific topic came up recently in the context of the Large Hadron Collider, and an argument given¹ at the Global Catastrophic Risks conference:

That we couldn't be sure that there was no error in the papers which showed from multiple angles that the LHC couldn't possibly destroy the world. And moreover, the theory used in the papers might be wrong. And in either case, there was still a chance the LHC *could* destroy the world. And therefore, it ought not to be turned on.

Now if the argument had been given in *just* this way, I would not have objected to its epistemology.

But the speaker actually purported to assign a probability of at least 1 in 1,000 that the theory, model, or calculations in the LHC paper were wrong; and a probability of at least 1 in 1,000 that, if the theory or model or calculations were wrong, the LHC would destroy the world.

After all, it's surely not so improbable that future generations will reject the theory used in the LHC paper, or reject the model, or maybe just find an error. And if the LHC paper is wrong, then who knows what might happen as a result?

So that is an argument—but to assign *numbers* to it?

I object to the air of authority given to these numbers pulled out of thin air. I generally feel that if you can't use probabilistic tools to shape your feelings of uncertainty, you ought not to dignify them by calling them probabilities.

The alternative I would propose, in this particular case, is to debate the general rule of banning physics experiments because you cannot be absolutely certain of the arguments that say they are safe.

I hold that if you phrase it this way, then your mind, by considering frequencies of events, is likely to bring in more consequences of the decision, and remember more relevant historical cases.

1. [sharov.me/r/5-16](https://www.sharov.me/r/5-16)
global-catastrophic-risks.com



If you debate just the one case of the LHC, and assign specific probabilities, it (1) gives very shaky reasoning an undue air of authority, (2) obscures the general consequences of applying similar rules, and even (3) creates the illusion that we might come to a different decision if someone else published a new physics paper that decreased the probabilities.

The authors at the Global Catastrophic Risk conference seemed to be suggesting that we could just do a bit more analysis of the LHC and *then* switch it on. This struck me as the most disingenuous part of the argument. Once you admit the argument “Maybe the analysis could be wrong, and who knows what happens then,” there is no possible physics paper that can ever get rid of it.

No matter what other physics papers had been published previously, the authors would have used the same argument *and made up the same numerical probabilities* at the Global Catastrophic Risk conference. I cannot be sure of this statement, of course, but it has a probability of 75%.

In general a rationalist tries to make their minds function at the best achievable power output; sometimes this involves *talking* about verbal probabilities, and sometimes it does not, but always the laws of probability theory *govern*.

If all you have is a gut feeling of uncertainty, then you should probably stick with those algorithms that make use of gut feelings of uncertainty, because your built-in algorithms may do better than your clumsy attempts to put things into words.

Now it may be that by reasoning thusly, I may find myself inconsistent. For example, I would be substantially more alarmed about a lottery device with a well-defined chance of 1 in 1,000,000 of destroying the world, than I am about the Large Hadron Collider being switched on.

On the other hand, if you asked me whether I could make one million statements of authority equal to “The Large Hadron Collider will not destroy the world,” and be wrong, on average, around once, then I would have to say no.

What should I do about this inconsistency? I’m not sure, but I’m certainly not going to wave a magic wand to make it go away. That’s like finding an inconsistency in a pair of maps

you own, and quickly scribbling some alterations to make sure they're consistent.

I would also, by the way, be substantially more worried about a lottery device with a 1 in 1,000,000,000 chance of destroying the world, than a device which destroyed the world if the Judeo-Christian God existed. But I would not suppose that I could make one billion statements, one after the other, fully independent and equally fraught as "There is no God," and be wrong on average around once.

I can't say I'm *happy* with this state of epistemic affairs, but I'm not going to modify it until I can see myself moving in the direction of *greater accuracy and real-world effectiveness*, not just moving in the direction of greater self-consistency. The goal is to win, after all. If I make up a probability that is not shaped by probabilistic tools, if I make up a number that is not created by numerical methods, then maybe I am just defeating my built-in algorithms that would do better by reasoning in their native modes of uncertainty.

Of course this is not a license to ignore probabilities that are well-founded. Any numerical founding at all is likely to be better than a vague feeling of uncertainty; humans are terrible statisticians. But pulling a number *entirely* out of your butt, that is, using a non-numerical procedure to produce a number, is nearly no foundation at all; and in that case you probably are better off sticking with the vague feelings of uncertainty.

Which is why my writing generally uses words like "maybe" and "probably" and "surely" instead of assigning made-up numerical probabilities like "40%" and "70%" and "95%." Think of how silly that would look. I think it actually *would* be silly; I think I would do worse thereby.

I am not the kind of straw Bayesian who says that you should make up probabilities to avoid being subject to Dutch books. I am the sort of Bayesian who says that in practice, humans end up subject to Dutch books because they aren't powerful enough to avoid them; and moreover it's more important to catch the ball than to avoid Dutch books. The math is like *underlying* physics, inescapably governing, but too expensive to calculate.

Nor is there any point in a ritual of cognition that *mimics* the surface forms of the math, but fails to produce systematically better decision-making. That would be a lost purpose; this is not the true art of living under the law.

Newcomb's Problem and Regret of Rationality

The following may well be the most controversial dilemma in the history of decision theory:



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A superintelligence from another galaxy, whom we shall call Omega, comes to Earth and sets about playing a strange little game. In this game, Omega selects a human being, sets down two boxes in front of them, and flies away.

Box *A* is transparent and contains a thousand dollars.

Box *B* is opaque, and contains either a million dollars, or nothing.

You can take both boxes, or take only box *B*.

And the twist is that Omega has put a million dollars in box *B* if and only if Omega has predicted that you will take only box *B*.

Omega has been correct on each of 100 observed occasions so far—everyone who took both boxes has found box *B* empty and received only a thousand dollars; everyone who took only box *B* has found *B* containing a million dollars. (We assume that box *A* vanishes in a puff of smoke if you take only box *B*; no one else can take box *A* afterward.)

Before you make your choice, Omega has flown off and moved on to its next game. Box *B* is already empty or already full.

Omega drops two boxes on the ground in front of you and flies off.

Do you take both boxes, or only box *B*?

And the standard philosophical conversation runs thusly:

ONE-BOXER: "I take only box *B*, of course. I'd rather have a million than a thousand."

TWO-BOXER: "Omega has already left. Either box *B* is already full or already empty. If box *B* is already empty, then taking both boxes nets me \$1,000, taking only box *B* nets me \$0. If box *B* is already full, then taking both boxes nets \$1,001,000, taking only box *B* nets \$1,000,000. In either case I do better by taking both boxes, and worse by leaving a thousand dollars on the table—so I will be rational, and take both boxes."

ONE-BOXER: "If you're so rational, why ain'cha rich?"

TWO-BOXER: "It's not my fault Omega chooses to reward only people with irrational dispositions, but it's already too late for me to do anything about that."

There is a *large* literature on the topic of Newcomblike problems—especially if you consider the Prisoner's Dilemma as a special case, which it is generally held to be. *Paradoxes of Rationality and Cooperation: Prisoner's Dilemma and Newcomb's Problem*¹ is an edited volume that includes Newcomb's original essay. For those who read only online material, Ledwig's PhD thesis summarizes the major standard positions.²

I'm not going to go into the whole literature, but the dominant consensus in modern decision theory is that one should two-box, and Omega is just rewarding agents with irrational dispositions. This dominant view goes by the name of "causal decision theory."

I'm not going to try to present my own analysis here. Way too long a story, even by my standards.

1. Richmond Campbell and Lanning Snowden, eds., *Paradoxes of Rationality and Cooperation: Prisoner's Dilemma and Newcomb's Problem* (Vancouver: University of British Columbia Press, 1985).

2. Marion Ledwig, "Newcomb's Problem" (PhD diss., University of Constance, 2000).

But it is agreed even among causal decision theorists that if you have the power to precommit yourself to take one box, in Newcomb's Problem, then you should do so. If you can precommit yourself before Omega examines you, then you are directly causing box *B* to be filled.

Now in my field—which, in case you have forgotten, is self-modifying AI—this works out to saying that if you build an AI that two-boxes on Newcomb's Problem, it will self-modify to one-box on Newcomb's Problem, if the AI considers in advance that it might face such a situation. Agents with free access to their own source code have access to a cheap method of precommitment.

What if you expect that you might, in general, face a Newcomblike problem, without knowing the exact form of the problem? Then you would have to modify yourself into a sort of agent whose disposition was such that it would generally receive high rewards on Newcomblike problems.

But what does an agent with a disposition generally-well-suited to Newcomblike problems look like? Can this be formally specified?

Yes, but when I tried to write it up, I realized that I was starting to write a small book. And it wasn't the most important book I had to write, so I shelved it. My slow writing speed really is the bane of my existence. The theory I worked out seems, to me, to have many nice properties besides being well-suited to Newcomblike problems. It would make a nice PhD thesis, if I could get someone to accept it as my PhD thesis. But that's pretty much what it would take to make me unshelve the project. Otherwise I can't justify the time expenditure, not at the speed I currently write books.

I say all this, because there's a common attitude that "Verbal arguments for one-boxing are easy to come by; what's hard is developing a good decision theory that one-boxes"—coherent math which one-boxes on Newcomb's Problem without producing absurd results elsewhere. So I do understand that, and I did set out to develop such a theory, but my writing speed on big papers is so slow that I can't publish it. Believe it or not, it's true.

Nonetheless, I would like to present some of my *motivations* on Newcomb's Problem—the reasons I felt impelled to seek a new theory—because they illustrate my source-attitudes toward rationality. Even if I can't present the theory that these motivations motivate ...

First, foremost, fundamentally, above all else:

Rational agents should WIN.

Don't mistake me, and think that I'm talking about the Hollywood Rationality stereotype that rationalists should be selfish or shortsighted. If your utility function has a term in it for others, then win their happiness. If your utility function has a term in it for a million years hence, then win the eon.

But at any rate, WIN. Don't lose reasonably; **WIN**.

Now there are defenders of causal decision theory who argue that the two-boxers are doing their best to win, and cannot help it if they have been cursed by a Predictor who favors irrationalists. I will talk about this defense in a moment. But first, I want to draw a distinction between causal decision theorists who believe that two-boxers are genuinely doing their best to win; versus someone who thinks that two-boxing is the *reasonable* or the *rational* thing to do, but that the reasonable move just happens to predictably lose, in this case. There are a *lot* of people out there who think that rationality predictably loses on various problems—that, too, is part of the Hollywood Rationality stereotype, that Kirk is predictably superior to Spock.

Next, let's turn to the charge that Omega favors irrationalists. I can conceive of a superbeing who rewards only people born with a particular gene, *regardless of their choices*. I can conceive of a superbeing who rewards people whose brains inscribe the *particular algorithm* of "Describe your options in English and choose the last option when ordered alphabetically," but who does not reward anyone who chooses the same option for a different reason. But Omega rewards people who choose to take only box *B*, *regardless of which algorithm they use to arrive at this decision*, and this is why I don't buy the charge that Omega is rewarding the irrational. Omega doesn't care whether or not you follow some particular ritual of cognition; Omega only cares about your predicted *decision*.

We can choose whatever reasoning algorithm we like, and will be rewarded or punished only according to that algorithm's choices, with no other dependency—Omega just cares where we go, not how we got there.

It is precisely the notion that Nature does not care about our *algorithm* that frees us up to pursue the winning Way—without attachment to any particular ritual of cognition, apart from our belief that it wins. Every rule is up for grabs, *except* the rule of winning.

As Miyamoto Musashi said—it's really worth repeating:

You can win with a long weapon, and yet you can also win with a short weapon. In short, the Way of the Ichi school is the spirit of winning, whatever the weapon and whatever its size.³

(Another example: It was argued by McGee that we must adopt bounded utility functions or be subject to “Dutch books” over infinite times. But: *The utility function is not up for grabs*. I love life without limit or upper bound; there is no finite amount of life lived N where I would prefer an 80.0001% probability of living N years to a 0.0001% chance of living a googolplex years and an 80% chance of living forever. This is a sufficient condition to imply that my utility function is unbounded. So I just have to figure out how to optimize *for that morality*. You can't tell me, first, that above all I must conform to a particular ritual of cognition, and then that, if I conform to that ritual, I must change my morality to avoid being Dutch-booked. Toss out the losing ritual; don't change the definition of winning. That's like deciding to prefer \$1,000 to \$1,000,000 so that Newcomb's Problem doesn't make your preferred ritual of cognition look bad.)

“But,” says the causal decision theorist, “to take only one box, you must somehow believe that your choice can affect whether box B is empty or full—and that's *unreasonable*! Omega has already left! It's physically impossible!”

3. Musashi, *Book of Five Rings*.

Unreasonable? I am a rationalist: what do I care about being unreasonable? I don't have to conform to a particular ritual of cognition. I don't have to take only box *B* because *I believe my choice affects the box, even though Omega has already left*. I can just ...take only box *B*.

I do have a proposed alternative ritual of cognition that computes this decision, which this margin is too small to contain; but I shouldn't need to show this to you. The point is not to have an elegant theory of winning—the point is to win; elegance is a side effect.

Or to look at it another way: Rather than starting with a concept of what is the reasonable decision, and then asking whether “reasonable” agents leave with a lot of money, start by looking at the agents who leave with a lot of money, develop a theory of which agents tend to leave with the most money, and from this theory, try to figure out what is “reasonable.” “Reasonable” may just refer to decisions in conformance with our current ritual of cognition—what else would determine whether something seems “reasonable” or not?

From James Joyce (no relation), *Foundations of Causal Decision Theory*:⁴

Rachel has a perfectly good answer to the “Why ain't you rich?” question. “I am not rich,” she will say, “because I am not the kind of person the psychologist thinks will refuse the money. I'm just not like you, Irene. Given that I know that I am the type who takes the money, and given that the psychologist knows that I am this type, it was reasonable of me to think that the \$1,000,000 was not in my account. The \$1,000 was the most I was going to get no matter what I did. So the only reasonable thing for me to do was to take it.”

Irene may want to press the point here by asking, “But don't you wish you were like me, Rachel?”

4. James M. Joyce, *The Foundations of Causal Decision Theory* (New York: Cambridge University Press, 1999), doi:10.1017/CBO9780511498497.

Don't you wish that you were the refusing type?" There is a tendency to think that Rachel, a committed causal decision theorist, must answer this question in the negative, which seems obviously wrong (given that being like Irene would have made her rich). This is not the case. Rachel can and should admit that she does wish she were more like Irene. "It would have been better for me," she might concede, "had I been the refusing type." At this point Irene will exclaim, "You've admitted it! It wasn't so smart to take the money after all." Unfortunately for Irene, her conclusion does not follow from Rachel's premise. Rachel will patiently explain that wishing to be a refuser in a Newcomb problem is not inconsistent with thinking that one should take the \$1,000 *whatever type one is*. When Rachel wishes she was Irene's type she is wishing *for Irene's options*, not sanctioning her choice.

It is, I would say, a general principle of rationality—indeed, part of how I *define* rationality—that you never end up envying someone else's mere *choices*. You might envy someone their genes, if Omega rewards genes, or if the genes give you a generally happier disposition. But Rachel, above, envies Irene her choice, and *only* her choice, irrespective of what algorithm Irene used to make it. Rachel wishes *just* that she had a disposition to choose differently.

You shouldn't claim to be more rational than someone and simultaneously envy them their choice—*only* their choice. Just *do* the act you envy.

I keep trying to say that rationality is the winning-Way, but causal decision theorists insist that taking both boxes is what *really* wins, because you *can't possibly do better* by leaving \$1,000 on the table ...even though the single-boxers leave the experiment with more money. Be careful of this sort of argument, any time you find yourself defining the "winner" as someone other than the agent who is currently smiling from on top of a giant heap of utility.

Yes, there are various thought experiments in which some agents start out with an advantage—but if the task is to, say, decide whether to jump off a cliff, you want to be careful not to define cliff-refraining agents as having an unfair prior advantage over cliff-jumping agents, by virtue of their unfair refusal to jump off cliffs. At this point you have covertly redefined “winning” as conformance to a particular ritual of cognition. *Pay attention to the money!*

Or here’s another way of looking at it: Faced with Newcomb’s Problem, would you want to look really hard for a reason to believe that it was perfectly reasonable and rational to take only box *B*; because, if such a line of argument existed, you would take only box *B* and find it full of money? Would you spend an extra hour thinking it through, if you were confident that, at the end of the hour, you would be able to convince yourself that box *B* was the rational choice? This too is a rather odd position to be in. Ordinarily, the work of rationality goes into figuring out which choice is the best—not finding a reason to believe that a particular choice is the best.

Maybe it’s too easy to say that you “ought to” two-box on Newcomb’s Problem, that this is the “reasonable” thing to do, so long as the money isn’t actually in front of you. Maybe you’re just numb to philosophical dilemmas, at this point. What if your daughter had a 90% fatal disease, and box *A* contained a serum with a 20% chance of curing her, and box *B* might contain a serum with a 95% chance of curing her? What if there was an asteroid rushing toward Earth, and box *A* contained an asteroid deflector that worked 10% of the time, and box *B* might contain an asteroid deflector that worked 100% of the time?

Would you, at that point, find yourself *tempted to make an unreasonable choice*?

If the stake in box *B* was something you *could not* leave behind? Something overwhelmingly more important to you than being reasonable? If you absolutely *had to* win—*really* win, not just be defined as winning?

Would you *wish with all your power* that the “reasonable” decision were to take only box *B*?

Then maybe it's time to update your definition of reasonableness.

Alleged rationalists should not find themselves envying the mere decisions of alleged nonrationalists, because your decision can be whatever you like. When you find yourself in a position like this, you shouldn't chide the other person for failing to conform to your concepts of reasonableness. You should realize you got the Way wrong.

So, too, if you ever find yourself keeping separate track of the "reasonable" belief, versus the belief that seems likely to be actually *true*. Either you have misunderstood reasonableness, or your second intuition is just wrong.

Now one can't simultaneously *define* "rationality" as the winning Way, and *define* "rationality" as Bayesian probability theory and decision theory. But it is the argument that I am putting forth, and the moral of my advice to trust in Bayes, that the laws governing winning have indeed proven to be math. If it ever turns out that Bayes fails—receives systematically lower rewards on some problem, relative to a superior alternative, in virtue of its mere decisions—then Bayes has to go *out the window*. "Rationality" is just the label I use for my beliefs about the winning Way—the Way of the agent smiling from on top of the giant heap of utility. *Currently*, that label refers to Bayescraft.

I realize that this is not a knockdown criticism of causal decision theory—that would take the actual book and/or PhD thesis—but I hope it illustrates some of my underlying attitude toward this notion of "rationality."

[Edit 2015: I've now written a book-length exposition of a decision theory that dominates causal decision theory, "Timeless Decision Theory."^{5,6} The cryptographer Wei Dai has responded with another alternative to causal decision theory, updateless decision theory, that dominates both causal and timeless decision theory. As of 2015, the best up-to-date discussions of these

5. Yudkowsky, *Timeless Decision Theory*.

6. Yudkowsky and Soares, "Functional Decision Theory: A New Theory of Instrumental Rationality" sharov.me/r/5-17



theories are Daniel Hintze’s “Problem Class Dominance in Predictive Dilemmas”⁷ and Nate Soares and Benja Fallenstein’s “Toward Idealized Decision Theory.”⁸]

You shouldn’t find yourself distinguishing the winning choice from the reasonable choice. Nor should you find yourself distinguishing the reasonable belief from the belief that is most likely to be true.

That is why I use the word “rational” to denote my beliefs about accuracy and winning—*not* to denote verbal reasoning, or strategies which yield certain success, or that which is logically provable, or that which is publicly demonstrable, or that which is reasonable.

As Miyamoto Musashi said:

The primary thing when you take a sword in your hands is your intention to cut the enemy, whatever the means. Whenever you parry, hit, spring, strike or touch the enemy’s cutting sword, you must cut the enemy in the same movement. It is essential to attain this. If you think only of hitting, springing, striking or touching the enemy, you will not be able actually to cut him.

7. Daniel Hintze, “Problem Class Dominance in Predictive Dilemmas,” Honors thesis (2014).

8. Nate Soares and Benja Fallenstein, “Toward Idealized Decision Theory,” Technical report. Berkeley, CA: Machine Intelligence Research Institute (2014), sharov.me/r/5-18.



Interlude: The Twelve Virtues of Rationality

The first virtue is curiosity. A burning itch to know is higher than a solemn vow to pursue truth. To feel the burning itch of curiosity requires both that you be ignorant, and that you desire to relinquish your ignorance. If in your heart you believe you already know, or if in your heart you do not wish to know, then your questioning will be purposeless and your skills without direction. Curiosity seeks to annihilate itself; there is no curiosity that does not want an answer. The glory of glorious mystery is to be solved, after which it ceases to be mystery. Be wary of those who speak of being open-minded and modestly confess their ignorance. There is a time to confess your ignorance and a time to relinquish your ignorance.

The second virtue is relinquishment. P. C. Hodgell said: “That which can be destroyed by the truth should be.”¹ Do not flinch from experiences that might destroy your beliefs. The thought you cannot think controls you more than thoughts you speak aloud. Submit yourself to ordeals and test yourself in fire. Relinquish the emotion which rests upon a mistaken belief, and seek to feel fully that emotion which fits the facts. If the iron approaches your face, and you believe it is hot, and it is cool, the Way opposes your fear. If the iron approaches your face, and you believe it is cool, and it is hot, the Way opposes your calm. Evaluate your beliefs first and then arrive at your emotions. Let yourself say: “If the iron is hot, I desire to believe it is hot, and if it is cool, I desire to believe it is cool.” Beware lest you become attached to beliefs you may not want.

The third virtue is lightness. Let the winds of evidence blow you about as though you are a leaf, with no direction of your own. Beware lest you fight a rearguard retreat against the evidence, grudgingly conceding each foot of ground only when forced, feeling cheated. Surrender to the truth as quickly as you



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1. Patricia C. Hodgell, *Seeker's Mask* (Meisha Merlin Publishing, Inc., 2001).

can. Do this the instant you realize what you are resisting, the instant you can see from which quarter the winds of evidence are blowing against you. Be faithless to your cause and betray it to a stronger enemy. If you regard evidence as a constraint and seek to free yourself, you sell yourself into the chains of your whims. For you cannot make a true map of a city by sitting in your bedroom with your eyes shut and drawing lines upon paper according to impulse. You must walk through the city and draw lines on paper that correspond to what you see. If, seeing the city unclearly, you think that you can shift a line just a little to the right, just a little to the left, according to your caprice, this is just the same mistake.

The fourth virtue is evenness. One who wishes to believe says, "Does the evidence permit me to believe?" One who wishes to disbelieve asks, "Does the evidence force me to believe?" Beware lest you place huge burdens of proof only on propositions you dislike, and then defend yourself by saying: "But it is good to be skeptical." If you attend only to favorable evidence, picking and choosing from your gathered data, then the more data you gather, the less you know. If you are selective about which arguments you inspect for flaws, or how hard you inspect for flaws, then every flaw you learn how to detect makes you that much stupider. If you first write at the bottom of a sheet of paper "And therefore, the sky is green!" it does not matter what arguments you write above it afterward; the conclusion is already written, and it is already correct or already wrong. To be clever in argument is not rationality but rationalization. Intelligence, to be useful, must be used for something other than defeating itself. Listen to hypotheses as they plead their cases before you, but remember that you are not a hypothesis; you are the judge. Therefore do not seek to argue for one side or another, for if you knew your destination, you would already be there.

The fifth virtue is argument. Those who wish to fail must first prevent their friends from helping them. Those who smile wisely and say "I will not argue" remove themselves from help and withdraw from the communal effort. In argument strive for exact honesty, for the sake of others and also yourself: the part

of yourself that distorts what you say to others also distorts your own thoughts. Do not believe you do others a favor if you accept their arguments; the favor is to you. Do not think that fairness to all sides means balancing yourself evenly between positions; truth is not handed out in equal portions before the start of a debate. You cannot move forward on factual questions by fighting with fists or insults. Seek a test that lets reality judge between you.

The sixth virtue is empiricism. The roots of knowledge are in observation and its fruit is prediction. What tree grows without roots? What tree nourishes us without fruit? If a tree falls in a forest and no one hears it, does it make a sound? One says, "Yes it does, for it makes vibrations in the air." Another says, "No it does not, for there is no auditory processing in any brain." Though they argue, one saying "Yes," and one saying "No," the two do not anticipate any different experience of the forest. Do not ask which beliefs to profess, but which experiences to anticipate. Always know which difference of experience you argue about. Do not let the argument wander and become about something else, such as someone's virtue as a rationalist. Jerry Cleaver said: "What does you in is not failure to apply some high-level, intricate, complicated technique. It's overlooking the basics. Not keeping your eye on the ball."² Do not be blinded by words. When words are subtracted, anticipation remains.

The seventh virtue is simplicity. Antoine de Saint-Exupéry said: "Perfection is achieved not when there is nothing left to add, but when there is nothing left to take away."³ Simplicity is virtuous in belief, design, planning, and justification. When you profess a huge belief with many details, each additional detail is another chance for the belief to be wrong. Each specification adds to your burden; if you can lighten your burden you must do so. There is no straw that lacks the power to break your back. Of artifacts it is said: The most reliable gear is the one that is designed out of the machine. Of plans: A tangled web breaks. A chain of a thousand links will arrive at a correct conclusion if

2. Cleaver, *Immediate Fiction: A Complete Writing Course*.

3. Antoine de Saint-Exupéry, *Terre des Hommes* (Paris: Gallimard, 1939).

every step is correct, but if one step is wrong it may carry you anywhere. In mathematics a mountain of good deeds cannot atone for a single sin. Therefore, be careful on every step.

The eighth virtue is humility. To be humble is to take specific actions in anticipation of your own errors. To confess your fallibility and then do nothing about it is not humble; it is boasting of your modesty. Who are most humble? Those who most skillfully prepare for the deepest and most catastrophic errors in their own beliefs and plans. Because this world contains many whose grasp of rationality is abysmal, beginning students of rationality win arguments and acquire an exaggerated view of their own abilities. But it is useless to be superior: Life is not graded on a curve. The best physicist in ancient Greece could not calculate the path of a falling apple. There is no guarantee that adequacy is possible given your hardest effort; therefore spare no thought for whether others are doing worse. If you compare yourself to others you will not see the biases that all humans share. To be human is to make ten thousand errors. No one in this world achieves perfection.

The ninth virtue is perfectionism. The more errors you correct in yourself, the more you notice. As your mind becomes more silent, you hear more noise. When you notice an error in yourself, this signals your readiness to seek advancement to the next level. If you tolerate the error rather than correcting it, you will not advance to the next level and you will not gain the skill to notice new errors. In every art, if you do not seek perfection you will halt before taking your first steps. If perfection is impossible that is no excuse for not trying. Hold yourself to the highest standard you can imagine, and look for one still higher. Do not be content with the answer that is almost right; seek one that is exactly right.

The tenth virtue is precision. One comes and says: The quantity is between 1 and 100. Another says: The quantity is between 40 and 50. If the quantity is 42 they are both correct, but the second prediction was more useful and exposed itself to a stricter test. What is true of one apple may not be true of another apple; thus more can be said about a single apple than about all the apples in the world. The narrowest statements slice deepest,

the cutting edge of the blade. As with the map, so too with the art of mapmaking: The Way is a precise Art. Do not walk to the truth, but dance. On each and every step of that dance your foot comes down in exactly the right spot. Each piece of evidence shifts your beliefs by exactly the right amount, neither more nor less. What is exactly the right amount? To calculate this you must study probability theory. Even if you cannot do the math, knowing that the math exists tells you that the dance step is precise and has no room in it for your whims.

The eleventh virtue is scholarship. Study many sciences and absorb their power as your own. Each field that you consume makes you larger. If you swallow enough sciences the gaps between them will diminish and your knowledge will become a unified whole. If you are gluttonous you will become vaster than mountains. It is especially important to eat math and science which impinge upon rationality: evolutionary psychology, heuristics and biases, social psychology, probability theory, decision theory. But these cannot be the only fields you study. The Art must have a purpose other than itself, or it collapses into infinite recursion.

Before these eleven virtues is a virtue which is nameless.

Miyamoto Musashi wrote, in *The Book of Five Rings*:⁴

The primary thing when you take a sword in your hands is your intention to cut the enemy, whatever the means. Whenever you parry, hit, spring, strike or touch the enemy's cutting sword, you must cut the enemy in the same movement. It is essential to attain this. If you think only of hitting, springing, striking or touching the enemy, you will not be able actually to cut him. More than anything, you must be thinking of carrying your movement through to cutting him.

Every step of your reasoning must cut through to the correct answer in the same movement. More than anything, you must think of carrying your map through to reflecting the territory.

4. Musashi, *Book of Five Rings*.

If you fail to achieve a correct answer, it is futile to protest that you acted with propriety.

How can you improve your conception of rationality? Not by saying to yourself, "It is my duty to be rational." By this you only enshrine your mistaken conception. Perhaps your conception of rationality is that it is rational to believe the words of the Great Teacher, and the Great Teacher says, "The sky is green," and you look up at the sky and see blue. If you think, "It may look like the sky is blue, but rationality is to believe the words of the Great Teacher," you lose a chance to discover your mistake.

Do not ask whether it is "the Way" to do this or that. Ask whether the sky is blue or green. If you speak overmuch of the Way you will not attain it.

You may try to name the highest principle with names such as "the map that reflects the territory" or "experience of success and failure" or "Bayesian decision theory." But perhaps you describe incorrectly the nameless virtue. How will you discover your mistake? Not by comparing your description to itself, but by comparing it to that which you did not name.

If for many years you practice the techniques and submit yourself to strict constraints, it may be that you will glimpse the center. Then you will see how all techniques are one technique, and you will move correctly without feeling constrained. Musashi wrote: "When you appreciate the power of nature, knowing the rhythm of any situation, you will be able to hit the enemy naturally and strike naturally. All this is the Way of the Void."

These then are twelve virtues of rationality:

Curiosity, relinquishment, lightness, evenness, argument, empiricism, simplicity, humility, perfectionism, precision, scholarship, and the void.

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